أدنوك البحرية
ADNOC Offshore

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ADNOC OFFSHORE
UMM SHAIF LONG TERM DEVELOPMENT PLAN 2

DOCUMENT TITLE:

FEED SCOPE OF WORK

FOR
US LTDP 2.0 PKG 2 DAS MIN FACILTIES TERMINAL PROJECT
PROJECT NO. 12026

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1	Sep-25	Issued for Review	Abdallah Farahat	Mohamed Almansoori	Jaber Al Shehhi	Ahmed Marei Al Radhi
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1. INTRODUCTION

1.1 Purpose of Document

This document provides a briefing to the ENGINEERING CONTRACTOR and outlines the Scope of Work for the FEED (Front-End Engineering Design) stage of Phase-1 of the "DAS Minimum Facilities for NASR CRUDE" project. It defines the required modifications, design basis and boundaries, deliverables, and general requirements for the FEED activities.

While this document sets the framework, it is the ENGINEERING CONTRACTOR's responsibility to conduct comprehensive FEED studies to meet the Project Objectives.

The document serves as a reference for the ENGINEERING CONTRACTOR—appointed by the COMPANY—who will be responsible for executing the FEED engineering scope. Although detailed, the document is not intended to specify every individual item or material, but rather to provide a high-level overview of the Scope of Work and its key components.

1.2 General Information

1.2.1 Use of Language

All documents relating to this SOW and all correspondence between the COMPANY and ENGINEERING CONTRACTOR shall be in English. Throughout this specification, the terms 'will,' 'may/can,' 'should' and 'shall/must,' when used in the context of actions by COMPANY / ENGINEERING CONTRACTOR, have specific meanings as follows:

Term	Meaning
Will	Is used normally in connection with action by COMPANY and/or nominated representative, rather than by ENGINEERING CONTRACTOR
May / Can	Is used where alternatives / actions are equally acceptable
Should	Is used where provision is preferred
Shall / Must	Is used where a provision is mandatory / vital

Table 1-1 Use of Language

1.2.2 Codes, Standards & Specifications

All technical options and recommended technical solutions shall comply with the relevant Legislation, Codes and Standards in accordance with the order of precedence discussed in Section 1.2.3. Any proposed technical deviations shall be managed under Project MOC Procedure.

The list of applicable Codes and Standards is provided in Appendix 1 of this document. All standards and specifications shall utilize the latest edition on the effective date of the award unless stated otherwise.

1.2.3 Order of Precedence

The ENGINEERING CONTRACTOR, including Sub-Contractor, shall comply with UAE Statutory Legislation and Regulation inclusive of ADNOC Standards, Applicable International Laws and Regulations, Project Specifications, COMPANY Specifications, Regulations, Best Practices and Standards, Guidance, and

Practice Documents and relevant International Codes and Standards. Where equal but conflicting requirements are identified between regulations, specifications, codes and standards, the order of precedence shall be as follows:

- UAE Legislation and Statutory Regulation inclusive of ADNOC HSE standards.
- ADNOC Group Practices and Standards (AGPS)
- PROJECT Documents.
- COMPANY Engineering Specifications, Standards and Procedures.
- International Codes and Standards.

Where conflict arises between various specifications, standards and procedures, the most stringent requirements shall govern. ENGINEERING CONTRACTOR/SUB-CONTRACTOR shall refer to any such conflict and any other matter which requires further interpretation, to the attention of the COMPANY and COMPANY's, in writing, interpretation shall be final.

Any exception / deviation due to technical constraints shall be submitted with justification as an itemized list for COMPANY acceptance. Any deviation not included in the list of deviations shall not be considered and is deemed to be in full compliance with all specifications.

Review and Approval of project deliverable documents shall not relieve ENGINEERING CONTRACTOR of their responsibility in supplying deliverables in full compliance with the technical documents as included in this SOW.

The ADNOC Group Practices and Standards (AGPS) shall be used for the design of the Project. Any reference to superseded COMPANY Standards or Code of Practice in the Select Stage documentation shall be updated by ENGINEERING CONTRACTOR in FEED stage.

The latest edition/revision of COMPANY Technical Standards Documents List, on the effective date of the award, shall be provided to the successful bidder.

1.2.4 Data Utilization

ENGINEERING CONTRACTOR shall not rely upon any SELECT Documents and other documents provided in this SOW package, which provide information related to any existing facility or SITE conditions. ENGINEERING CONTRACTOR is required to satisfy itself with the site conditions and must undertake appropriate SITE visits and investigations to verify data relevant to existing facilities and site conditions and complete its design for Procurement and Installations. ENGINEERING CONTRACTOR’s reliance on any such information is at ENGINEERING CONTRACTOR’s own risk.

The exceptions to the above are the following, which shall be considered relied upon information for Engineering CONTRACTOR to complete its design:

- Subsurface data, including production profiles and compositions.

- Island and pipeline surveys, including new soil surveys to be provided prior to FEED.
- Metocean and Bathymetry data.
- New Artificial Island and New WHT locations.
- PROJECT Battery limits.
- Overall, Field Development Strategy.

1.2.5 Abbreviation

Abbreviation	Description
ADNOC	Abu Dhabi National Oil Company
AGPS	ADNOC Group Practices and Standards
AI	Artificial Island
API	American Petroleum Institute
ARP	Asset Replacement Plan
EEMUA	Engineering Equipment and Materials Users Association
FEED	Front End Engineering Design
FID	Final Investment Decision
DC	Drilling Centre
DSP	Decision Support Package
GBS	Gravity Based Structure
IEC	International Electrotechnical Commission
LTDP	Long Term Development Plan
MBD	Thousand Barrel per Day
MMBPD	Million Metric Barrel Per Day
MOC	Management Of Change
MP	Medium Pressure
NFPA	National Fire Protection Association
PIP	Process Industry Practices
RP	Riser Platform
RTOC	Real time Operating Centre
SW	Sea Water
TSI	Telecom System Integrator
UAE	United Arab Emirates
UDR	Under-Developed Reservoir
US	Umm Shaif
USGIF	Umm Shaif Gas Injection Platform
USSC	Umm Shaif Super Complex
UZ	Upper Zakum
WHP	Well Head Platform
WHT	Well Head Tower

Abbreviation	Description
WI	Water Injection
WIP	Water Injection Platform
WIU	Water Injection Unit

Table 1-2 Abbreviations

1.2.6 Definitions

Term	Definition
COMPANY	The party so specified in CONTRACT AGREEMENT, its successors, and assignees
EFFECTIVE DATE	Date when CONTRACT is deemed to have come into force, as is specified in CONTRACT AGREEMENT
ENGINEERING Contractor	Entity engaged to execute the FEED SERVICES as indicated in this SCOPE OF WORK
GOODS	Any and all things, including but not limited to materials and equipment (including spare parts) required for and to be incorporated in the WORK
INSTALLATION(S)	The new facilities and/or the modified and refurbished sections of existing facilities including tie-ins which constitute the project as described in this Scope of Work
Mechanical Completion	The stage when all WORK has been erected and installed in accordance with this Scope of Work and all drawings, SPECIFICATIONS, applicable codes, and standards and tagging and completeness of components and packages have been checked, verified and ready for pre-commissioning.
MARINE SPREAD	All offshore marine units, including all VESSELS and BARGES, fully mobilized and complete with all personnel, plant, equipment, special tools, and consumables necessary to perform the WORK.
MODULE	Minimum independent structure that may constitute, alone or in combination with other modules, a process stand-alone structure, such as PAR - Pre-assembled (Pipe) Rack PAU - Pre-assembled Unit VPU - Vendor Package Unit PAS - Pre-Assembled Structure VAM - Vendor Assembled Module PAM - Pre-Assembled Manifold Module PAB - Pre-Assembled Building
MWS	Marine Warranty Surveyor appointed by COMPANY or COMPANY's insurers
PMC	Project Management Consultant appointed by COMPANY to provide project management consultancy services
PROJECT	US LTDP 2.1/2.2 -DAS Minimum Facilities for NASR Crude (Phase 1)
READY FOR START UP (RFSU)	Means the stage at which all non-reservoir fluid systems are commissioned and operational, all other systems are commissioned ready for the introduction of reservoir fluids, and all systems' READY FOR START-UP Certificates have been issued in accordance with COMPANY'S Guideline for Pre-commissioning, Commissioning and Hand-over of Projects A0- PSD-VAP-GDL-004
SITE	Offshore or onshore locations where the systems, plant and INSTALLATIONS are to be permanently constructed, erected, installed, tested, operated, and maintained.

Term	Definition
SUBCONTRACTOR(S)	All persons including VENDOR(S), firms, partnerships, companies, or a combination thereof (not being employees of CONTRACTOR), to whom any part of the WORK has been subcontracted by CONTRACTOR (including any contracts novated or assigned by COMPANY to CONTRACTOR), and the successors and assignees of such persons, firms, partnerships, companies, or a combination thereof.
TPA	Third Party Inspection Agency
VENDOR(S)	All persons, firms, partnerships, companies, bodies, entities, or a combination thereof, including sub-VENDORs and suppliers, who are providing GOODS (including any of the foregoing whose purchase orders are novated or assigned by COMPANY to CONTRACTOR) and the successors and assignees of such persons, firms, partnerships, companies, bodies, entities, or a combination thereof.
WORK	Includes all work and services to be performed and GOODS, services, and other things to be provided by CONTRACTOR or SUBCONTRACTOR(S) to design, procure, construct, erect, install, commission, and test the INSTALLATIONS to satisfy all of CONTRACTOR's obligations under CONTRACT.

Table 1-3 Definitions

2. PROJECT DESCRIPTION

2.1 ADNOC Offshore

ADNOC Offshore is an operating company in charge of five oil and gas upstream concessions for Abu Dhabi Offshore: Umm Shaif/Nasr, Lower Zakum, Upper Zakum, Umm Lulu/Sarb, and Umm Al Dalkh, refer Figure 1.

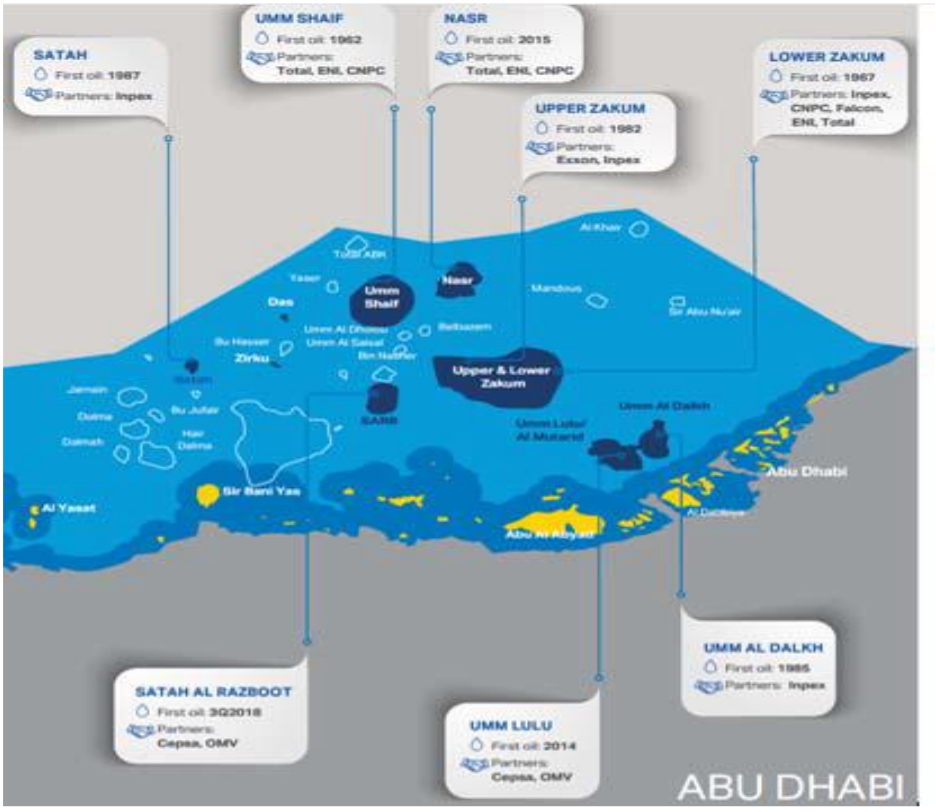


Figure 1: ADNOC Offshore Concession Map

The current crude oil and condensate production is exported via two Oil terminals located on DAS island and Zirku Island. Part of the gas produced is sent to LNG terminal located on Das Island, whilst the rest is sent onshore for local consumption. New long term development plans are being introduced with the objective to increase production to contribute to ADNOC's long term target of 5.0 MMBD by YE2027.

2.2 DAS ISLAND UMM SHAIF EXISTING FACILITIES

2.2.1 Overall Facilities

The partially degassed oil, gas, and condensate from US, Zakum, Bunduq, and Total ABK fields are transferred to Das Island through MOLs, condensate lines, and MGLs. The stabilized oil from Zakum Plant (ZKP), Umm Shaif Plant (USP), Bunduq Plant (BQP), and Total ABK fields are routed to the 13 Crude Oil Storage (COS) tanks whereas stabilized condensate from CTU is routed to a condensate tank. The gas separated from the different pressure stages, having been processed in these plants are delivered to ADNOC LNG and OAG process area. Part of it is used as fuel gas. Produced water separated from separators, spheroids and Crude Oil Storage tanks is treated in Sour Water De-Oiling Plant (SWDP) where oil content

of water is reduced to an acceptable level before being discharged into the monopod. Loading pumps transfer processed crude oil through metering facilities to tankers.

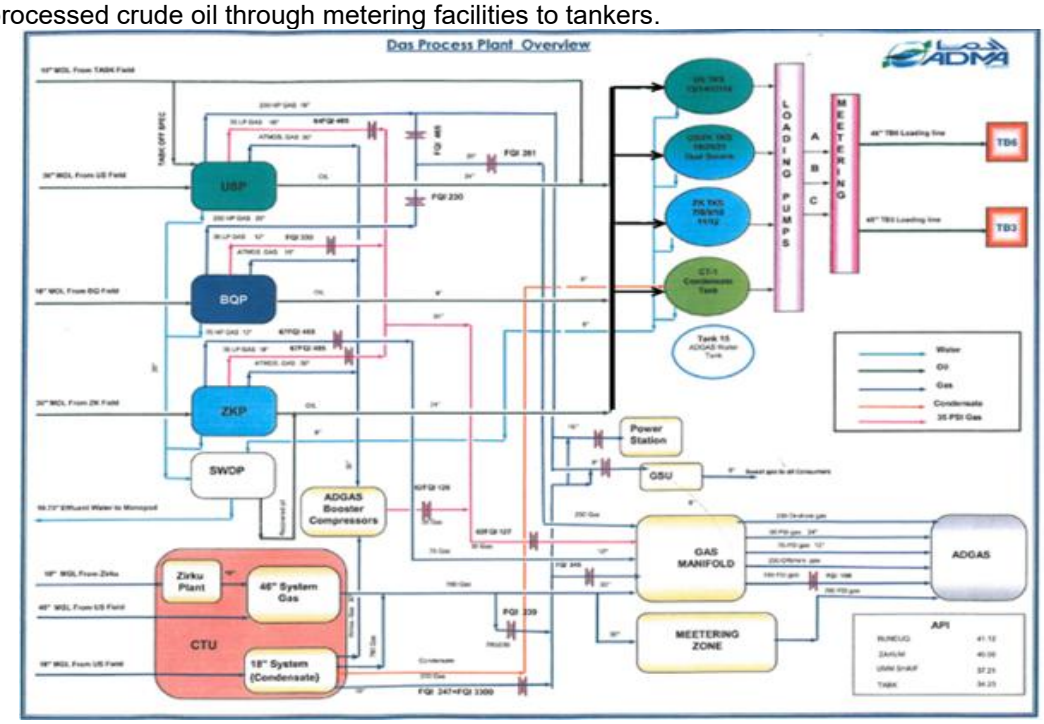


Figure 2: Das Process Plant Overview

2.2.2 US Process Plant

The USSC 36" MOL coming arrives at Das to US plant, refer to Figure 10. The plant comprises of the following:

- 7 HP Separators, 2nd stage, operating at pressure 250 psig.
- 7 LP Separators as 3rd stage operating at pressure 40 psig.
- 2 Spheroids (S4 and S5) and 1 common Spheroid with ZK (S3) as 4th stage operating at ATM pressure.
- 4 Cold stripping towers (one standby common for both Umm Shaif and Zakum).

After separation trains (comprising of HP, LP separators and Spheroids/4th Stage Separators), the crude is routed through heat exchangers into the flash zone of the respective cold stripping towers. The sour crude percolates down through the tower trays against a counter flow of sweet gas either from the 75-psig gas stream from 2nd stage Zakum separators or from the Gas Sweetening Unit (GSU4), thus stripping almost all its dissolved H_2S to the stripping gas. Subsequently, the crude H_2S content is reduced to 0.006% by weight, or less. The crude is then pumped to their corresponding storage tanks.

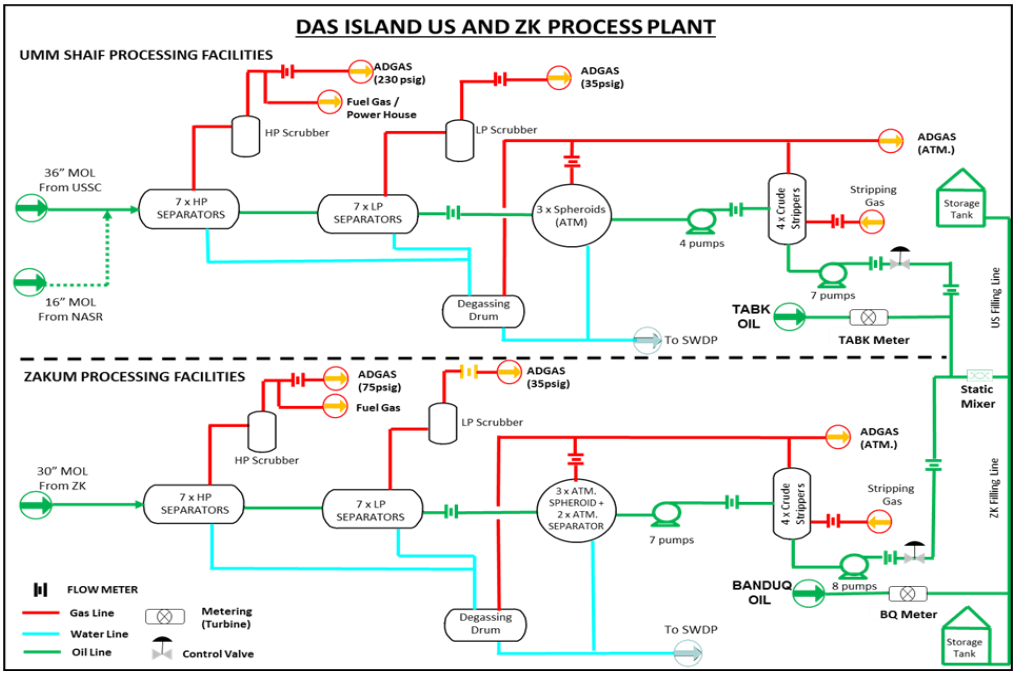


Figure 3: US Process Plant Overview

Nasr crude is processed both in US and ZK Plants. Nasr crude is preferentially directed to US Plant if flow is higher than 75 MSTBD. Stabilized Total ABK crude is exported to Das through 10" line and passes through the metering facilities (turbine meters with proving facilities) before combining with the 24" US blend stream.

2.2.3 Crude Oil Storage Tanks

The crude oil from all processing plants flows to Storage Tanks Nos. 7, 8, 9, 10, 11, 12, 13, 14, 17, 18, 19, 20 and 21. Modifications made for Tank No.17 & Tank No.12 for handling condensate production from the CTU. Condensate is normally received and stored in Storage Tank No. CT-1.
Paraffinic Naphtha, a by-product from ADNOC LNG area is spiked to Das crude oil during the off-spec conditions. Off-spec condensate is spiked into the upstream of US plant for processing.

2.3 US LTDP-2 Background

This project was split into two stages, where stage 1 (LTDP-2.0) aimed at ramping up production to 390 MBD, with the focus on developed and underdeveloped reservoirs using current facilities (i.e. high-pressure radials and ESPs.

The project aims to ramp-up US field production from 275 MBD to 390 MBD from undeveloped/developed reservoirs through existing facilities (HP Network & ESPs). The scope includes:

- 5 new WHTs & 49 new wells to reach 390 MBD from Thamama & A1 reservoirs.
- Installation of new WIP to cater for US and NASR fields growth and initiate ARP for existing facilities.
- Leverage on adequacy of USSC existing facilities.

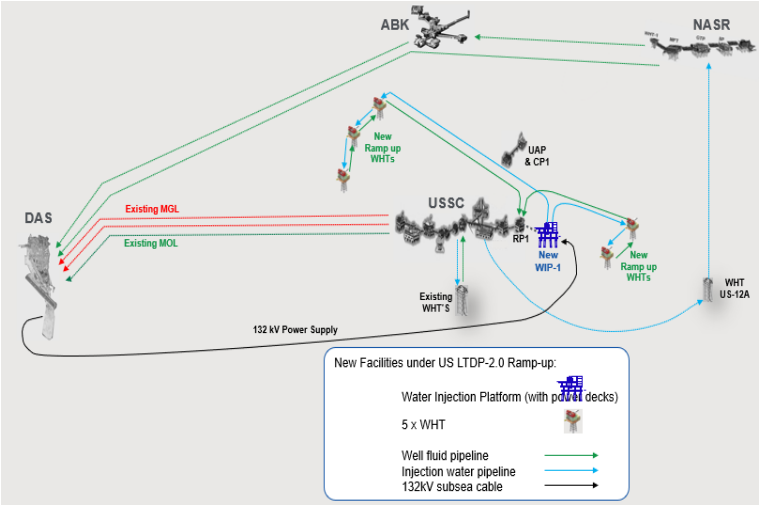


Figure 4: US LTDP-2.0 Project Schematic

Stage 2 is LTDP-2.1/2.2 initially aimed to sustain production of 390 MBD, and is based on adding new WHTs, replacing USSC facilities with new Combined Island(s) and replacing US DAS facilities with new facilities. After undergoing several phases, project scope has been revised following a Value Improvement Initiative (VIIs) workshop. Project revised scope includes select and design stages, where the following two concepts are assessed:

- Option-1B 1 Combined Island (Process and Drilling Facilities), along with 15 new WHTs (oil producers and water injectors).
- Option-1C 2 Islands (1 Combined Island (Process and Drilling Facilities) and 1 Drilling Island), along with 10 new WHTs (oil producers and water injectors).

Including side studies of:

- Early MP (MPP) study
- Alternate Scheme for processing NASR Crude at DAS Island.

The US revised production and injection profiles aim to reach a production of 390 MBD by YE2027 (LTDP-2.0) and 430 MBD by YE2030 (LTDP-2.1/2.2) with the business plan of ADNOC AiP5&5.5 oil production targets as depicted in Figure 5 below.

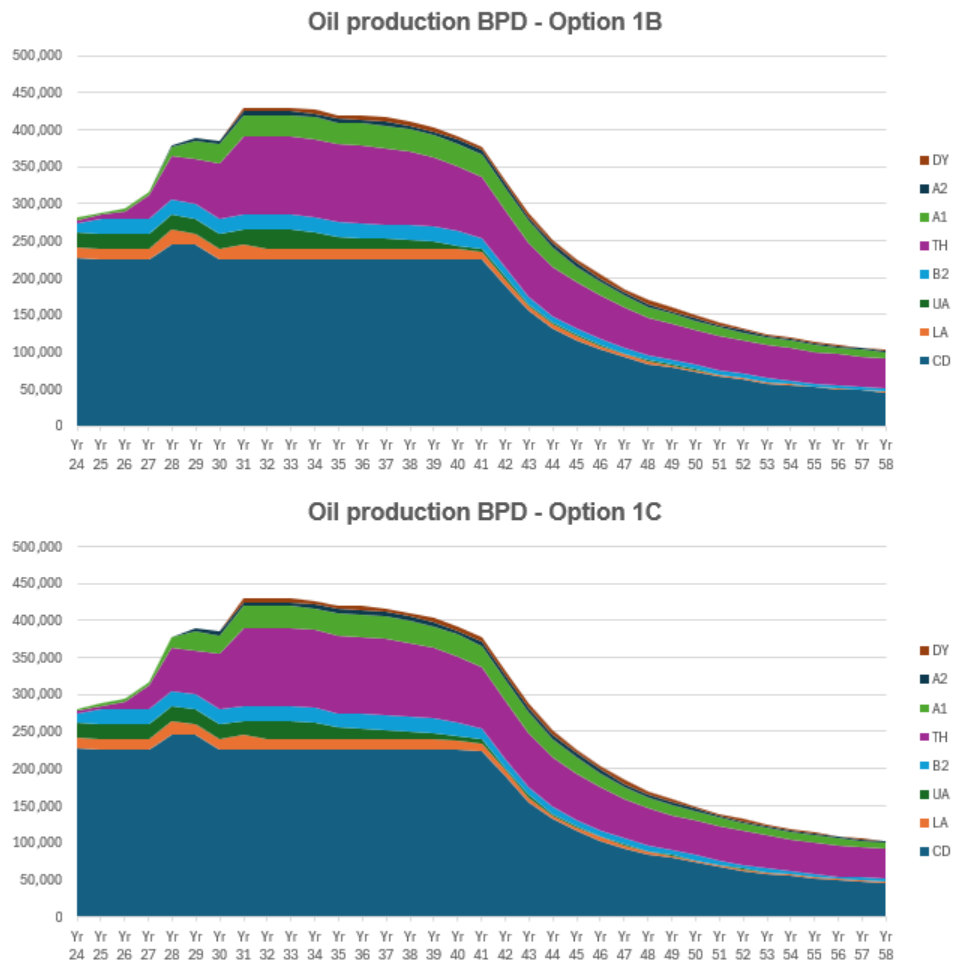


Figure 5: Oil Production Profiles for Development Options 1B & 1C

2.3.1 US LTDP Phase-2.1/2.2 Facilities

The US LTDP Phase-2.1 shall include the installation of the following facilities:
New Combined (Process and Drilling) Island:

- Wells (slots for OP-GL & WI wells) and Drilling facilities
- Riser platform(s) for incoming and outgoing pipelines and cables.
- Pipeline Pig barrels and pig handling facilities.
- MP Train A Facilities (modularized)
- Export oil, gas, and LP water injection subsea pipelines from Island to USSC.
- Bridges for connected Riser Platforms and Flares
- SW Treatment & Injection Units A & B (modularized)
- PW Treatment Facilities Unit A (modularized), and disposal wells
- Electrical Sub-stations, LERs
- Firewater, potable water, service water, instrument air, nitrogen, chemical and utility packages

- Associated Utilities with ullage to account for LTDP Phase-2.2 (modularized)
- Building, Living Quarters
- Areas for CICPA facilities, Laydown, construction camps, workshops etc.

Modifications at USSC:

- Topside modifications to Riser Platform RP-1 / IGP-HAP/ NGTP / CSP/ Bridges etc.
- Routing LTDP1 and LTDP2.0 Thamama/A1 wells from RP-1 to Combined Process and Drilling Island via new subsea pipeline.
- Tie-ins for export oil and gas pipelines, and LP water injection pipeline from Combined Process and Drilling Island.

WHTs and Pipeline/Cables Subsea Networks:

- New WHTs (for Diyab, Thamama, Arab C/D/B2, and Araej Reservoirs).
- New MP Production subsea pipelines network
- New water injection subsea pipelines network for new WHTs.
- New MFT with common radial pipelines direct to/from Combined Island and WHTs (new and existing).
- Rerouting of US55 subsea pipeline around Combined Island

Modifications at Das Island:

- New Stabilization Train with tie-ins as required (initially to process Nasr crude, and later to be converted for combined US/Nasr crude).
- Provision for LTDP Phase-2.2 tie-ins with facilities built during LTDP Phase-2.1.
- Tie-ins on common facilities

The US LTDP Phase-2.2 shall include the installation of the following facilities:

On new Combined (Process and Drilling) Island from Phase 2.1:

- Lift gas compression (modularized)
- MP Facilities Trains B & C (modularized)
- SW Treatment & Injection Facility Unit C (modularized)
- PW Treatment Facilities Units B & C (modularized), and disposal wells

WHTs and Pipeline/Cables Subsea Networks:

- Additional New WHTs
- Additional well-fluids production pipelines.
- Additional LP water injection pipelines (serving new WHTs only).
- Gas lift pipelines to WHTs and MFT

Modifications at Das Island:

- Three additional new Stabilization Trains with tie-ins as required to complete the Asset Replacement Plan (ARP)

2.4 DAS MINIMUM FACILITIES FOR NASR CRUDE (PHASE 1)

In order to accommodate AiP5 mandates, new minimum facilities comprising of Crude Charge Heater, Steam Trap Package, Crude Stripper/Stabilizer & Product Pumps are to be installed within Zakum Process plant as shown in the figure below:

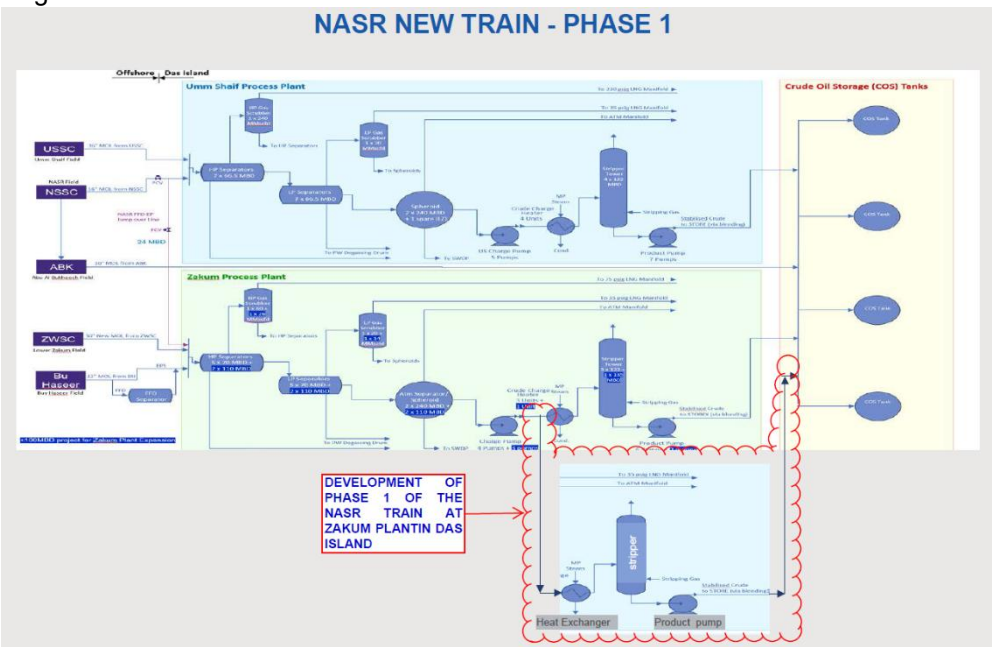


Figure 6: Proposed Minmum DAS FACILITIES

Details for the required facilities are provided in section 10.1

2.5 PROJECT OBJECTIVES

The objective of the FEED Stage is to perform discipline wise FEED stage activities and produce FEED Stage deliverables for “DAS Minimum Facilities for NASR CRUDE” PHASE 1

The following key project objectives shall be met;

- Health and Safety: ENGINEERING CONTRACTOR shall comply with relevant COMPANY and industry codes and practices for health and safety of employees and all other persons who may be affected directly or indirectly by the project.
- Environment: WORK shall comply with COMPANY’s high performance standards for protection of the environment, as well as UAE environmental regulations.
- Quality: ENGINEERING CONTRACTOR shall ensure that WORKS shall meet all the prescriptive quality standards and best industry practices.
- AI and Autonomous:- ENGINEERING CONTRACTOR shall ensure that the facilities are AI enabled and Autonomous by design. Autonomous scope covers domains, including, but not limited to Operations, Maintenance, HSE, Integrity & Inspection and Emissions. For Digital and AI requirements, please refer to Attachment-5 of this document.

- Cost: Deliver a quality product at a competitive price.
- Flawless execution: ENGINEERING CONTRACTOR shall ensure flawless design and procurement works and smooth transition to Construction & Commissioning works completion to achieve RFSU on time.

2.5.1 Key Milestones

The Project Schedule is critical and to achieve the project completion, some Milestones (Key Dates) have been defined by the COMPANY and the rest to be indicated by the FEED CONTRACTOR. The Project Schedule developed by the ENGINEERING CONTRACTOR shall include, but not limited to, the following:

Sr. No.	Milestone Description	Months from Commencement Date
1	Kick-off meeting	One [1] Week
2	Issue Project Execution plan and Planning Package	15 Days
A.1	Site visit report submission (IFR)	30 Days
A.2	HAZOP	2 Months
A.3	30% Model review	2.5 Months
A.4	60% Model review	4 Months
A.5	90% model review	4.5 months
A.6	Issue Project Cost Estimate and Project Implementation (EPC) Schedule (IFR)	5 months
A.7	Issue SOW for EPC Works (IFR)	5 months
A.8	RFT for LLIS	5 months
A.9	TBE for LLIs	8.5 months
A.10	Project Completion	9 Months

Milestone Description	Start Date	End date
FEED (Tendering)	Dec 2025	Feb 2026
FEED (Services)	Mar 2026	November 2026

Table 2-1 DAS Minimum Facilities for Nasr Key Milestones

2.5.2 Project Completion Schedule

The project completion schedule follows a structured, phased approach with clearly defined timelines for each stage. The Front-End Engineering Design (FEED) phase will be executed under a separate contract and is planned for a duration of nine months. During this period, all necessary engineering groundwork will be developed in addition to LLI TBE to support the subsequent phases. In parallel, the Invitation to Tender (ITT) process for the Engineering, Procurement, and Construction (EPC) phase is scheduled to commence in September 2026. This ITT phase will run for approximately eight months, encompassing bid preparation, submission, evaluation, and finalization. The EPC contract is expected to be awarded by May 2027. Following the award, the EPC execution phase will begin and is planned to last for 18 months, covering detailed engineering, procurement, construction, and pre-commissioning activities. Upon completion of the EPC works, a two-month commissioning period will be undertaken to ensure all systems are tested, verified, and ready for operational handover.

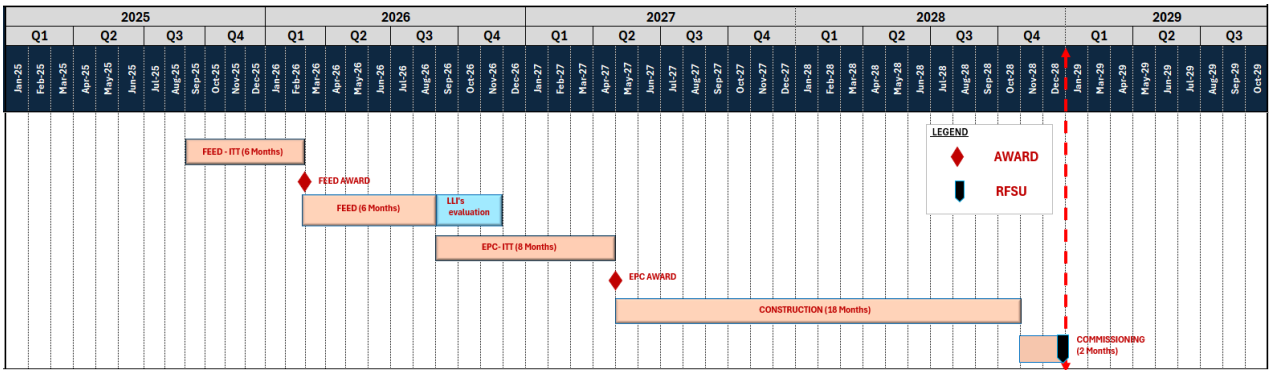


Figure 7: US LTDP-2.1/2.2 DAS Minimum Facilities Overall Schedule

2.6 Parties Involved

The FEED stage will involve the following parties:

ADNOC OFFSHORE (COMPANY)	Abu Dhabi National Oil COMPANY, owner and operator of facilities and in charge of the project
ENGINEERING CONTRACTOR	The ENGINEERING CONTRACTOR undertaking overall FEED & detail design engineering WORKS as per this scope of work
EPC CONTRACTOR	COMPANY appointed CONTRACTOR to carry out Detailed ENGINEERING ,Procurement, Construction & Commissioning works
PMC	Resources sourced from PMC part of Integrated project management team of COMPANY
TPA	Third Party Authority for inspection and testing services

Vendor/Suppliers	Persons, firms, partnerships, companies, bodies, entities or a combination thereof who are providing goods
SUBCONTRACTOR	CONTRACTOR appointed third party service CONTRACTS for specialist study/service such as CRAS, HSE, RBI, Transient study, etc.

2.7 Delineation of Scope of Work

COMPANY'S intent is to execute this Project using FEED ENGINEERING CONTRACT, together with COMPANY's integrated project management team.

The general functional project organization chart is presented below.

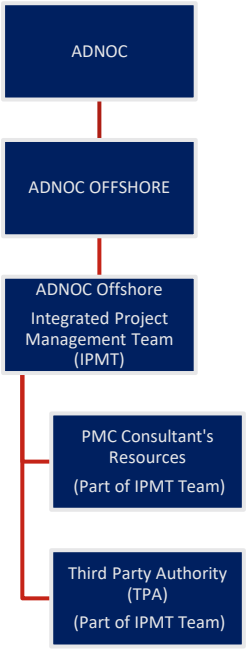


Figure 8: general functional project organization chart

2.7.1 Integrated Project Management Team (IPMT)

COMPANY has assembled an integrated Project Management Team which is responsible for overall management of this Scope of Work. The IPMT may comprise personnel sourced from COMPANY, a third-party Project Management Consultant (PMC), third party authority (TPA) sourced from manpower supply/service CONTRACTS.

2.7.2 ENGINEERING CONTRACTOR

The following description in this section is general in nature. The definitive scope of work, risk, interface management and other contractual responsibilities are contained in other parts of the contract documents.

ENGINEERING CONTRACTOR shall be responsible for ALL WORK related to this project;

- Implementation of all scope in accordance and compliance with COMPANY HSE/QMS/Risk

management standards.

- Completion of the FEED, ENGINEERING CONTRACTOR shall be responsible for ALL WORK related to this project;
- Prepare the SOW for Detailed Engineering, Procurement, Construction and Commissioning tender scope of work covering Project Management, Project Controls, Fabrication, Installation, Testing & Commissioning, As-built and Handover.
- Prepare SOW of Detailed Engineering associated with methodologies for onshore fabrication, and offshore fabrication associated with the new facilities and tie-ins/ modifications to the existing facilities
- Prepare SOW for the detailed engineering, procurement and installation of AI and Autonomous technologies that fit within the scope of this project. For AI and autonomous technologies, refer to attachment-5 of this document. The AI technologies shall include , but not limited to the Digital twin, Advance process Control (APC), Robotics, AR/VR etc. In addition to attachment-5, Contractor proposes any additional AI technologies not listed in attachment-5, that may contribute to autonomous operations and offer tangible improvements in safety, efficiency, reliability and cost-effectiveness.
- The AI and Autonomous operations is rapidly evolving. Contractor Shall ensure that the most current and emerging AI technologies – whether in development phase, pilot phase , or fully matured are considered in this project. The facilities shall be AI enabled and fully autonomous at Commissiong and Go-live stage.
- Management of all internal and external interfaces. Co-ordination and follow up with all PARTIES involved and/or associated with project scope of work directly or indirectly.
- Appoint approved third party services as required for specialist study/service.
- Handover of ENGINEERING dossier, lessons learned register and close out report.
- Compliance to COMPANY COVID 19 testing, vaccination and quarantine requirements for its personnel prior to mobilization to offshore site for any WORK.

2.7.3 Third Party Authority (TPA)

COMPANY may appoint an Independent Third Party Authority (TPA who will act as an independent inspection agent, responsible for ensuring ENGINEERING CONTRACTOR'S QA/QC compliance.

3. SCOPE OF WORK

The scope of work shall include but not limited to the following:

a) FEED engineering including process studies, social impact assessment (SIA) screening/study and HSE studies. ENGINEERING CONTRACTOR shall perform all basic design and engineering for the facilities as described. ENGINEERING CONTRACTOR shall be responsible for validating the selected scheme to ensure viability with the PROJECT objectives. Engineering Contractor shall ensure that the Artificial intelligence (AI) solutions and technologies applicable for the scope of work, shall be considered during engineering design phase so that that the Facilities SHALL be AI enabled and Autonomous by design

b) Appoint approved third party services as required for specialist study/work.

c) Preparation and submission of Scope of Work, Schedule and Cost Estimate for Detailed Engineering, Procurement, Construction & Commissioning and seek approval from COMPANY.

d) ENGINEERING CONTRACTOR's Project team shall include but not limited to following positions:

- Project Manager
- Project Engineer
- Planning & Control Engineer
- Engineering Manager
- Discipline Engineers
- Interface Engineer
- Cost Estimate Engineer
- QA/QC Engineer

ENGINEERING CONTRACTOR shall assess the SOW and propose the detail project organization chart and shall provide CVs for all proposed positions to COMPANY for review/approval.

e) Provide CMMS documentation for registration of new facility items, asset register as per COMPANY procedure for CMMS Requirement for Asset Register Data Build-up Project Stage. Procedure (A0-OP-R-PRO-005) and templates will be provided later. s

f) Provide documentation for equipment registration in CMMS with all required mandatory documents as per APPENDIX-E of A0-IG-Z-CP-001 (COP for Integrity Assurance in Projects).

g) Project document control / management and its compliance to specification for Information Management in Projects (A0-ENG-I-SP-001) including MDR templates.

h) Provide project close out report including lessons learned register.

i) Management of all internal and external interfaces, Co-ordination and follow up with all PARTIES involved and/or associated with project scope of work directly or indirectly.

j) Prepare and arrange project risk assessment session on monthly basis. Project risk management process shall include preparation, identification, assessment, mitigation, management and reporting as per ADNOC Project Risk Management Guideline and risk matrix. The Project Risk Register shall be updated monthly until all items are satisfactorily closed (or reduced to an ALARP level).

k) Overall Project Quality, HSE, RISK management and compliance with COMPANY's respective applicable standards.

l) Dead leg register with all required documentation.

m) Prepare or modify existing Operation, Maintenance and Integrity (OMI) Assurance Plan/OMI Philosophy and participate in the preparation of Operations Readiness Plan (ORP).

n) Prepare or modify existing operating manual/operating procedures/start-up procedures, etc.

o) Prepare MR for equipment and materials procurement including technical/commercial bid evaluation.

p) Perform criticality rating study, ensure compliance with Criticality Rating Specification (AGPS-SD-QAQC-001) and Procurement Inspection and Certification Requirements in Projects (AGPS-SP-QAQC-003).

q) Provide project close out report including lessons learned register.

4. INTERFACE WITH OTHER PROJECTS & STUDIES

4.1 Key Umm Shaif Projects

The key projects are different project phases though all will be in operation prior to the commissioning and start-up of the US LTDP-2.1/2.2.Gas Cap Phase-1 Project. The project aims to produce and inject 600 MMSCFD that will yield around 50 MBD of condensate. 3 new gas wellhead towers with 8 new gas producers, 6 gas injectors and an Environmental Condensate Separation and Compression Platform are located offshore close to existing USGIF. Condensate will be exported to Das Island for stabilisation in a stand-alone Condensate Treatment Unit (CTU). Some brownfield modification will be carried out on existing accommodation platform UAP1 and in Das Island.

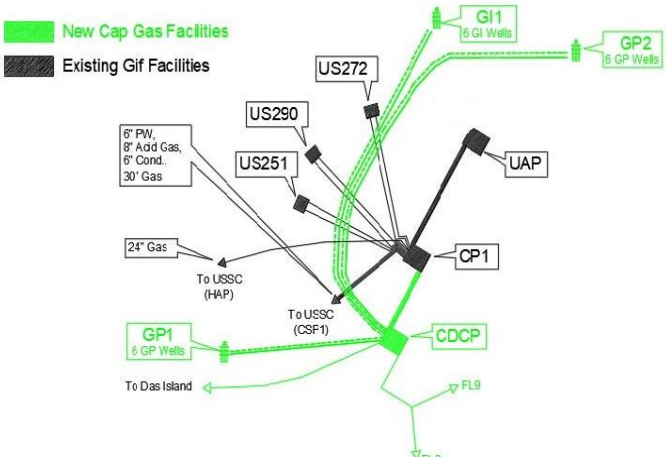


Figure 9: Gas Cap Phase-1 Project Schematic

4.1.1 US LTDP-1 Project

The project aims to extend the 275 MBD plateau until 2028 with Three WHTs. Two (HP1 & HP2) WHTs to sustain Arab C&D plateau and introduce the IRWI to mitigate gas front advance whilst the 3rd WHT (HP3) is equipped with ESP as artificial lift to unlock potential of under-developed Thamama & A1 reservoirs and use an Early Production Scheme to reduce subsurface uncertainties.

The project also includes the expansion of US production/water injection radials network to eliminate current bottlenecks limiting potential from weak reservoirs and mitigate risk linked to GOR behaviour (current/future). Existing Sub-sea cable network of this project is considered for augmentation to feed some of the new WHTs.

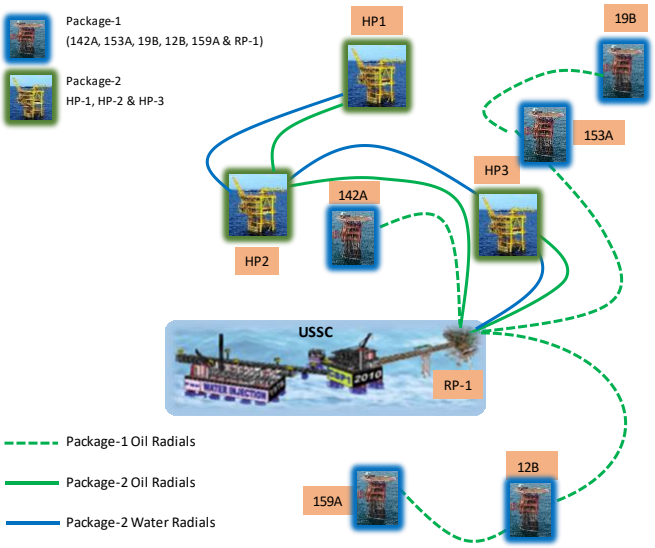


Figure 10: US LTDP-1 Project Schematic

4.1.2 US LTDP-2.0 Project

The project aims to ramp-up US field production from 275 MBD to 390 MBD from undeveloped/developed reservoirs through existing facilities (HP Network & ESPs). The scope includes:

- 5 new WHTs & 49 new wells to reach 390 MBD from Thamama & A1 reservoirs.
- Installation of new WIP to cater for US and NASR fields growth and initiate ARP for existing facilities.
- Leverage on adequacy of USSC existing facilities.

Existing Sub-sea cable network of this project is considered for augmentation to feed some of the new WHTs.

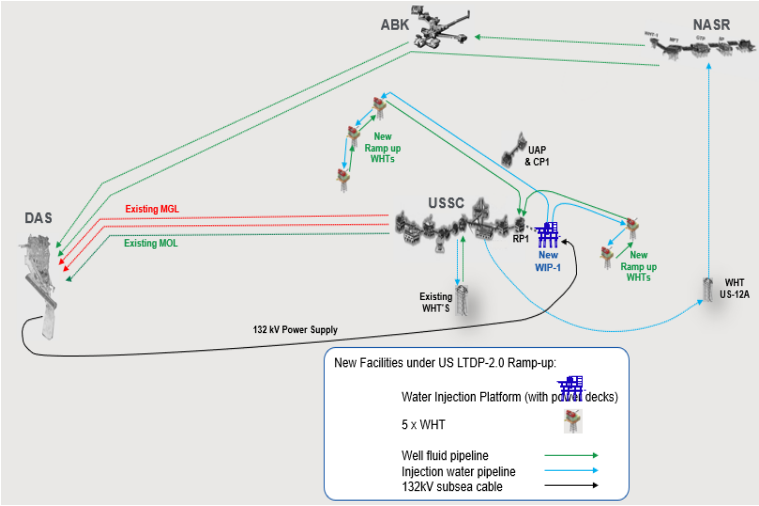


Figure 11: US LTDP-2.0 Project Schematic

4.2 Management of External Interfaces

4.2.1 General Requirements

ENGINEERING CONTRACTOR shall be responsible for the management and co-ordination of all external interfaces associated with the WORK. These interfaces cover engineering, procurement and construction activities where tie-ins or information are required or where other contractors are directly involved in any aspect of WORK.

Identification, definition and obtaining details of all interface conditions shall be ENGINEERING CONTRACTOR'S responsibility. ENGINEERING CONTRACTOR shall prepare all design, planning, scheduling or other work required in respect of interface matters.

ENGINEERING CONTRACTOR shall comply with COMPANY requirements with respect to contacts with parties involved in interfacing with the WORKS. ENGINEERING CONTRACTOR is required to make direct contact with suppliers, COMPANY Project Management team, third party authority and/or other EPC Contractors.

ENGINEERING CONTRACTOR shall define all interfaces with other parties and further detail the attribute of each interface as follows by means of interface agreement:

- Description of interface

- Information/data requestor and supplier
- Priority
- Details of data interface
- Critical Date to Finalize Details

ENGINEERING CONTRACTOR shall inform COMPANY representatives to attend any or all meetings arranged to discuss interfacing matters on priority. ENGINEERING CONTRACTOR shall prepare minutes of such meetings, arrange for signature by, and issue to all relevant parties.

4.2.2 Interface Coordination Procedure

ENGINEERING CONTRACTOR shall prepare, submit for approval and implement a detailed interface coordination procedure to cover for the external interface activities as defined above.

The Interface Coordination Procedure shall cover all design issues.

The Procedures shall include but not be limited to the following:

- Parties, Relationship & Responsibilities (COMPANY, other EPC Contractor, TPA, Construction & Commissioning CONTRACTOR)
- Project Organization
- Project Telephone & e-mail Directory
- Project Communications (Letters, e-mails, interface agreements, etc.)
- Project Meetings (Schedule, Agenda, attendees.)

ENGINEERING CONTRACTOR shall identify within 2 weeks after the EFFECTIVE DATE what information it requires COMPANY to provide, including the latest dates by which such information is required to allow CONTRACTOR to properly manage its interface issues.

ENGINEERING CONTRACTOR shall liaise and work with all other parties to ensure adequate exchange of information occurs across the interfaces involving the tie-ins to existing systems. COMPANY will facilitate the exchange of information as required.

4.2.3 Interface with Other Projects / Facilities

- ENGINEERING CONTRACTOR shall liaise with other Projects which are currently either ongoing or recently finished to ensure that there is a seamless integration of new facilities with the existing facilities.
- EP ENGINEERING CM CONTRACTOR's responsibility includes corrective actions required at interfaces with any other ongoing COMPANY projects and other OPCO's projects at DAS island. To that effect ENGINEERING CONTRACTOR shall request COMPANY for the ongoing interface PROJECT details at appropriate time and verify interfaces prior to submission of PROJECT drawings and documents.

- ENGINEERING CONTRACTOR shall develop the Interface management plan including interface register covering all interface projects

- Following are the interface projects identified. However, if any new projects planned to be executed before the start of this project, COMPANY will communicate to the successful bidder. ENGINEERING CONTRACTOR shall request an alignment meeting with COMPANY PMT/respective project teams at early stage to avoid any impact on design deliverables.

- ENGINEERING CONTRACTOR to update and add other projects identified during ENGINEERING stage and any associated interface related requirement shall be captured as part of project scope of work.

SR. NO	LIST OF ONGOING PROJECTS	PROJECT STAGE
1	LTDP PHASE-1 FOR LOWER ZAKUM FIELD- PACKAGE 5	EPC
2	NASR FULL FIELD DEVELOPMENT PROJECT - PACKAGE-3	EPC
3	Bu-Haseer Full Field Development Project	EPC
4	CRITICAL PIPING REPLACEMENT Project	EPC
5	FIEP – FACILITIES INTEGRITY EXECUTION PROJECT	EPC
6	COS TANKS INLET PIPING UPGRADE Project	EPC
7	Metering System Improvement at Das	EPC
8	AUTOMATIC FIRE SUPPRESSANT SYSTEM SUBSTN & LER	EPC
9	LTDP – Crude Oil Storage Tanks (All Packages)	EPC
10	LPS-Lightning Protection Schemes-Das Island	EPC
11	Project Lightning (BOOT)	EPC
12	Project Lightning-EPC for Grid Power INTEGRATION	EPC
13	IGDE2 (ADNOC LNG Scheme)	EPC
14	IGDE2 – FGTU part of IGDE2(ADNOC Gas / ADNOC LNG execution-61% AOFF share)	EPC
15	NASR 100MBD Adequacy studies of US & LZ plants	FEED
16	FEED FOR PAGA SYSTEM AT DAS	FEED
17	AIR EMISSION MONITORING SYSTEM	EPC

SR. NO	LIST OF ONGOING PROJECTS	PROJECT STAGE
18	Relief Valves Discharge to ATM	FEED
19	Das Flare Rationalization	FEED/EPC
20	Long term Development of Umm Shaif including Cap Gas Development	SELECT
21	Umm Shaif LTDP-2	SELECT
22	NMGL Project	EPC
23	EPC For Critical I&C Systems Upgradation at Das Island	EPC
24	OTHERS	SELECT/FEED/ EPC

5. SCOPE OF PROJECT

This section outlines the scope of services required for carrying out FEED engineering, of DAS Minimum facilities for Nasr Crude (Phase 1).

5.1 ENGINEERING CONTRACTOR's responsibilities

ENGINEERING CONTRACTOR shall be accountable for executing all the WORK for DAS Minimum facilities for Nasr Crude (Phase 1) and will act as a single focal point for the execution of the WORKs within the project boundaries. The responsibilities shall include but not limited to:

5.1.1 Project Management

ENGINEERING CONTRACTOR shall perform all Project Management activities required for the WORK, including but not limited to the following;

- General Project management, control and administration.
- Provide suitably qualified and experienced personnel who, for all key positions, shall be members of its permanent staff and CVs shall be subject to COMPANY approval..
- Management of interfaces with all parties that are affecting the scope of work.
- Preparation and implementation of Project Execution Plan (PEP) that covers discipline Engineering Execution Plan detailing the requirements of SCOPE. The Engineering Discipline Execution Plan shall address issues such as:

  a. Specify HOME OFFICE location.
  b. Major technical activities.
  c. Organization
  d. Roles and responsibilities/ Authority Matrix/ Signatory List.
  e. CVs of Key Personnel, CVs of Leads in Functional roles and Senior Engineers (15 years' experience in Oil and Gas with minimum 5 years experience in Brownfield Offshore Projects).
  f. Approval authorities.
  g. Histogram.
  h. Detailed list of deliverables.
  i. Detailed list of deliverables with RACI identification against each deliverable.
  j. Include a Master Document Register linked to a Level 3 Engineering Schedule by the identification of deliverables with each activity of the schedule.
  k. A matrix structure showing deliverable type, revision sequence, review/ approval requirements, squad checking to/from other disciplines (IDC), etc.
  l. Engineering and design methodologies.
  m. Software and software validation.
  n. Discipline scheduled activities/ logic with other disciplines.
  o. Key milestone dates – discipline/ project.
  p. Key discipline "freeze" points.
  q. Discipline internal check (DIC) and inter discipline check (IDC).

r. CTRs/ man-hour budgets/ progress reporting.
s. Discipline quality activities – including quality assurance and technical assurance requirements
t. Change management procedures
u. Intra- and Inter-discipline communication
v. Interface management – including procurement
w. Project Interface with other EPC/Contractors if applicable

- Preparation and implementation of project planning package. As a minimum, the PEP shall address the engineering, procurement services, procurement, material storage/short term preservation, materials management and supervision/management of fabrication, installation plans, pre-commissioning and commissioning, management of change, technical/site queries, etc.
- Preparation and implementation of project specific procedures for coordination, communication, interface management, procurement, document review/control, monitoring, Quality/HSE/Risk management, management of change, progress reporting, invoicing, etc.
- Coordinate with COMPANY to identify key engineering personnel who shall be required to provide support and continuity during Procurement, Construction and Commissioning phases of project.
- Preparation and presentation of progress reports. Preparation of project presentations as required including minutes of meeting.
- Organize and execute Project Risk Assessment workshops with COMPANY participation.
- ENGINEERING CONTRACTOR shall perform technical/quality audit on a regular basis to cover its own work and advise COMPANY the results of such audits together with the appropriate action taken and any other recommendation for COMPANY consideration.
- ENGINEERING CONTRACTOR shall prepare, provide and implement procedures for exercising discipline and control over the WORK.
- ENGINEERING CONTRACTOR shall clearly define ALL the PARTIES involved, their joint and individual responsibilities, and split of work, interfaces and office locations for each key project team member.

5.1.2 Use of Engineering Study Report

Engineering Study reports below are prepared for Package IV 'NASR Crude Alternative Scheme (Details about this package is provided in section 10.1.1.2) and are provided for reference only:

- Attachment 1 : Engineering Report (CED Job No.: C1292)
- Attachment 2 : Uweinat Cap Gas to ABK - TABK Study Report (TR21-TMT-ADNOC-MEM-0003)

5.1.2.1 Preliminary MTO Estimates

MTOs provided by COMPANY are for guidance only and are provided without prejudice to FEED CONTRACTOR's obligations.

ENGINEERING CONTRACTOR shall remain solely responsible for developing all MTOs required for project WORKs.

5.1.3 FEED Engineering Activities

The design activities performed by ENGINEERING CONTRACTOR in executing the WORK shall include but not be limited to:

1. Site survey, data collection, including review and as-built verification of all relevant COMPANY drawings and documents.

2. Retrieval of existing drawings, documents from COMPANY engineering document management system (EDMS).

3. Prepare FEED engineering documents, specifications, data sheets, MR for Long lead items, drawings, study reports, constructability/SIMOPS/Shutdown optimization review reports, etc., in accordance with COMPANY, national and international design codes, standards, policies, procedures and industry best practice.

4. All FEED Engineering unless otherwise APPROVED by COMPANY shall be performed in the Engineering HOME OFFICE. Furthermore all Engineering activities shall be completed in the HOME OFFICE from where they commenced and shall not be transferred to another HOME OFFICE without COMPANY'S APPROVAL.

5. Prepare a fully checked set of discipline design calculations for all aspects of the design. All calculations shall be prepared in consistent format. All calculations shall be packaged to allow clear demonstration of application of design bases, applicable codes, methodologies, utilized equations, references to other documents, input/outputs and allocated design margin. Relevant references shall be copied and included in the packaging. QA front sheets shall demonstrate the checking function and also designate originator, checker and approver.

6. Develop, as part of the overall Quality Plan established during the Engineering mobilization phase, a rigorous, demonstrable Discipline Engineering Technical Assurance program that incorporates scheduled activities into the work program. These activities shall be designed to add value to and assure the technical integrity of the design package.

7. Develop a new Technical Query Register at start of engineering to clarify technical issues.

8. Incorporate all discipline checking, third party audit requirements into the overall Project Validation Plan.

9. Prepare Basis of Design documents (all disciplines) and Design Philosophies, as required during the Engineering phase.

10. Prepare MRs for all equipment, materials, bulk items, etc., required to complete the works and obtain COMPANY's approval.

11. Conduct P&IDs Review with COMPANY.

12. Liaise with and provide input to other disciplines and departments.

13. The term "FEED Engineering" referred herein shall mean, as a minimum, the production of all documents as per A0-ENG-Z-GDL-001 (Guideline for project deliverables) and issued for

Design for Design, as deemed appropriate by COMPANY. ENGINEERING CONTRACTOR shall submit a deliverables list for each discipline for COMPANY review/approval.

14. Conduct and complete technical safety/HSE studies.

15. Close out of all actions in the SHEAMS register developed during the detail design.

16. Prepare for and arrange SIMOPS workshop, constructability review workshop, SHUTDOWN optimization workshops and close out of actions if any.

17. Confirm structural integrity for in-service conditions.

18. Preparation of piping layout, plot plans, structural arrangement drawings, etc.

19. Conduct HAZOP and close out of actions.

20. Conduct a SIL assessment, verification and validation for all the safety instrumented functions (SIFs) associated with new instrumentation catering to the modifications scope and the existing SIFs where there are modifications, tie-ins and based on HAZOP recommendations.

21. Squads check other disciplines' deliverables (including interface packages) in accordance with the discipline execution plans and documentation matrix.

22. Seek COMPANY approval for class 1 and 2 documents. All documents and drawings shall be submitted in manageable quantities and logical sequence. ENGINEERING CONTRACTOR shall prepare and submit a Document Submittal Plan.

23. Demonstrate that system design and equipment selections have reduced associated safety risks to ALARP (As Low As Reasonably Practicable) level.

24. Create and manage a HOLDS Register to identify and close out uncertainties as the FEED Engineering progress. The HOLDS Register shall include cross reference to all impacted documentation where the actual HOLD appears.

25. Attend discipline and project meetings, including those requested by COMPANY.

26. Produce discipline progress and other reports as required and submit to COMPANY.

27. Prepare and arrange project risk assessment/management session on monthly basis. Project risk assessment is a critical and mandatory component of project management across all phases of a project lifecycle. Project risk assessment shall be performed through qualitative and quantitative analysis; this should be selectively applied as applicable

28. Project risk management process shall include preparation, identification, assessment, mitigation, management and reporting as per ADNOC Project Risk Management Guideline and risk matrix. The Risk Register shall be updated on a monthly basis until all items are satisfactorily closed (or reduced to an ALARP level).

29. Develop and implement an interface management program to facilitate information flow and change control across all interfaces both within ENGINEERING CONTRACTOR'S organization and external to ENGINEERING CONTRACTOR'S team.

30. Interface and coordination with other COMPANY projects and other EPC Contractors. ENGINEERING CONTRACTOR shall submit an Interface Management Plan/Procedure and responsible for management of ALL interfaces.

31. Interface and coordination with COMPANY appointed third parties (e.g., TPA, etc.)

32. Liaise with external parties (e.g. VENDORS, other CONTRACTORS, Construction & Commissioning CONTRACTOR, and COMPANY) as necessary and in accordance with project procedures.

33. Develop or modify existing operating manual /procedures of systems and equipment and any other procedures as advised by COMPANY.

34. Provide native and PDF format files of all deliverables. PDF format shall be created from the native file and be fully searchable utilizing the latest version of Adobe Reader and shall not be created from scanning and saving files.

1) Provide native/data base of all software (Smart 3D , Smart P&ID and instrumentation, Smart Electrical, Caesar II, SACS, etc.). Any software not mentioned here but used during the project execution then its database in native format shall be provided to COMPANY as well.

35. Software listed in Section # 9.10 shall be used to develop, design and perform the WORK.

36. Development of following registers

    a. Lock open and Lock close.

    b. Register of safety related devices (RSRD).

    c. HSECES list.

    d. Dead leg register with mitigation measures.

37. Undertake the Specialist Study Work as mentioned in the scope. ENGINEERING CONTRACTOR shall demonstrate, to COMPANY'S satisfaction, that it has the "in-house" capability to undertake this specialist study work including the provision of competent specialists and any necessary specialist software. Otherwise, within the Scope of Work and ENGINEERING CONTRACTOR'S Lump Sum Price, ENGINEERING CONTRACTOR shall appoint and manage a specialist SUBCONTRACTOR to undertake this specialist work. For each of these studies, ENGINEERING CONTRACTOR shall:

    - Prepare a scope and methodology statement for review and approval by COMPANY before the work commences.
    - Undertake the study work.
    - Present outcomes to COMPANY and prepare a study report for each study.

38. ENGINEERING CONTRACTOR shall be deemed to have allowed for preparation of all deliverables (process studies/specialist studies, reports, technical models/3D model, data sheets, MR, TBE, vendor documents review, , etc.) to complete project WORKS in all respect.

39. ENGINEERING CONTRACTOR shall provide Cost estimate to ±15% accuracy in accordance with COMPANY requirement.

40. ENGINEERING CONTRACTOR shall Prepare Close-Out Report.
41. Any service required but not specifically defined in this Scope or Project definition but is deemed as essential by the Company for proper completion of the project.

5.1.4 Project Standards, Specifications, Guidelines and Procedures

ENGINEERING CONTRACTOR shall comply with UAE legislation and environmental regulations, ADNOC standards, Codes of Practice, Specifications, Procedures, Guidelines, applicable international codes and standards

The latest versions and amendments of all UAE legislation and environmental regulations, ADNOC standards, Codes of Practice, Specifications, Procedures, Guidelines, applicable international codes and standards on the EFFECTIVE DATE of contract, shall be used.

Where equal but conflicting requirements are identified between regulations, codes and standards, the order of precedence shall be:

a. UAE Statutory legislation and regulation inclusive of ADNOC standards and codes of practice.

b. ADNOC Standards.

c. ADNOC Offshore Standards.

d. International codes and standards.

5.2 Company Responsibilities

COMPANY shall be responsible for the following:

1. Nomination of COMPANY Representative and setting up COMPANY Project Management Team.

2. Provision of physical access to COMPANY existing facilities and fields.

3. Inputs/data for Basis of Design.

4. Provision of relevant ADNOC/COMPANY standards, procedures, guidelines, specifications, formats, templates. Upon award, COMPANY will issue in a CD an update of all the relevant ADNOC/COMPANY standards, procedures, guidelines, specifications, formats, templates.

5. Overall project schedule plan.

6. CV approval of ENGINEERING CONTRACTOR's/its SUBCONTRACTOR key personnel prior to their mobilization.

7. Facilitating access to enable site visits, at relevant locations including letter of Assistance for enabling CONTRACTOR to obtain security passes. COMPANY assistance is limited to providing letters.

8. Provision of information related to interfacing projects.

9. Provision of COMPANY TPA personnel
10. Review and approval of engineering/design, VENDOR, Construction & Commissioning deliverables as per deliverables review matrix to be developed by ENGINEERING CONTRACTOR.
11. Approval of technical bid evaluations/commercial bid evaluation.
12. Approve completion certificates as required as per conditions of contract.

6. PROJECT MANAGEMENT

6.1 General Requirements

ENGINEERING CONTRACTOR shall perform ALL project management service required for the WORK, including, but not limited to the following:

1. Preparation of organization charts.
2. Provision of ENGINEERING CONTRACTOR's Project Management Team acceptable to COMPANY to manage the implementation of the WORK.
3. In order to fulfill the obligations detailed under the contract terms, ENGINEERING CONTRACTOR shall provide suitably qualified and experienced personnel who, for all key positions, shall be members of its permanent staff or that of its parent or subsidiary and CVs shall be subject to COMPANY approval.

Following key personnel shall be assigned to the project:

- Project Manager
- Engineering Manager/Engineering Co-ordinator
- Lead/Senior Project Engineer
- Interface manager/Interface Engineer
- Project Control Manager / Engineer
- Cost Estimating Engineer
- Contracts Manager/Engineer
- Lead Discipline Engineer(s) /Senior Engineers
- Designers/Drafting resources
- HSE Manager/ Engineer
- QA-QC Manager / Engineer
- Construction Manager / Engineer
- Procurement Manager / Engineer
- Buyer / Expeditor

- Document Controller

4. Coordinate with COMPANY to identify key engineering personnel who shall be required to provide support and continuity during construction and commissioning phases of project.
5. Prepare project coordination, quality assurance, HSE, risk, interface management, document control, materials storage/preservation/management, inspection and testing procedures, and fabrication/construction/commissioning strategy.
6. Preparation and presentation of progress reports. Preparation of project presentations as required including minutes of meeting.
7. Organize and execute Project Risk Assessment workshops with COMPANY participation.
8. ENGINEERING CONTRACTOR shall perform technical/quality audit on a regular basis to cover its own work
9. ENGINEERING CONTRACTOR shall prepare, provide and implement procedures for exercising discipline and control over the WORK.
10. ENGINEERING CONTRACTOR shall clearly define ALL the PARTIES involved, their joint and individual responsibilities, and split of work, interfaces and office locations for each key project team member.

6.2 Co-ordination with Company

ENGINEERING CONTRACTOR shall pay particular attention to the following:

- Timely supply of engineering documents and information.
- Obtaining, reviewing and comment on final VENDOR/ supplier dossiers.
- Obtaining VENDOR information for timely completion and submission of SPIR forms.
- Preparation of Project Asset Register as per COMPANY formats in compliance with COMPANY Guideline.
- Submittal of as-built redline mark-ups and final as-built documentation.

6.2.1 Health Safety and Environmental Impact Assessment (HSEIA) Consultant

ADNOC Group Guidelines on Health, Safety and Environmental Impact Assessment dictate that an independent HSEIA is to be carried out at each stage of a Project.

ENGINEERING CONTRACTOR shall undertake all studies and consolidate all relevant documentation in this regard such that the HSEIA Consultant may carry out their review independently, to ensure that the project fully complies with the ADNOC Guidelines. FEED CONTRACTOR'S scope of work shall include adjusting its design to adhere to the requirements of HSEIA.

6.2.2 Third Party Authority (TPA)

COMPANY appointed Independent Third Party Authority (TPA will act as an independent inspection agent, responsible for ensuring ENGINEERING CONTRACTOR'S QA/QC compliance, COMPANY Auditors.

COMPANY may elect to appoint other third party auditors to review the performance of ENGINEERING CONTRACTOR, if in COMPANY'S opinion alone, such audits are necessary to demonstrate ENGINEERING CONTRACTOR'S work efforts are meeting HSE, quality, schedule or cost performance objectives.

6.3 Project Control Requirements

Refer to Company project control requirements - A0-PSD-PCT-PRO-004 Rev.1 document.

7. QUALITY MANAGEMENT SYSTEM REQUIREMENTS

7.1 QMS – GENERAL

- CONTRACTOR's Quality Management System (QMS) and QMS of their (Sub-Contractors/ Consultants/ Vendors/ OEM - as applicable) shall cover the Organizational structure, processes interactions, resources, and procedures and shall include all planned and systematic actions necessary to assure and ensure that all project Quality, specified project requirements and objectives are fulfilled.
- CONTRACTOR's Quality Management System (QMS), and QMS of their (Sub-Contractors/ Consultants/ Vendors/ OEM - as applicable) shall comply with the requirements of Quality Standards ISO 9001 or ISO 29001 (latest edition) and shall be certified by an Internationally Accredited Certifying Authority, the ISO Certification shall be valid for the entire duration of the project and extend to all performing areas (Locations/ Offices) of the CONTRACTOR's Organization and applicable to the project scope of work being undertaken
- The Scope of Works shall be carried out only by the (Locations / Offices) of CONTRACTOR that are registered in ADNOC Approved Contractor List. All activities and services associated with the scope of project performed by CONTRACTOR's (Sub-Contractors/ Consultants/ Vendors / OEM as applicable) shall be approved by the COMPANY.
- The use of Sub-CONTRACTORs for part of the SCOPE OF WORK is not permitted for any services / functions without specific and written approval of the COMPANY.
- CONTRACTOR's Quality Management System shall ensure that all actions and activities that impact upon quality are systematically planned of the start of the activity. That adequate review of contract and other pertinent documents is performed in order to conduct a full analysis of tasks and requirements prior to the commencement of work.
- CONTRACTOR shall develop and implement Project's Quality Management System in line with AGPS-SD-QAQC-001 (Contractors QAQC Requirement) supplemented by A0-Q-PQ-SP-002 (Specification for Requirements for Projects Contractor Quality System), in addition to ISO 10006 (Guidelines for Quality Management in Projects) or equivalent.
- CONTRACTOR's Project Quality Management System and associated documents to be implemented in the project shall be subject to review by the COMPANY. All applicable documentation shall be reviewed for the need of any revision before initiating or implementing any change and revised documentation shall be resubmitted for COMPANY review for comment and subsequent approval.
- CONTRACTOR's Project Quality Management System (QMS) shall address all applicable reference to all project functions, such as Governing Regulations, project specifications, ADNOC / COMPANY

Engineering Standards, Codes of Practice, Specifications, Procedures, Manuals and Project General Specifications as referenced in the contract are fully incorporated and complied with.

- CONTRACTOR Project Quality Management System (QMS) CONTRACTOR shall incorporate and compliance with all requirements of Quality Assurance and Quality Control as specified in:
  - AGPS-SD-QAQC-001 - Quality Management Requirements for Contractors in Projects)
  - AGPS-SP-GN-001 - Criticality Rating Specification.
  - AGPS-SP-QAQC-002 - Inspection and Certification Requirements in Project Procurement
  - AGES-GL-13-002 - Positive Material Identification of Equipment and Piping
  - AGES-SP-13-003 - Traceability of Shop & Field Piping Materials.
  - A0-Q-PQ-SP-001- EPC Contractor Quality Personnel Requirements.
  - A0-Q-PQ-SP-003 - Specification for QA/QC requirements for construction works.
  - A0-Q-PQ-PRO-001 - Procedure for Project Non-conformity Management.
  - A0-Q-PQ-PRO-002- Procedure for Prevention and Control of Counterfeit Material.
  - A0-Q-PQ-PRO-005 - Procedure for Assessment and Approval of Stockist Material.
  - A0-Q-PQ-PRO-006- Procedure for Project Specific Prequalification of Subcontractors & Vendors.
  - A0-Q-PQ-PRO-007- Request for Inspection-RFI requirement in Construction Phase
  - A0-Q-PQ-GDL-001- Guideline for Project Quality Plan.
  - A0-Q-PQ-GDL-003- Guideline for Preservation Inspection in Construction Phase of Projects.
  - A0-Q-PQ-GDL-004- Guideline for Critical Quality Risk Identification and Mitigation during Project Execution.
- CONTRACTOR shall seek clarification from COMPANY regarding any conflict between standards or confirm their interpretation of the requirements stated in standards, which are considered unclear, however COMPANYs interpretation shall be binding.
- CONTRACTOR shall provide all necessary resources including adequate and competent personnel and all facilities to meet the requirements of the project and meet COMPANY satisfaction and relevant stakeholder expectations.
- CONTRACTOR shall provide adequate facilities, needed to achieve conformity to product quality in terms of buildings, workspace, associated utilities, process equipment (both hardware and software) and any supporting services, and shall ensure that the work environment in which these activities shall be undertaken shall achieve conformity to product requirements.
- CONTRACTOR is solely responsible for achieving the required standards of work and is responsible for implementing all contractual requirements.

7.2 PROJECT MANAGEMENT COMMITMENT

1. CONTRACTOR's Project Management Commitment is very important when implementing the Project Quality Management System and is the key factor in ensuring that the organization remains committed to its Quality Objectives.
2. CONTRACTOR's Project Management shall provide all necessary resources including adequate and competent personnel and all facilities to meet the Quality requirements of the Scope of Work and/or Purchase Requisition / Purchase Order and, meet COMPANY satisfaction and relevant stakeholder expectation and shall provide a full support to their Quality Function and involved personnel in the Project and ensure their full independency and non-influenced by any other functions.
3. CONTRACTOR's Quality personnel shall have sufficient authority to implement the Project Quality Plan and take corrective measures whenever necessary to maintain the desired level of quality and shall have independent communication and reporting function to the CONTRACTOR's Corporate Quality and to higher management to ensure independency and freedom of communication of CONTRACTORs Quality Function in the project.
4. CONTRACTOR's Project Quality Manager shall have direct functional reporting to the Project Manager as well as the CONTRACTOR's Top Management responsible for Quality and have the necessary authority for ensuring that the Project Quality requirements are implemented and maintained to COMPANY satisfaction. All Quality Personnel (including JV / Consortium if any) shall be reporting functionally to the designated Project Quality Manager of the CONTRACTOR
5. The authority of CONTRACTOR's Quality Personnel shall not replace or reduce the responsibility of other departments concerning their contribution to the attainment of the project quality requirements and objectives.
6. CONTRACTOR's Project Management shall provide all essential support to their Quality Personnel to implement the Audit program effectively and to close all raised findings is due agreed time as necessitated.

7.3 PROJECT QUALITY PLAN

1. CONTRACTOR's Project Quality Plan is the base document for assurance of Quality in Projects as they specify Quality practices, resources, sequence of activities and record how the CONTRACTOR will ensure that the standard of work including all functions / activities in the Project complies with COMPANY requirements. The Project Quality Plan shall address all applicable reference to all project functions, such as Governing regulations,
2. ADNOC/ COMPANY Standards, Specifications, International Standards & norms, etc. CONTRACTOR and (their Sub-CONTRACTORs as applicable) shall develop Project Specific Quality Plan in line with A0-Q-PQ-GDL-001 (Guideline for Project Quality Plan) with full consideration of AGPS-SD-QAQC-001 - Quality Management Requirements for Contractors in Projects) and ISO 10005 (Quality Management-Guidelines for Quality Plans) latest edition.
3. CONTRACTOR's Project Quality Plan shall be submitted within two weeks from the EFFECTIVE DATE of award of Contract as per the Project Deliverable List and prior to start of work for COMPANY review and approval. CONTRACTOR's Project Quality Plan shall be a stand-alone document and not included as part of any other documents.
4. Where Joint Ventures are involved, specific detail shall be provided to demonstrate how the individual Quality Management System and technical management systems will interface and efficiently work together. Where this is applicable, relevant procedures shall be provided.

5. Quality related Risks shall be determined and controlled to provide confidence that intended results will be achieved and undesired effects will be prevented or minimized.
6. Quality objectives for the project shall be stated; the Project Quality Objectives specified in this Scope of Work are the Minimum to be addressed, however CONTRACTOR shall further develop project specific quality objectives and agree with the COMPANY Quality. The monitoring and measurement processes and tools to determine achievement of these objectives shall be included or referenced in the CONTRACTOR Project Quality Plan.
7. CONTRACTOR's Project Organogram including Project Quality Organogram shall contain Quality key positions by name, lines of authority, and reporting relationships. CONTRACTOR to issue the project organization chart including the QA/QC charts as a standalone deliverable with unique document control number to avoid frequent revisions on main strategic deliverable e.g., PEP, PQP, etc. in case of any update.
8. The Project Quality Plan shall be verified for adequacy and effectiveness by CONTRACTOR Project Quality Lead / Manager. However, the changes to the Quality Plan shall be verified for impact, adequacy, and effectiveness by CONTRACTOR Project Quality Lead / Manager and shall be submitted to the COMPANY Quality for review and approval.
9. Requests to deviate from the approved Project Quality Plan shall be submitted formally to COMPANY Quality for review and approval, the requests shall follow the approved & established Project Specific Procedures.
10. The Project Quality Plan shall be approved and endorsed by CONTRACTOR's Project Manager and Project Quality Lead / Manager to demonstrate CONTRACTORs management commitment to Quality, identify and utilization of processes for the Quality Management System, understanding their interactions and interrelationships and engaging qualified resources for project realization.

7.4 PROJECT QUALITY POLICY, QUALITY OBJECTIVES AND (KPIs)

1. CONTRACTOR shall establish a Project Quality Policy and Project Quality Objectives in alignment with COMPANY Quality Policy and Objectives, and they shall be understood and communicated with all project personnel.
2. CONTRACTOR shall establish Project Specific Quality Objectives and Key Performance Indicators (KPIs) to be focused / aligned across all process functions and embedded in all strategic project deliverables and project delivery objectives without any conflict.
3. Key Performance Indicators (KPIs) shall be specified by CONTRACTOR for Project Quality, and shall be measurable, SMART and in line with the guidelines as specified in AGPS-SD-QAQC-001 - Quality Management Requirements for Contractors in Projects) Appendix-D.
4. CONTRACTOR's (KPIs) shall measure, monitor, and report the Project Key Quality Performance Indicators (both leading and lagging) on monthly basis to COMPANY to ensure targets are achieved and submit results of monitoring and measurement to facilitate subsequent corrective and preventive action analysis. In case KPI's are not met, CONTRACTOR shall perform trend analysis and propose immediate mitigation actions for COMPANY approval.

7.5 QUALITY DELIVERABLES

1. CONTRACTOR shall develop all necessary Project deliverables as Identified by COMPANY in GPE-GE-GDL-001 (Standard Engineering Deliverables for Projects Guideline) and FEED Deliverables List to cover all aspects of the execution of Contract based on ADNOC/ COMPANY standards and/or internationally accepted codes/practices. These shall include planning, co-ordination and execution

procedures, organizational interfaces with Company & Sub-contractors/Vendors (where applicable) and internal responsibilities together with reporting and limits of authority.

2. An overall register of all deliverables, including those planned, shall be produced and maintained up to date. This register shall be reviewed by the COMPANY who will decide that sufficient procedures are identified for the control of any activity. COMPANY shall be consulted to identify procedures to be submitted to their review and approval.
3. CONTRACTOR ensure developing and issuance of the following Quality Deliverables as a minimum:

- *Project Quality Plan.*
- *Project Quality KPIs and KPIs Tracking Register.*
- *Project Quality Risk Management*
- *Project Quality Organization chart.*
- *Project Audit procedure, Audit Plan and Audit Reports.*
- *Project information management procedure (Document Control/ Records Control)*
- *Project Non-conformance Control procedure, Reports and NCR tracking register.*
- *Criticality Rating Procedure and Matrix.*
- *Inspection strategy Procedure /Inspection co-ordination procedure.*
- *Procurement and Inspection procedure describing TBE and MR processes.*
- *Key Personnel CV's (Quality).*
- *ISO accreditation for CONTRACTOR from all performing office.*
- *ISO accreditation for Sub-Contractors, OEM and Vendors (as applicable).*

7.6 QUALITY SYSTEM REVIEW AND AUDIT

1. The Project Quality System shall be subject to regular review and audit by the CONTRACTOR and COMPANY to ensure continued suitability, adequacy, and compliance with project requirements and objectives.
2. CONTRACTOR shall establish a comprehensive Audit program covering all project phases, all performing (Locations / Offices) and extended to Sub-CONTRACTOR's/ Vendors as applicable. in addition, AD-HOC audits shall be performed for any process deficiency encountered during execution of the project.
3. CONTRACTOR shall develop a project audit procedure in line with the requirements addressed in AGPS-SD-QAQC-001 - Quality Management Requirements for Contractors in Projects) and ISO 19011 (Guidelines for Auditing Management Systems) and submit be submitted for COMPANY review and approval.
4. Audits shall be planned to cover all activities as applicable in the Project SOW, including all project disciplines and sub-disciplines relevant to the project and carried out in accordance with a schedule (progress wise, on each milestone, system wise) agreed with the COMPANY.
5. CONTRACTOR shall plan and carry out Audits at least at 30%, 60%, and 90% of Project Execution Progress as a minimum and shall invite COMPANY to witness such audits and COMPANY may attend at its own discretion. CONTRACTOR shall submit their audit plan to COMPANY within two weeks days of contract award. CONTRACTOR's Auditing Plan should be issued as a standalone documented deliverable to avoid frequent revisions in Project Quality Plan in case of any updates.
6. CONTRACTOR shall make regular planned audits on their own establishment and shall invite COMPANY specifically to witness such audits, and the COMPANY may attend at its own discretion. CONTRACTOR shall notify COMPANY formally, providing two (2) weeks notification along with Audit

plan to the COMPANY regarding actual date and location for all the audits to be undertaken to permit COMPANY representation.

7. CONTRACTOR's Quality Lead / Manager shall be responsible to ensure that Audits are attended to cover performing (Locations / Offices) of CONTRACTOR that are Approved by COMPANY for the execution of Project Scope of Work.
8. CONTRACTOR Audit personnel shall be certified to ISO-9001and approved by the COMPANY; they shall have no direct responsibility for the activities to be audited. Audit personnel shall have sufficient authority and organizational freedom to make the audit process meaningful & effective. Auditors shall have the relevant education, experience, and training to assure competence for performing assigned tasks.
9. CONTRACTOR's Quality Lead / Manager or his delegated representative, approved by COMPANY Quality Lead, shall undertake all audits, and shall have the authority to rectify identified procedural non-conformances and/or deficiencies.
10. CONTRACTOR's Audit Reports shall be issued to the COMPANY not more than 2 weeks after the audit. The audit report shall include a detailed audit frame report, relevant correspondence, details of the Non-conformance, proposed Corrective & Preventive Actions and closeout date.
11. Where deficiencies are identified, follow-up audits shall be required to address them and confirm close out/compliance.
12. COMPANY intends to perform Audits on the CONTRACTOR at approximately 15%, 45%, and 75% Project Progress during contract execution to verify compliance and effectiveness of the Quality Management System. Further, the COMPANY reserves the right to perform unscheduled audits of all functions / locations / offices of the CONTRACTOR.
13. COMPANY will provide two (2) weeks advance notification to the CONTRACTOR regarding actual date and location for the audits and the CONTRACTOR shall ensure the availability / participation of their personnel based on the Audit Plan. COMPANY will issue the Audit Reports in two (2) weeks' time from the completion of Audit, for the action of the CONTRACTOR.
14. The CONTRACTOR shall aid COMPANY representatives and provide all information and full access to their Quality systems as required by the COMPANY. CONTRACTOR shall provide copies of all audit reports and corrective actions to COMPANY and shall maintain a non-conformance, corrective and preventive action register available for review.
15. When non-conformities are found, corrections of inadequate actions are required. CONTRACTOR shall respond with the appropriate corrective action immediately along with evidence of rectification. All non-conformances shall be closed within three (3) weeks of identification or COMPANY must approve the closeout date.
16. CONTRACTOR shall track all Audit findings through tracking register and status of closeout actions shall be reported in Monthly progress report's Quality section.

7.7 QUALITY RESOURCES

1. CONTRACTOR shall define adequate Quality Resources which are required to ensure effective implementation of Quality function throughout the project without lapses. CONTRACTOR shall maintain sufficient Inspection, tests, and Quality Assurance personnel, independent of their other divisions/work processes to ensure adequate implementation of project QMS.
2. CONTRACTOR shall select and mobilize Quality Manpower/ Resources as specified in A0-Q-PQ-SP-001(EPC Contractor Quality Personnel Requirements), and AGPS-SD-QAQC-001 - Quality Management Requirements for Contractors in Projects). CONTRACTOR shall provide details of

qualifications and experience of the job holders as proof of their competence, suitability, and eligibility; this section should give in general terms the allocation of Quality functions and other relevant responsibilities of the various management or department heads.

3. CONTRACTOR shall provide logical, well-structured, adequately staffed project specific Quality organization chart. CONTRACTOR's Project Quality Organogram shall specify Key positions of Quality Personnel by name, lines of authority, and reporting relationships.
4. CONTRACTOR proposed Quality Personnel Curriculum Vitae's (CV) shall be submitted for COMPANY review / approval. CV shall be accompanied by valid copy(s) of their Qualifications and Certifications. COMPANY reserves the right to interview any of the CONTRACTOR personnel via face-to-face at the expense of the CONTRACTOR. The decision of the COMPANY regarding acceptance or rejection of the proposed personnel shall be final and binding upon the CONTRACTOR even if the candidate has previous experience with the COMPANY. Approval and interview records shall be maintained for all the Key Personnel by the CONTRACTOR.
5. The CONTRACTOR's organization shall depute committed Project Quality Lead / Manager, responsible for the coordination, planning, implementation, and review of the Quality Management System. In addition, CONTRACTOR shall provide adequate & competent Quality Personnel to support the Project Quality Lead / Manager for the duration of the Project.
6. All Quality personnel (including JV / Consortium if any/ Sub-Contractor) shall be reporting functionally to the designated CONTRACTOR's Project Quality Lead / Manager.
7. Project Quality Lead / Manager shall have independent functional reporting to the Project Manager as well as the CONTRACTOR's Top Management responsible for Quality as direct function.
8. CONTRACTOR's Project Quality Lead / Manager shall hold a university degree in Engineering or equivalent, and an internationally recognized certification in auditing IS0 9001:2015 or equivalent with a min. of 20 years of experience (15 years for Project Quality Lead) in oil & gas projects of which minimum five (5) years in managing Quality functions in Offshore Oil & Gas Projects similar to the Project scope of work. PQM should have experience in handling quality assurance and quality control in the project and possess strong coordination / communication skills and very good knowledge of spoken and written English language.
9. Senior Quality Assurance Engineer / Quality Assurance Engineer shall hold a university degree in Engineering or equivalent with a minimum of 12 years of experience (10 years for Quality Assurance Engineer) in Oil & Gas Projects of similar to Project scope of work and have experience in handling both quality assurance and quality control functions in the project. He shall hold lead auditor certificate in accordance with ISO 9001:2015 or equivalent recognized internationally, must have adequate experience in auditing. He should possess coordination / communication skills and very good knowledge of spoken and written English language.
10. Typical roles and responsibilities of CONTRACTOR's Quality Personnel shall be, as a MINIMUM:

    - *Accountable for the implementing of the Contractual Quality requirements.*
    - *Accountable for the establishment and implementation of the Project Quality Plan, Procedures, Work Instructions, and Inspection & Test Plans, as applicable.*
    - *Focal point for all project quality related matters.*
    - *Establish the Project Quality KPIs with cross functional teams and continuously monitor and audit the implementation of the Project Quality Management System. Manage, correct, analyze, and prevent reoccurrence of non-conformances; and record the lessons learnt.*
    - *Manage, correct, analyze, and prevent reoccurrence of non-conformances; and extract the lessons learnt.*

- *Communicate, coordinate, and report to COMPANY Quality Representative of all the Quality Function related matters.*
- *Shall be responsible for review of Technical Deliverables and providing inputs (e.g., Project Specification, Data Sheets, Material Requisition's, Technical Bid Evaluation (TBE), Technical Queries, Concession Requests, etc.). The review shall be performed on ALL revisions until the approval of the Project Deliverable.*
- *Develop and review specifications and procedures.*
- *Implementation of Audit program against approved Project Audit Plan.*
- *Follow up NCR close-out and maintain NCR tracking register.*
- *Monitor the implementation and effectiveness of the project quality system.*

11. CONTRACTOR shall retain the records of Competence, Interview / Approval of the Quality Personnel of CONTRACTOR, Sub-CONTRACTOR, Consultant etc. as applicable and further submit the updated records for COMPANY information.
12. CONTRACTOR shall provide a mitigation plan for shortage in manpower resources. However, any planned demobilization, replacement, or even long leave of any of CONTRACTORs approved Quality Personnel shall be communicated officially through letter at least one month in advance with proper justification along with proposed replacement.

7.8 CRITICALITY ASSESSMENT / INSPECTION CLASS

1. CONTRACTOR shall develop a Project Criticality rating and Inspection procedure, as well as develop an Inspection Criticality matrix at the early stage of the project as soon as the Material and equipment list is available and submit for COMPANY approval.
2. Inspection criticality matrix shall be established for the entire procurement process. Criticality rating (CR) procedure shall derive Inspection level of engineered equipment, packages, and bulk materials, based on AGPS-SP-GN-001 (Criticality Rating Assessment in Projects), and AGPS-SD-QAQC-001 (Quality Management Requirements for Contractors in Projects), and the same shall be the means for developing the overall inspection strategy.
3. The results of Criticality assessment shall be used to develop the minimum quality system and quality control requirements to be applied during design, procurement, construction, and commissioning activities for each level of criticality. The assigned Criticality rating and Inspection level shall be specified on the individual equipment Data sheets, Material Requisitions, and Inspection and Test plans.
4. Inspection points shall be compliant with and in accordance with COMPANY AGPS-SP-QAQC-002 requirements, however if COMPANY specification or code does not cover any activity or item, then, inspection points shall be at the discretion of COMPANY Quality Representative.
5. The requirements for source inspections for all procurement items shall be clearly established, based on the Criticality Assessment requirements. as per COMPANY specifications. COMPANY reserves the right to increase the Minimum Inspection Class already established on the Project if either the VENDOR recommended by the CONTRACTOR for a particular requisition has had a not-so satisfactory performance in the past or the VENDOR executing the Purchase Order is not meeting the specified performance requirements in the purchase order. The change in Inspection Class to the next highest level for the purchase order shall be without additional cost to COMPANY.

7.9 DESIGN, DEVELOPMENT AND ENGINEERING

1. CONTRACTOR shall develop Engineering Execution Plan for the Project providing a strategy for development and progression of the design and Engineering works.
2. Control of Design Inputs procedure shall be established, and records shall be maintained for adequacy and completeness, Technical Data, Design Criteria, Contractual Information, etc.
3. Control of Design Verification procedure shall be established, and records shall be maintained for constructability; Discipline Check (DC), Interdisciplinary Check (IDC), etc., CONTRACTOR's Inter Discipline Check (IDC) and Discipline Check (DC) systems shall be in place to ensure that engineering deliverables are checked by interacting disciplines for correctness and shall be reviewed and approved by COMPANY.
4. CONTRACTOR shall ensure that the IDC Matrix for the Project Deliverables along with the DC/IDC Procedure is submitted for review and approval. CONTRACTOR shall apply only the approved DC/IDC Procedure & Matrix on the Project. CONTRACTOR DC/IDC shall be performed till the document has been approved by COMPANY. The Procedure shall provide for an adequate review period for Project Deliverables taking into consideration the number of documents to be reviewed in a specified time. COMPANY doesn't accept review cycles of 2 or 3 days specifically prior to Issued for Review (IFR) submission, as this has resulted in numerous COMPANY comments and delays in the subsequent revisions, in the recent past. The CONTRACTOR shall ensure that ALL the required Discipline / Departments are identified in the IDC Matrix, and no exceptions are made ie., at the discretion of the Discipline Lead etc.
5. Control of Design Review procedure shall be established, and records shall be maintained to assess the basis of the design and confirm its safety and installation aspects, HAZOP Report, etc.
6. Control of Design Validation procedure shall be established, and records shall be maintained to ensure that design needs and contractual requirements are met; Approved for Design Drawings (AFD), Final Project Report, etc.
7. Control of Design Output procedure shall be established, and records shall be maintained; Drawings, Specs, Data Sheets, BOQ etc. and control of Design Change procedure shall be established, and records shall be maintained, Design Modification, Design Change Request, Key Document Change Notice etc.
8. CONTRACTOR Shall maintain a Controlled list of their approved disciplines specifying their names, abbreviations, and their signature forms for proper and ease tracking of SDC/IDC process, this list shall be communicated with COMPANY for information and provided whenever updated.
9. CONTRACTOR shall develop all necessary deliverables/ procedures in accordance with GPE-GE-GDL-001 Standard Engineering Deliverables for Projects Guideline, and Feed Master Deliverables List, to cover the entire scope of project and related activities in line with COMPANY Engineering Standards and/or internationally accepted codes and practice.
10. All deliverables/ procedures shall be project specific and shall be subjected to COMPANY review and approval, CONTRACTOR's Corporate procedures shall not be acceptable by COMPANY as project deliverables/ procedures. CONTRACTOR's Deliverables / Procedures shall include planning, co-ordination and execution procedures and organizational interfaces with COMPANY and internal responsibilities together with reporting and limits of authority.
11. An Engineering Document Register that lists the project engineering drawings and documents shall be developed and submitted for COMPANY Review/ Approval. Documents which are required for review and comments by the COMPANY shall be identified at the start of the project through markup

of the deliverable register. A separate procedure shall be produced by the Contractor to control this requirement.

12. CONTRACTOR shall perform Technical Peer Review (Design Cold Eye Review), as part of the Project Quality Management System and Identify the same in the Project Quality Audit Plan. ISO 9001:2015 specifies Design and development of products and services in Clause 8.3, and Subclause 8.3.4 Design and development controls covers the controls in your design process, including design reviews, and sets out standards for the review process. The objective of auditing the design and development process is to determine whether it is managed and controlled to enable products and services to meet their intended use and specified requirements.
13. CONTRACTOR shall submit the Technical Bid Evaluation (TBE) Procedure for review and approval of COMPANY. TBE for Long Lead Items (LLI) shall be performed on BIDDERs (MANUFACTURERs) in the COMPANY Approved Vendor List (AVL), unless otherwise approved by COMPANY to include Non-AVL BIDDERs.
14. COMPANY may require during the execution of the project that the CONTRACTOR for effective control of any work, if required develop additional procedures.

7.10 PROCUREMENT AND SUBCONTRACT CONTROL

1. CONTRACTOR shall establish a Project Procedure for Procurement and Subcontract describing the Responsibilities of all stakeholders during Procurement Services i.e.., CONTRACTOR, COMPANY, VENDOR / BIDDER and EPC CONTRACTOR. The Procedure shall specify approval of the Inquiry Requisition, Selection of VENDOR's/ BIDDER's/ Sub-Contractor, RFQ, Evaluation of Bids, and Approval of Technical Bid Evaluation by CONTRACTOR and COMPANY. The Procedure shall include the Templates like the Material Requisition, VENDOR Data Requirements List (VDRL), Technical Bid Evaluation (TBE) etc. CONTRACTOR shall develop the Project Procurement Procedure based on COMPANY Project Procurement Procedure, A0-PSD-CPPA-GDL-002.
2. Documents for procurement of material, equipment, and services, shall include, but not be limited to the following:

- ***Material Requisitions***; and shall include all the necessary Specifications, Datasheets, Instrumentation / Control requirements, Electrical requirements, Piping and Structural requirements, Criticality Rating & Inspection Class of Material / Equipment, Certification Type, and Vendor Document Requirements.
- CONTRACTOR shall ensure that the Sub-components are identified for the Packages / Skids in the Purchase Requisition and Specify the Criticality Rating, Inspection Class & Material Certification requirements.
- Reference to applicable project codes & standards to the purchase requisitions.
- Listing of VENDOR documents such as instructions, procedures, drawings, specifications, inspection and test plans, inspection and test records, and VENDOR quality assurance (QA) records to be prepared, submitted, or made available for CONTRACTORs and/or COMPANY review or approval.
- Technical Bid Evaluation (TBE) shall be performed on Long Lead Items (LLI), as agreed with COMPANY. TBE shall be issued by the CONTRACTOR only when the Material Requisition is Issued for Approval (IFA) and at least Approved with Comments by COMPANY, as a Minimum. Technical Bid Evaluation (TBE) Report summary shall include the Product Group Number, Product Group Description, Vendor Number, Vendor Name, Manufacture Number, Manufacture Name, and Manufacture Country.

- *The following "List of attachments" shall be part of the Technical Bid Evaluation (TBE) Report,*

  a) *Deviation list with CONTRACTOR's response/disposition for each BIDDER, and space for COMPANY review response.*
  b) *Experience list (as applicable)*
  c) *List of clarifications with response*

3. *Status of COMPANY approval of applicable documents that include the Material Requisition, Datasheets, Specifications, P&ID's etc. This is to understand the basis of the evaluation ie., which revisions of the documents* were issued to the BIDDERS.
4. CONTRACTOR shall apply AGES-SP-13-001 (Criticality Rating Specification) in establishing the Criticality Rating of the Project Material / Equipment and apply AGPS-SP-QAQC-002 (Procurement Inspection and Certification Requirement in Projects) in determining the Inspection Class and Certification Requirement of the Project Material / Equipment.
5. CONTRACTOR shall incorporate and implement A0-Q-PQ-PRO-005 (Procedure for Assessment and Approval of Stockist Material), and A0-Q-PQ-PRO-006 (Procedure for Project Specific Prequalification of Subcontractors and Vendors) in Procurement and Inspection Procedure.
6. CONTRACTOR is responsible for ensuring and assuring that all goods and services procured/ contracted are from ADNOC approved list and are in accordance with the contract requirements. If the COMPANY revokes approval of any Supplier/Vendor/Sub-contractor, then upon notification from COMPANY all further orders/contract with such Supplier shall be cancelled immediately, and the COMPANY shall not bear any financial or schedule implication.
7. CONTRACTOR shall perform adequate assessment of Sub-Contractor/ OEM/Vendor listed in AVL prior to engage in the project. As well as their Quality system is assessed and appraised to ensure that it is suitable and able to satisfy the requirement for the goods or service to be procured.
8. CONTRACTOR's quality system and deliverables shall assure and ensure the Sub-contractor or Vendor operates a certified quality management system and has established necessary quality plans / procedures / work instructions / method statements etc. to execute their scope of work, the Sub-contractor, and Vendor's Project Quality Plans (PQPs) shall be reviewed by CONTRACTOR's Quality Manager and approved by the Contractor's Project Management.
9. A listing of all proposed Sub-contractors and Suppliers is submitted to the COMPANY for approval along with project procurement plan prior to any order or contract placement. Company may decide to audit the proposed Sub-contractors and Vendors prior to any order or contract placement.
10. Procurement and sub-contract documents shall contain all the applicable accurate and approved technical data, test, inspection, verification and delivery and final documentation requirements including asset registration as per COMPANY standards, Quality and inspection requirements shall be incorporated in the purchase orders to comply with the criticality of the items to be supplied.

7.11 INFORMATION and DATA MANAGEMENT

1. CONTRACTOR shall develop Information Management Procedure define the control of documentation and data process in line with A0-ENG-I-SP-001 (Specification for Information Management in Projects) to demonstrate to the COMPANY that timely distribution of the required documents to project personnel is performed. It is highly recommended to have EDMS (Electronic Data Management System) to be utilized by CONTRACTOR for the project.
2. The Information Management Procedure shall specify and/or include:

- Responsibilities for documentation control,
- Identification and control of project document deliverables,
- Document numbering system,
- Document review and distribution matrices,
- Correspondence control and distribution,
- Control of documents of external origin and distribution, and document transmission,
- Control of document review and approval,
- Methodology for the receipt, control and approval of Sub-contractor/Vendor documents,
- How document confidentiality will be maintained.

3. Information Management Procedure shall define that All project documentation shall be controlled to ensure they are at all times, legible, accurate, complete and indexed, available at the place of work or operation, and revised, when necessary, promptly and effectively, fully traceable, Easily and promptly retrievable, authorized by the appropriate manager, and suitably identified and withdrawn when obsolete.
4. documents can be revised once the changes are approved by the COMPANY through appropriate change approval process, if Contractor/Vendor made changes without Company approval, it is considered as non-conformance, and the affected activities shall be reinstated to the original project specification requirements,
5. Documentation includes the information relevant to design, purchase, construction, inspection, testing, installation, commissioning, operate and maintain the plant as well as all the as-built drawings, dossiers, technical data sheets, TPA documentation etc. identified by COMAPNY by COMPANY in GPE-GE-GDL-001 (Standard Engineering Deliverables for Projects Guideline), and FEED Deliverables List.
6. Information Management Procedure shall define Handover and feed dossiers content and distribution requirements.

7.12 MANAGEMENT OF CHANGE/ CONCESSIONS.

1. CONTRACTOR shall develop a Project Management of Change (MOC) procedure; whereby any proposed changes to baseline documentation are carefully reviewed to establish all resultant effects on all other disciplines before approval of change is given. This procedure shall be developed by the Contractor in consultation with COMPANY and shall be in line with the latest ADNOC / COMPANY Change Management Procedure; A0-ENG-Z-PRO-002 (Procedure for Projects Technical Query), A0-ENG-Z-PRO-006 (Concession Requests), A0-ENG-Z-PRO-003 (Procedure for Concession Requests) and A0-PSD-CPPA-PRO-001 (Project change management procedure).
2. CONTRACTOR shall ensure that all changes are identified, assessed by Competent persons, and approved by COMPANY, and are effected/implemented in a controlled manner which ensures that intent and a safe working environment is maintained during and after the process of change.
3. CONTRACTOR Quality System shall ensure that Change requests are processed when technical integrity, design, construction, safety and environment, schedule and/or cost would be seriously affected without them. Changes can also be requested by the COMPANY and shall be only approved by relevant COMPANY Technical Authority.
4. Project Management of Change Procedure shall also be applicable to all VENDORs, Sub-CONTRACTORs, Consultants, or any other entity involved in delivery of a part or full project.

5. Where COMPANY approval of original design/design documents is required, i.e., Class 1 Documents, COMPANY approval of design deviation is also required. All such changes shall be controlled by entering it in the Management of Change Register and periodically, status of each change shall be provided to the COMPANY.
6. All requests raised by CONTRACTOR shall clearly reflect the originator, reviewer and approver names, designations, and signatures.
7. CONTRACTOR shall maintain a controlled Register addressing all MOC/CR/RFI/TQ classified, and register shall be properly tracked and updated throughout the project's lifetime.

7.13 INTERFACE CONTROL

1. CONTRACTOR shall ensure that all interfaces between this PROJECT, other ongoing applicable PROJECT's existing operating facilities are identified through the review of relevant hazards and any potential risks are assessed and managed.
2. CONTRACTOR shall develop a procedure to identify interfaces and control associated risks, the procedure shall be submitted for COMAPNY approval.
3. The CONTRACTOR shall maintain a controlled Interface Register, it shall be properly tracked and updated throughout the project's lifetime.

7.14 RISK MANAGEMENT CONTROL

1. CONTRACTOR shall establish a Project risk management procedure in accordance with COMPANY "Project Risk Management Guidelines" and Project Risk Assessment guidelines (ADNOC Group VAP Guidelines) and "HSE Risk Management Procedure". Project Risk Assessment Workshops shall be carried out on agreed intervals, and Workshop shall be chaired by a third-party chairman, proposed by CONTRACTOR, and approved by COMPANY.
2. The CONTRACTOR shall maintain a controlled Risk Management Register comprising Quality Risks, it shall be properly tracked and updated throughout the project's lifetime.

7.15 NONCONFORMANCE CONTROL

1. CONTRACTOR shall establish & implement Project Specific Nonconformance management Procedure for controlling identifying, reporting, and disposing nonconforming products/services. The Procedure shall be developed in line with A0-Q-PQ-PRO-001 (Procedure for Project Non-conformity Management) using Company NCR template.
2. When non-conformities are found, corrections of inadequate procedures are required. The CONTRACTOR shall respond with the appropriate corrective action immediately along with evidence of rectification. Necessary correction action shall be proposed by NC recipient within 5 working days of receiving NC and a CLOSE-OUT agreed with COMPANY however it shall not exceed six (6) weeks unless otherwise specifically agreed with COMPANY for exceptional NC's, the NCR Procedure of the CONTRACTOR shall address this requirement.
3. Copies of all CONTRACTOR non-conformance/corrective action reports shall be submitted to COMPANY within 5 working days of the date of issue of the non-conformance / corrective action report. Corrective action for all Major Non-conformance shall be approved by COMPANY.
4. Any Nonconformities identified during the project, CONTRACTOR shall perform root cause analysis prior to start of rectification and shall notify the failures to COMPANY. COMPANY reserves the right to participate in the root cause analysis investigation.

5. CONTRACTOR shall maintain a Register for NCR / Deficiency Report's for monitoring of Close-Out. The Register shall be updated and submitted on a basis agreed between the CONTRACTOR and COMPANY.
6. COMPANY has in place a dedicated Nonconformity Management System to record, communicate and monitor nonconformities on the Project; the CONTRACTOR and its Sub-CONTRACTOR's shall register on the portal, https://pqcncm.adnocoffshore.ae/ immediately on award of the Contract. This is a mandatory requirement, and the CONTRACTOR shall immediately report to COMPANY of any Technical Issues regarding the registration in the portal.

7.16 QUALITY MEETINGS, REPROTING AND REVIEWS

1. Project Quality Meeting shall be held every week or on a basis agreed between CONTRACTOR and COMPANY; it should be organized and by CONTRACTORs Project Quality Lead / Manager and attended by COMPANY Quality representative and shall be focused on Quality Objectives/Audit Program / KPI, Concerns related to the project deliverables, and NCR close out status etc.
2. Minutes of Meeting (MOM) shall be issued within 2 working days from the date of completing the meeting to COMPANY quality for review and endorsement.
3. CONTRACTOR's Project Quality Lead / Manager shall escalate the major issues for his Project Management for immediate resolution.
4. CONTRACTOR shall Report Quality either separately either through Quality Report or Project Weekly Progress Report (weekly/ Monthly), Quality Report shall contain the followings as a minimum:

- *Quality deliverables approval status.*
- *Summary of KPI statistics, Risk action register close out statistics, lessons learned action register statistics.*
- *Summary of all internal and external audits carried out, reported against the audit schedule.*
- *List of non-conformance and corrective action reports, the status and the planned dates for follow up and close-out.*
- *Major areas of concern or weakness identified in the Quality System and details of any investigations to prevent the recurrence of failures.*

7.17 QUALITY IMPROVEMENT

1. CONTRACTOR shall continually seek to improve the effectiveness of its QMS through the use of the quality policy (quality objectives), audit results, analysis of data, risk matrix/mitigating, corrective and preventative actions and management reviews.
2. CONTRACTOR should review the effectiveness of this process and take the necessary steps to continually improve the project organization and the processes used to support effective project delivery.
3. It is recommended that the CONTRACTOR's project team carry out a documented review of the project performance, highlighting experience from the project that can be used by other projects.

7.18 Handover

1. CONTRACTOR shall establish a handover procedure in accordance with COMPANY Guideline Document (A0-PSD-VAP-GDL-004) and contractual requirements. Final Dossier Handover Procedure, A0-ENG-D-PRO-003 shall be applied for Dossier Handover.

2. Upon completion of the FEED and approval of all the deliverables by COMPANY, CONTRACTOR shall submit to COMPANY, FEED design dossier (s) that will include all the, drawings, documents, deliverables finalized FEED including documents from Vendor(s) and Sub-CONTRACTOR. The FEED dossier index shall be discussed and agreed with the COMPANY prior to compilation. The dossier shall contain all the FEED deliverables including PROJECT management deliverables and segregated per discipline.
3. Deliverables as listed in the Master Document Register (MDR) and the following documents, as a minimum, shall be prepared by CONTRACTOR and submitted to COMPANY:

PROJECT Management Documents shall include but not limited to:

a) Project Quality Plan
b) Project Execution Plan
c) Integrity Management Plan
d) Quality Control Procedures
e) Overall Coordination Procedure
f) Change Control Procedure
g) Project Procurement Plan & Procurement Procedure
h) Criticality Rating Procedure and Matrix
i) Inspection Management Procedure
j) Document Control and Management Procedure
k) Long Lead Items List
l) EPC Scope of Work
m) PROJECT SCHEDULE
n) Cost Estimate

Design Manual shall include but not be limited to:

o) Introduction
p) Summary
q) Statement of Requirement (SOR)
r) FEED Report
s) FEED Close Out Report
t) List of the following:
- SPECIFICATIONS
- Basis of design and design criteria
- Description of systems
- PROJECT philosophies segregated per discipline
- Drawing index
- Document index
- All data sheets segregated per discipline
- All drawings segregated per discipline
- All calculation notes segregated per discipline
- All studies
- Vendor(s) documents and drawings
- SUBCONTRACTOR documents and drawings
- Technical Queries
- Concession Requests

4. Any assumptions made in the design analysis must be properly explained and justified. All the input, output and summary results shall be in a format such that they can be traced and reviewed effectively. Wherever possible CONTRACTOR shall include sketches with the required information for clear understanding.

8. RISK MANAGEMENT

8.1 Risk Identification, assessment and mitigation

Project risk identification, assessment, mitigation and management is a critical and mandatory component of project management across all phases of a project lifecycle. Project risk assessment shall be performed through qualitative and quantitative analysis; this should be selectively applied as applicable.

As per COMPANY risk management standard, Risk Identification for this project will be done using brainstorming sessions based on the qualitative and quantitative (if required) parameters. Participants in the Risk Identification workshop will cover all subjects of potential risks such as contractual, engineering, procurement, fabrication, construction, offshore installation and commissioning works.

ENGINEERING CONTRACTOR shall;

a) Prepare and arrange project risk assessment session/workshop on monthly basis.
b) Prepare Project Risk Register, maintain and update it on monthly basis for the entire duration of the project until all items are satisfactorily closed (or reduced to an ALARP level).
c) The risks shall be classified as per ADNOC Project Risk Management Guideline and risk matrix
d) Arrange risk review meeting on monthly basis or as agreed with the COMPANY's Project Management Team to review the progress or actions taken by the risk owners.

8.2 Risk monitoring, reporting and review

After the implementation of mitigation strategies, residual risks need to be actively monitored until they are reduced down to ALARP level (taking into account that the cost of further risk reduction may be excessive compared to the benefit associated). This means systematic, continuous tracking and evaluation of the effectiveness and appropriateness of the risk mitigation strategies, techniques, and actions established within the Risk Register.

ENGINEERING CONTRACTOR shall:

- Provide COMPANY all available information regarding the status of each risk and its exposure
- Allow monitoring of the progress and the efficiency of all undertaken risk control actions
- Provide regular updates (monthly) of the Risk Register;
  - To ensure that risk mitigation actions have been undertaken by the Risk Owner and new actions have been identified where appropriate
  - To assess the effect of those actions on the probability and impact on overall project objectives
  - To amend the risk scores where appropriate and update the qualitative analysis

9. GENERAL ENGINEERING SCOPE OF WORK

This section outlines a high level summary of COMPANY's requirements for this scope of work.

9.1 General

1. Supply of Manpower, services, and Supervisions to carry out Scope of work (WORK) which includes Site surveys and Data collection, Front End Engineering Design (FEED) including Project Management and Control

2. ENGINEERING CONTRACTOR shall be fully responsible for day-to-day informal coordination with COMPANY departments and ensure timely execution of the project.

3. Without limitation, ENGINEERING CONTRACTOR shall be responsible for Multi-discipline Engineering, Design, Project Management and Support Services

4. Prepare the comments register (CRS) with ENGINEERING CONTRACTOR response to COMPANY comments, resolve and perform all necessary work for the action points based on resolution agreed by COMPANY.

5. Perform Engineering/Design in accordance with COMPANY, National, International Design Codes, Standards, Policy & Procedures. However, in absence of these, Industry Best Practices shall be followed by ENGINEERING CONTRACTOR.

6. Site survey, data collection including review of the collected data and documentation, confirmation of proposed tie-ins or identification of alternate tie-ins (if better) to the existing facilities and where necessary verification and updating applicable COMPANY drawings to "As-built" status. Further to data provided in contract documents, ENGINEERING CONTRACTOR shall identify and perform site surveys to confirm and / or collect data and retrieve documents as may be required to carry out the WORK.

7. Material of Construction of Bulk Items, Equipment, Pipe, Piping-Bulk items, Piping Speciality items, valves and Instruments shall be finalized.

8. Review and endorse the concept design and subsequently develop the same to detailed level of engineering.

9. To optimize / improve upon the design and minimize / eliminate shut down requirement.

10. Prepare/update Design Basis, Project Specifications & Standard Drawings.

11. Development of Tie-ins philosophies, shutdown requirements and shut down durations.

12. ENGINEERING CONTRACTOR shall carry out and be responsible for mandatory HAZOP review(s) Workshop involving Third Party Chairman and implement all action items within his scope without additional cost and schedule impact.

13. Preparation of review reports and review close out reports.

14. Carryout SIL assessment workshop for the shutdown functions as required based on the outcome of HAZOP. Prepare SIL study report and implement all recommendations and actions in the Engineering deliverables. ENGINEERING CONTRACTOR shall appoint and sponsor Certified Functional Safety

Engineer as SIL Chairman who will be from Specialist Third party approved by COMPANY. CV of chairman will be approved by COMPANY.

15. Comply with COMPANY HSE processes (refer section-3 – HSE), PHSER, PHA, Project Risk Assessment etc.

16. Preparation of Multi discipline drawings / documents.

17. Preparation of Quality plan in compliance with ISO 10005 “Quality Management Guidelines for Quality Plan”, COMPANY A0-Q-PQ-SP-002, A0-Q-PQ-GDL-001 and in line with ISO 9001

18. Material Requisition for Long Lead Item(s)(LLI) based on COMPANY vendor list.

19. Assistance in providing ENGINEERING deliverables to Independent Project Review (IPR) agency, participation in IPR review session (to be conducted by COMPANY appointed team) and Updation of engineering deliverables to close out of all applicable action items.

20. Submission of all deliverables as per Scope of Services for review and approval from COMPANY.

21. Preparation of a demolition package/drawings for applicable Project scope that are to be replaced/modified.

22. Weekly, Monthly progress review meetings and the sponsors meetings shall be held on monthly basis at COMPANY’s office. ENGINEERING CONTRACTOR shall prepare MOM for all such meeting held.

23. Presentations to COMPANY.

24. Preparation and submission of all specifications related to design and installation and datasheets.

25. Study of any underground existing facilities to avoid clashes with new facilities being proposed as part of project.

26. Engineering & design as required for modification of existing facilities shall be studied and clearly defined by ENGINEERING CONTRACTOR without any ambiguity to perform construction activities without study and/or redesign of the facilities.

27. ENGINEERING CONTRACTOR shall note that all modification to the existing Electrical, Telecom and control systems/equipment will be carried out by OEM or OEM authorized agent under overall contract to maintain the originality of the equipment. ENGINEERING CONTRACTOR shall appoint OEMs to perform the necessary site visits (accompanied by ENGINEERING CONTRACTOR) and adequacy checks of the existing system. ENGINEERING CONTRACTOR shall develop the modification specifications/work packages based on OEM Inputs.

28. Technical & HSE Review(s) with COMPANY Representation.

29. ENGINEERING CONTRACTOR shall provide Chairman and Scribe for all Technical /HSE reviews and prepare the respective reports and close-outs. CV for chairman shall be submitted for COMPANY’s approval.

30. Project Management Services, Planning, Reporting, Controls & Administration.

31. Document Management including provision of electronic documents and drawings compatible with COMPANY format.

32. Overall Project Schedule in accordance with COMPANY requirement.

33. Preparation and submission of all calculations.

34. Implementation of actions arising out of COMPANY Project Safety Review(s) and technical reviews/workshops.

35. Preparation of Project Quality Plan and Procedures in line with latest ISO 9001 guidelines & COMPANY procedure.

36. ENGINEERING CONTRACTOR shall be fully responsible for coordination with his subcontractors as well as any third parties, etc., to ensure timely execution of this Contract.

37. Further to data provided in this document, ENGINEERING CONTRACTOR shall identify and collect data during site survey and document retrieval as may be required, to carry out the complete scope of work as detailed in this document.

38. ENGINEERING CONTRACTOR shall clearly identify the need & scope of performance test(s) required for System & panel equipment and include the same in the Material Requisitions.

39. ENGINEERING CONTRACTOR shall perform discipline-wise design reviews, and the findings shall be incorporated in the engineering Dossier.

40. The ENGINEERING CONTRACTOR shall develop the detail design deliverables to Issued for Design status and then as-built status considering design developments, site survey findings, constructability, vendor data, etc.

41. The ENGINEERING CONTRACTOR shall perform all design related activities to complete the project in all respects, e.g., design calculations, simulations, installation engineering, development of specifications, drawings, 3D models, MTOs, MRs , etc.

42. The ENGINEERING CONTRACTOR shall perform necessary process simulations as mentioned in the detailed process scope of work.

43. The ENGINEERING CONTRACTOR shall be responsible for the design of all systems including, but not limited to, collection of all relevant data and drawings for all existing facilities, verification of the existing PFDs and P&IDs, and interface with COMPANY disciplines.

44. It is the responsibility of ENGINEERING CONTRACTOR to integrate the design with any other ongoing projects and perform any offshore site surveys to prepare core documents for the FEED stage. ENGINEERING CONTRACTOR shall include enough site surveys to familiarize and identify suitable interface points based on the ongoing projects.

45. ENGINEERING CONTRACTOR shall develop PDMS 3D models per COMPANY standard, of all piping, structure, electrical, instrumentation and controls, Telecommunication, safety, mechanical systems, etc., within the project scope.

46. ENGINEERING CONTRACTOR shall provide services such as preparation/updating of material requisitions, technical and commercial evaluation of bids, vendor document/drawing review, etc., of various items in accordance with the project requirements.

47. ENGINEERING CONTRACTOR's compliance to COMPANY standards

A0-ENG-Z-GDL-001	Project Deliverables
A0-ENG-Z-GDL-002	Design and Engineering Assurance in Projects
A0-ENG-Z-PRO-001	Procedure for Engineering Review
Section 17 of this SOW	Compliance with all applicable standards - Appendices

48. ENGINEERING CONTRACTOR shall perform safety studies per the SOW and the requirements of ADNOC standard. It is the responsibility of ENGINEERING CONTRACTOR to incorporate all action items arising out of such reviews or studies into the detail engineering.

49. Software listed in Section 9.10 shall be used to develop, design and perform the WORK.

50. COMPANY latest standards shall be followed while developing drawings and documents.

51. Project document control / management and its compliance to specification for Information Management in Projects (A0-ENG-I-SP-001) including MDR/VDR templates.

52. Handover of final documentation in compliance to Final Dossier Handover Procedure (A0-ENG-D-PRO-003).

9.2 Design Engineering and Services

Based on the Select study, deliverables as outlined in the following sections (as a minimum) shall be developed as part of FEED Engineering scope of service. This section and sub-sections cover the minimum design requirements for all engineering disciplines for carrying out the Front-End Engineering & Design activities by the ENGINEERING CONTRACTOR.

a) Facilities shall be designed for a safe, reliable, continuous, and uninterrupted operation. The facilities shall have stable operation and be capable of safely withstanding transient variations like start up, shutdown, upset conditions.

b) Facilities shall be designed for fully autonomous operations that require no human intervention during normal operations and upset situations. Facilities shall be self-sufficient leveraging on AI to optimize the process conditions, detect the anomalies, perform the analysis and take corrective action without manual human intervention.

c) FEED ENGINEERING shall be developed for all the recommendations arising out of the Select / and from various HSE and Technical Reviews.

d) ENGINEERING CONTRACTOR to note that all the recommendations made during the Select stage technical reviews/workshops are for action during FEED engineering stage of the project

e) Each engineering discipline shall be responsible for producing sufficient design detail commensurate with the level of engineering definition required for FEED Engineering.

f) ENGINEERING CONTRACTOR shall produce all drawings/documents identified under the list of deliverables section and those identified in this section as a minimum and submit to COMPANY for approval.

g) The facilities shall meet the requirements of latest COMPANY specifications, international Standards and Recommended Codes or Practice. ENGINEERING CONTRACTOR should notify COMPANY of any conflicts or discrepancies between standards and should seek COMPANY approval for precedence wherever applicable.

h) ENGINEERING CONTRACTOR shall ensure that the selected systems are adequately engineered and suitable for the intended services.

The deliverables listed for engineering scope in Attachment-1 and as outlined in the following sections (as a minimum) shall be developed by ENGINEERING CONTRACTOR. Any additional deliverables to fully define the ENGINEERING scope shall also be prepared by ENGINEERING CONTRACTOR as part of project scope development & Engineering package preparation scope.

- Drawings/document are not limited to certain quantity, it is ENGINEERING CONTRACTOR's responsibility to prepare required no. of P&ID's and other drawings/document for all the systems as required to define the design requirement clearly.

- ENGINEERING CONTRACTOR to provide all the deliverables including: various design studies, ENGINEER Design Packages, adequacy studies, demolition drawings, modification drawings, layouts) to meet requirements of Contract scope.

- ENGINEERING CONTRACTOR shall perform all the work to complete ENGINEERING scope and provide deliverables even not listed in SOW to meet Contract requirements.

- Furthermore, in view of the interface with existing operating facilities and ongoing project, due consideration shall be given and necessary close coordination / liaison with COMPANY site teams for the tie-ins & modification requirement in the existing systems at different locations / facilities. OEM shall be consulted for validation and approval by respective systems vendors at no extra cost to COMPANY.

- Adequacy Assessment shall be carried out for all utilities as required. ENGINEERING CONTRACTOR shall assess and confirm the utility loads for all new equipment at each location and ensure that the existing systems can meet the new demand without jeopardizing availability and continued safe operation of existing systems. If existing systems found to be inadequate in any way, then the ENGINEERING CONTRACTOR shall recommend modifications or alternatives at an early stage after contract award, and issue recommendations for review and approval by COMPANY and implement thereafter.

- ENGINEERING CONTRACTOR shall prepare modification specifications as required for the existing systems for the affected areas /facilities as well as for all the associated interfacing systems to accommodate new tie-ins.

- The modification specifications and update of existing drawings / document for each system/ facilities shall be separate for each location/facilities to clearly define the modification requirement.

- The services include FEED engineering of which all aspects of the Design shall be reviewed in stages before submission of the final & compiled Engineering Report for approval.

- ENGINEERING CONTRACTOR shall prepare all engineering documentation required as per the Scope of Services.

- Within two calendar (2) weeks of Effective Date, ENGINEERING CONTRACTOR shall prepare and submit for COMPANY's approval, a detailed project document register identifying and classifying (Approval, Review & Comment, Information) all documents to be provided by ENGINEERING CONTRACTOR.

- ENGINEERING CONTRACTOR 's own Engineering Procedures, optimized to suit the requirements of the Contract, shall be submitted to COMPANY for Approval.

- For the modifications to the facilities, in general, similar standards, plant specifications, philosophy, material, codes and practices to those used for the original design shall be used as applicable without compromising quality and performance of the facility, however ENGINEERING CONTRACTOR shall make recommendations for improvements.

- Documentation packages shall in general conform to the same numbering systems, symbols, tagging philosophy, etc. as defined by COMPANY, to allow integration of the new documents with existing manuals.

- ENGINEERING CONTRACTOR shall identify all existing drawings & documentation required to accomplish the scope of the Project. Upon ENGINEERING CONTRACTOR 's request COMPANY will make available electronic copy, either CADD file or scanned image depending on availability, of each existing drawing as required by ENGINEERING CONTRACTOR.

- It is the responsibility of ENGINEERING CONTRACTOR to review and verify correctness on site of applicable existing drawings including As-built drawings that are relevant to and impact on the design and modifications proposed and update where necessary to reflect the as-built condition in the drawings.

- If the existing As-built drawings could not be retrieved/traced, ENGINEERING CONTRACTOR shall prepare/generate necessary drawings as new. ENGINEERING CONTRACTOR shall physically verify the status of existing installation and update the As-Built drawings with mark-up as required for modifications. This shall also be applicable for demolition work (if any) required for the project.

- Existing Drawing Verifications by ENGINEERING CONTRACTOR: It is emphasized that the drawings provided with the RFP in pdf / tiff format may not give true correct AS BUILT status at site. Therefore, ENGINEERING CONTRACTOR shall site verify and revise / re-produce the relevant drawings conforming to the requirements stipulated in COMPANY Drawing Office practice. (D.O.P).

- Furthermore, in view of the interface with existing facilities, utilities, etc., due consideration should be given where interfacing the new facilities with the existing installation particularly in the areas of piping, electrical, and Instrument equipment.

- ENGINEERING CONTRACTOR shall ensure and verify that the interface / tie-ins with existing facilities and shall not jeopardize the performance of existing facilities.

- Contract scope does not define all the interfaces. identification of all interfaces shall be ENGINEERING CONTRACTOR's scope and responsibility.

ENGINEERING CONTRACTOR shall take note and act upon the following:

- The specifications, data and existing documents supplied by COMPANY are primarily intended to provide technical information. They do not necessarily reflect the proposed commercial/contractual relationships between COMPANY & ENGINEERING CONTRACTOR.
- It is ENGINEERING CONTRACTOR's responsibility pursuant to the Conditions of Contract to ensure that the specifications they prepare shall reflect those relationships properly.
- It is the responsibility of ENGINEERING CONTRACTOR to produce complete and acceptable studies, documentation, drawings, specifications and designs. If these are not acceptable to COMPANY, ENGINEER may receive "comment" from COMPANY. Such "comment' or "approval" in no way relieves ENGINEERING CONTRACTOR of their responsibility to produce complete acceptable studies, documentation and designs.
- Under no circumstances shall ENGINEERING CONTRACTOR interpret such "comment" as in any way an instruction relieving them of their obligations and Warranties under the Contract, unless it has been mutually agreed upon in specific instances.
- ENGINEERING CONTRACTOR shall be responsible for carrying out all corrective / supplementary or follow up Engineering design work as and when required for incorrect or incomplete design
- ENGINEERING CONTRACTOR 's Design work shall comply fully with the local law and statutory requirements, project specifications, COMPANY instructions and internationally recognized codes of practice.

9.3 Project Specifications

ENGINEERING CONTRACTOR shall be responsible for producing specifications for all materials, coatings, CP systems, packages, equipment and piping system components. Any other additional specifications shall also be produced, as necessary.

ENGINEERING CONTRACTOR shall provide a complete set of Project Specifications for all disciplines (including the COMPANY specifications which are recommended to use for the project without amendment) required for the PC phase of the project.

ENGINEERING CONTRACTOR shall also prepare Material Requisitions, Technical Specifications and Datasheets for each item to be procured and installed as part of PROJECT. All inspection requirements shall be clearly identified.

9.4 Long Lead Items (LLI)

Definition of Long Lead Items for the purposes of this PROJECT shall be all items / equipment whose delivery duration from placement of purchase order to receipt at site is more than 6 months including all valves of size 12" and above as well as all ESD valves. Any item falling under this criterion and all items whose delivery schedule is critical for achieving implementation schedule will be treated as Long Lead Items.

ENGINEERING CONTRACTOR shall develop detailed Material Requisition for Quotation (MRQs) for all Long Lead Items identified during ENGINEERING stage to allow placement of orders in the early stage. Material Requisitions are to be prepared in accordance with COMPANY Template /Format for MRQ, ZO-TS-Z-03010-MRF.

ENGINEERING CONTRACTOR shall facilitate COMPANY in issuing these Material Requisitions to COMPANY approved vendors. Accordingly, ENGINEERING CONTRACTOR shall prepare the bidders list based on COMPANY approved vendors list (standard bidders list).

Material Requisitions shall include Project Specifications, COMPANY Specifications & Addendums (as required), data sheets, drawings, battery limit information, tie-in details, requirements on delivery, specific VDRL, List of Manuals, Performance Test Criteria, Tests, Training, Spares and Procedures (as applicable). Material Requisitions shall be clear, accurate and complete.

9.5 Technical Bid Evaluation Of Long Lead Items

Minimum five technical bids analysis shall be performed to achieve minimum three commercially acceptable vendors for early award. Recent Lessons Learned in other Projects shall be implemented.

*Note: COMPANY to advise number of commercially acceptable bidders / vendors required during execution of ENGINEERING PACKAGE stage

-CONTRACTOR shall evaluate the formal technical offers received from Vendors, and shall submit the revised technical bid evaluation. Technical Bid Evaluation shall cover major aspects of design, driver selection, materials of construction, auxiliaries, acceptable technical deviations / exceptions etc. Multi-disciplinary review shall be conducted (including multi-disciplinary approval from COMPANY on sought deviations).

9.6 Site Survey and Data Collection

The services to be provided by ENGINEERING CONTRACTOR shall include, but not be limited to the following:

9.6.1 Data Collection

Data collection by ENGINEERING CONTRACTOR shall comprise as a minimum, but not be limited to, the following activities:

1) Verify and ensure availability of all requisite data to carry out the WORKS.

2) Verify availability of all COMPANY supplied drawings/ documents.

3) Verify data contained in the collected documents and clarify with COMPANY any inconsistencies identified.

4) Identify shortfalls, (if any) and collect/obtain the same from COMPANY through requisition/request as per COMPANY procedure(s).

5) Review and as-built verification of the applicable COMPANY drawings is the responsibility of ENGINEERING CONTRACTOR. COMPANY is not responsible for the accuracy of the drawings or other documents provided to ENGINEERING CONTRACTOR.

9.6.2 Site Survey

Site survey shall mean visits to survey existing facilities at US and Zakum processing facilities at the beginning of design and/or during detail design engineering.

ENGINEERING CONTRACTOR shall mobilize specialist teams of discipline engineers and/or designers to survey the existing facilities as necessary to complete design and engineering with the following objectives:

1) Familiarization with COMPANY's relevant existing facilities, their operation, codes of practice, systems of work and regulations.

2) Physically verify, obtain necessary measurements and become thoroughly familiar with sites.

3) If required, collect all the relevant data, documents and drawings which could not be sourced from COMPANY's Abu Dhabi offices.

4) Verify conditions and physical state of the relevant existing equipment and systems as applicable with emphasis on critical equipment and procedures that ensure a continuous and safe production of oil and gas. For this activity, COMPANY personnel will be available at their normal locations and during their normal office working hours for discussions with ENGINEERING CONTRACTOR's personnel.

5) Within one week of completion of the site survey, ENGINEERING CONTRACTOR shall issue a site survey report for COMPANY review / acceptance.

6) Site survey shall be conducted in accordance with COMPANY requirements, including but not limited to personnel offshore training, required authorization and security pass, quarantines.

ENGINEERING CONTRACTOR shall, at the start of engineering, undertake site visits/surveys as necessary to visually inspect existing facilities, to review the location of the new facilities/space requirements, tie-in locations, and hook-up with existing facilities.

ENGINEERING CONTRACTOR shall submit to COMPANY site visit/survey agenda to define the scope of survey by location, disciplines required, as well as the tentative manpower and schedule. ENGINEERING CONTRACTOR shall note that mobilization of its personnel for site surveys is subject to accommodation availability.

Detailed reports shall be provided within one week from the completion of site visit. Any problem areas noted as a result of site visit surveys and the mitigation solution shall be brought to the immediate attention of COMPANY. Further site visits shall be arranged as necessary during the course of the design before finalization of the drawings. The project space and other site requirements shall be identified and notified to COMPANY, for the purpose of securing the facility as "RESERVED". In addition, ENGINEERING CONTRACTOR shall confirm that the proposed requirements do not interfere with other ongoing or future projects.

9.7 Plant item Tag, Drawing /Document Numbering

- All drawings & documents numbering in DAS shall comply with ADNOC Offshore standard, A0-ENG-D-PRO-001 - Part B.

- Drawings and documents shall not be renumbered late in the project. All numbering/renumbering shall be done at the beginning of the project and reflected in the MDR accordingly.
- New Tags in the DAS facilities, shall follow the Numbering Procedure (Part B) - A0-ENG-D-PRO-001 and such request will be raised with COMPANY's ES- D&D
- All final submission from ENGINEERING CONTRACTOR to align & comply with the COMPANY's, Final Dossier Handover Procedure A0-ENG-D-PRO-003.
- In case of inconsistencies found in the data in terms of file naming, document numbering or the metadata of the register, ENGINEERING CONTRACTOR shall rectify and resubmit the final dossier to fulfill COMPANY requirements of EIMS register.
- ENGINEERING CONTRACTOR to provide list in the form of table for all drawing to be superseded and tags or plant items to be demolished for each discipline.
- Upon final handover of the project, all data will be loaded in COMPANY'S setup of EIMS. The soft copy of the final dossier shall be handed over along with filled EIMS register.
- In case of conflict between this paragraph and another one in the SOW, this part shall be considered as the correct one, and ENGINEERING CONTRACTOR shall highlight to COMPANY CED-Engineering Services Department for concurrence.

9.8 Execution Plan

ENGINEERING CONTRACTOR shall prepare a detailed plan for the ENGINEERING PACKAGE giving due consideration to the PROJECT requirements. The Plan shall also include, but not be limited to the following:

a) Scope of Work

b) Overall Objectives

c) Organization including roles and responsibilities

d) Staffing Plans

e) Milestone Schedule

f) Overall ENGINEERING PACKAGE Schedule and its Basis

g) ENGINEERING PACKAGE Implementation Schedules identifying program for all Discipline and area wise deliverables.

h) Management Information Systems.

i) Stakeholder Engagement Strategy and plan

j) Project Risk Register

k) Project Risk Management Plan

l) Contracting Strategy and Procurement Plan

m) Long Lead Items

n) Project Assurance Plan

o) Project Organization Chart

p) Risk and Mitigation Plan

q) Engineering MTO

The Execution Plan shall be submitted for COMPANY review and approval.

9.9 CONTRACTOR's Sub-contracts

Where ENGINEERING CONTRACTOR intends to subcontract any portion of the SERVICES, the provisions of the Articles of AGREEMENT shall apply.

The cost of any other subcontracted service is deemed included in estimated Lump-sum cost of PROJECT.

In the case of reimbursable subcontracts, COMPANY approval will be required for bidders list, scope of services, technical evaluation, commercial evaluation, and award.

For subcontracts included within the ENGINEERING CONTRACTOR's target Lump-sum cost COMPANY approval will still be required for bidders list, scope of services and technical evaluation as a minimum the following subcontractors have been identified for the following project phase:

a) UG and Geotechnical Survey

b) Safety Studies

9.10 Engineering Software

Below table shall be read in conjunction with COMPANY Engineering Requirements and is subject to approval prior to use for the PROJECT. ENGINEERING CONTRACTOR shall issue the full listing of applications / software in the IT Automation Plan that is intended to be utilized for the SERVICES.

Engineering software (latest versions) listed below shall be used. Native files (electronic) of all programs/simulations shall be provided to enable COMPANY to run additional cases, when required.

ENGINEERING CONTRACTOR'S in-house software or spread sheets cannot be used for the design calculations unless they were validated and submitted to COMPANY for approval, prior to implementation.

COMPANY shall review and agree the software and versions submitted by ENGINEERING CONTRACTOR in IT Automation Plan. ENGINEERING CONTRACTOR shall investigate fully if any additional software must be employed during PROJECT and the same shall be included and provided accordingly.

PROJECT FUNCTION	SOFTWARE APPLICATION
GENERAL DOCUMENTATION	Microsoft Office 2010 Suite
HSE	
PHA, HAZOP and PHSER	PHA Pro
SIL	SIFPro , exSILentia or equivelant
Bow-Tie	Bow-Tie XP / THESIS software
Firewater hydraulics (steady state and transient)	PIPENET

PROJECT FUNCTION	SOFTWARE APPLICATION
FERA /QRA / HC and H2S Dispersion	DNV PHAST / SAFETI (Last Edition) / FLACS CFD
F&G mapping	Detect3D or equivalent
Flare radiation study	Flaresim
PROJECT CONTROLS	
Scheduling	Primavera P6 8.2 or later version
Cost Estimation	Aspen Capital Cost Estimator (ACCE), @Risk Software
PIPING	
2D CAD	Auto Cad /GTX Raster
Pipe Stress Analysis	CAESAR II (Latest Version)
3D Model	SmartPlant 3D
3D model review	Navisworks
INSTRUMENTATION & CONTROL SYSTEMS (1)	
Sizing Calculations	INSTRUCALC
Instrument Database & Deliverables	SmartPlant Instrumentation
SIL Assessment	exSILentia (1)
PROCESS	
Steady State Hydraulic Analysis	PIPESIM or TGNet/ TLNet or Synergi Pipeline Simulator software (SPS)
Flare and vent system design software	FLARENET
Heat exchangers optimization	HTRI (Heat Transfer Research, Inc.)
Process Simulation System / Physical Properties (all process systems)	HYSYS / ProTreat
Column Hydraulic Rating Software	K.G. TOWER or ENGINEER default approved by COMPANY
Surge and separator vessel sizing / analysis/ line sizing	Approved Excel Spread Sheets

PROJECT FUNCTION	SOFTWARE APPLICATION
PFDs, UFDs, P&IDs	AutoCAD/GTX Raster, SmartPlant P&ID
SURVEY	
Alignment Sheets and Chart Generation	Auto CAD
Structural	
Structural Analysis	SACS
For all spread sheets & calculations	Microsoft Excel or Equivalent
For 2D Drawings	AUTOCAD
CSA	
Steel and concrete structures analysis and design	STAADPro (Bentley)
Isolated, combined and raft foundation design	MAT3D (Dimension Solutions Inc.)
2D CAD	AutoCAD
3D Model	SP3D SMART PLANT
CORROSION	
Electronic Corrosion Engineering	ECE5.5
ELECTRICAL	
Power System Study	ETAP 20 or higher (latest version to be used)
Lighting Calculation	Diallux
Electrical Database	SmartPlant Electrical
MECHANICAL	
Pressure Vessel Design and Analysis	PV Elite or equivalent (COMPANY approved)
Finite Element Analysis	ANSYS or equivalent (COMPANY approved)
Tank Design	AME Tank
TELECOM	
PAGA Coverage software	CADNAA or Equivalent

PROJECT FUNCTION	SOFTWARE APPLICATION
CCTV Coverage software	JVSG or Equivalent

9.11 Engineering Design Applications

SMART Engineering Design Systems

Smart Plant Engineering Design suites from Hexagon (Smart P&ID, Smart Instrumentation, Smart Electrical and Smart 3D) are applicable and shall be implemented for the project scope (Greenfield & Brownfield with DAS Island)

Smart P&ID

CONTRACTOR shall utilize Hexagon's Smart P&ID during this project to develop the P&ID database and generate the applicable project deliverables specified in this scope of work in compliance with "Specification for implementation of SmartPlant P&ID, A0-ENG-I-SP-002 Rev.0" and DOP requirements.

The Piping and Instrument Diagrams shall be prepared and maintained in Smart PID (SPPID) in accordance with the Project's standard symbology and formats; this shall include the Vendor P&ID as well the applicable software versions for SmartPlant P&ID will be provided by COMPANY during the start of the Project execution.

Reference data Change management shall be maintained by seeking COMPANY's Approval. Any exceptions and deviations incurred during execution shall be documented in a procedure.

Reference SEED Database for Smart PID which will include Symbols; Labels; Data dictionary templates; report formats to be used for the preparation of SmartPlant P&ID database will be provided by COMPANY. Further Customization required to meet the project requirements shall be done by the CONTRACTOR with prior approval from COMPANY.

Smart P&ID Project setup procedure which shall cover Oracle database server, Workstation hardware requirements, Network philosophy, Oracle database setup details (server and clients), Reference Data implementation plan, Plant-Area-Unit Hierarchy (PAU) structure, Procedure to resolve the inconsistencies., Backup & Disaster recovery procedure, Data maintenance procedure, Data quality check procedure and Database handover procedure.

Border templates and title blocks for the P&IDs shall be customized to match COMPANY border standards All applicable data for Pipe run, Equipment and Instrument shall be filled for the extraction of Line List and Equipment List.All lines shall have flow directions clearly shown.

Equipment description, Tag. No. and design parameters shall appear either near the top or near the bottom of the sheet directly above or below Equipment symbol
CONTRACTOR is required to ensure that P&ID instrument tags have association with the appropriate instrument loops.

The Layout of P&IDs shall allow for future additions and shall minimize crossing lines. Vertical lines shall be broken when it crosses with Horizontal line. Secondary Lines shall be broken when crossing Primary line. Utility lines shall be broken when crossing Process lines. Instrument lines shall be broken when crossing flow lines.

Piping Specification changes shall be clearly shown. Insulation on Piping and Equipment shall be shown. Heat tracing requirements shall also be indicated.Supply Responsibility shall be clearly shown to indicate

vendor scope.The Reports for Line List Equipment List and Instrument List shall be extracted from SP P&ID as per Company provided template.

CONTRACTOR shall upgrade SPPID data to Company approved version prior to final handover
CONTRACTOR shall be responsible for eliminating any inconsistencies in the upgrade process to the SPPID version approved by the Company.
CONTRACTOR shall take responsibility, to validate the data, after upgrade, to the SPPID version approved by the Company. Any inconsistency noticed during the upgrade shall be documented and provided to Company.

CONTRACTOR shall use the Smart P&ID “Re-Create P&ID from Database” functionality for each drawing before final handover to COMPANY.

CONTRACTOR shall ensure that all the Tags are consistent across the Smart Tools. CONTRACTOR shall extract reports to ensure the consistency between the Smart Tools’ databases.
CONTRACTOR shall ensure that all deliverables submitted to COMPANY can be extracted from the restored backups

Prior to data handover CONTRACTOR shall ensure that SPPID drawings and data are not corrupt and do not contain inconsistencies. All SPPID inconsistencies shall be resolved by the CONTRACTOR prior to handover to COMPANY.

CONTRACTOR shall submit a database consistency and health check to the SPPID version approved by the COMPANY. Any inconsistency noticed during the upgrade shall be documented and provided to COMPANY SMART P&ID DELIVERABLES

The final deliverables shall be in full compliance as specified in Company’s Specification “Specification for implementation of Smart P&ID, A0-ENG-I- SP-002 Rev.0 and COMPANY CADD Manual and Numbering Procedure, Plant Backup file, Item TagFormat.xml

Screen shot showing Folder structure used Customized DLL files along with source code and documentation Readme file explaining contents/description of files, version of Smart P&ID, Smart Engineering Manager, Version of Oracle, Oracle Instance Name, Oracle Character Set, Oracle Table space name. Final SPPID Implementation plan.Final SPPID Customization Document. Final SPPID Deviations and Exceptions Document.

Smart Electrical
Hexagon’s Smart Electrical (SEL) shall be implemented for the generation of the database and issue of Electrical deliverables.

The software versions for Smart Electrical will be provided by COMPANY during the execution of the project. However, CONTRACTOR shall comply with the requirement as specified in the Company document “Specification for implementation of Smart Electrical, A0-ENG-I-SP-005” as a minimum. SEL version mentioned in “A0-ENG-I-SP-005” is not applicable for this project.
COMPANY will provide the required Version during the time of project award

The Reference data Change management shall be maintained by seeking COMPANY's Approval. Reference data includes all items in SEL Reference data folder including Data dictionary, Rules, Options Manager

Settings, Formats, Filters and Insulation Manager Settings. All changes (additions, modifications and deletions) of SEL Reference Data shall be managed and tracked by documenting it in a procedure
Smart Electrical Project setup procedure which shall cover Oracle database server, Hardware requirements for Workstations, Network philosophy, Oracle database setup details, Reference Data implementation plan, Plant-Area-Unit Hierarchy (PAU) structure, Procedure to resolve the inconsistencies. Backup & Disaster recovery procedure, Data maintenance procedure, Data quality check procedure and Database handover procedure.

CONTRACTOR shall ensure that all the Tags are consistent across the Smart Tools. CONTRACTOR shall extract reports to ensure consistency between Smart Tools' databases.

he following List of Documents shall be produced using Smart Electrical include, but are not limited to:

- Single Line Diagrams (SLD)
- Electrical Equipment List
- Cable Schedule
- Load List
- Electrical Equipment Data Sheets
- Smart Electrical Deliverables

CONTRACTOR shall ensure that SEL data does not have any data inconsistencies before final delivery. CONTRACTOR shall adhere to the following guidelines for final SPEL deliverables:

- Plant Backup file
- Screen shot showing Folder structure used Customized DLL files along with source code and documentation
- Readme file explaining contents/description of files, version of Smart Electrical, Smart Engineering Manager,
- Version of Oracle, Oracle Instance Name, Oracle Character Set, Oracle Table space name
- Database, file system and all other requirements as stated in document no. A0-ENG-I-SP-005 and as per
- EPC deliverable list.

All drawings and reports within or extracted, using SEL, shall be updated with as built information.
CONTRACTOR shall ensure that all deliverables submitted to Company can be extracted from the restored backups.

Contractor shall upgrade SPEL data to Company approved version.CONTRACTOR shall be responsible for eliminating any inconsistencies in the upgrade process to the SEL version approved by the Company.

CONTRACTOR shall submit a database consistency and health check report before and after the upgrade. CONTRACTOR shall take responsibility, to validate the data, after upgrade, to the SPEL version approved by the Company. Any inconsistency noticed during the upgrade shall be documented and provided to Company.

CONTRACTOR shall facilitate the Company in uploading the data set to the Company's site.

CONTRACTOR shall resolve any issues, which may arise when Company is evaluating the data, until the

'Provisional Final Acceptance Certification' is obtained.

Contractor shall provide all documents concerned with SPEL, in hardcopies, PDF and native editable electronic formats (MS-Word, MS-Excel, etc).

The final deliverables shall be in full compliance as specified in Company's Specification "Specification for implementation of Smart Electrical, A0-ENG-I-SP-005" Rev.0 and COMPANY CADD Manual and Numbering Procedure

Smart 3D

CONTRACTOR shall utilize Hexagon's Smart 3D software during this project for the creation and management of the 3D model and generate the applicable project deliverables specified in this scope of work in compliance with Specification for implementation of Smart 3D, A0-ENG-I-SP-004 Rev.0 and COMPANY CADD Manual (A0-ENG-D-PRO-002) Numbering Procedure (A0- ENG-D-PRO-001)

he main objectives and requirements related to the 3D Model are outlined below:

The applicable software versions for Smart 3D will be provided by COMPANY during the start of the Project execution. S3D version stated in document" A0-ENG-I-SP-004 Rev.0" is not applicable.

The 3D modelling system shall be able to produce required documents such as Material take-offs, Plot Plan, Elevation and Sectional drawings, and Reports. CONTRACTOR is required to further customize SP3D software to meet the AGREEMENT requirements for 3D Model, drawings and reports etc.

The 3D modeling system shall be able to produce required documents such as Material take-offs, Plot Plan, Elevation and Sectional drawings, Isometric drawings and Reports

CONTRACTOR shall submit Progress Measurement Procedure for 3D Modeling assigning weightages to different activities discipline-wise

The development of the 3D models shall be ensured to have as a minimum all requirements and aspects of Safety, Operations, Maintenance, Constructability, Accessibility and Human Factor Engineering

The models shall be used for the various design, safety and constructability reviews during the engineering and construction phase of the project

CONTRACTOR shall perform regular clash checks and report on the resolutions of the clashes. The frequency of these clash checks shall be agreed with COMPANY

All equipment and vendor packages shall be based on input from selected VENDORS.

CONTRACTOR shall ensure that all the Tags are consistent across the Smart EDS. CONTRACTOR shall extract reports to ensure the consistency between the databases.

CONTRACTOR shall ensure access to equipment, instruments and valves etc, as per PROJECT Specifications.

CONTRACTOR shall ensure that Equipment /Piping arrangements/routings are consistent with Safety/Process requirements

provide input into constructability.

Establish and validate the mechanical handling system.
To reduce the rework during construction to an absolute minimum by using interference checking and data integrity.

During the project phase, CONTRACTOR shall provide access to S3D database to COMPANY personnel via Citrix or direct network to ensure the availability and quality of the data. CONTRACTOR shall submit the complete database backup to COMPANY on a monthly basis for review. For the above-mentioned data transfer to COMPANY, CONTRACTOR shall host a FTP site to transfer the database backup and access shall be provided to COMPANY personnel.

CONTRACTOR shall extract the following drawings from SP3D database as a minimum.

Discipline	Drawings
Electrical	Cable Tray Layouts
Electrical	Electrical Power Layouts
Electrical	Lighting Layouts
Instrument	F&G Detector Layout
Instrument	Cable Routing Plan
Instrument	Instrument Location Layout
Telecom	Cable Tray Layouts
Telecom	Equipment Layouts
Telecom	Equipment Plot Plan
Telecom	UG Trench Plot Plan
Structure	Pile Layouts
Structure	Foundation Layouts
Structure	Platform Details - Horizontal & Vertical Vessels
Structure	Pipe Support Marking Plan
Structure	Framing Plan with Support Details
Structure	Structural Framing Plan, Elevation & Sections

Structure	Structural Grating Plans
Structure	Structural Module Shipping Drawing
Piping	Plot Plan
Piping	Area Plot Plan
Piping	Piping General Arrangement Drawing
Piping	Piping Isometrics
Piping	Equipment Nozzle Orientation Drawings

Discipline Specific Requirements

CONTRACTOR shall coordinate with COMPANY to finalize extent of Vendor(s) data to be incorporated for equipment, piping, civil, electrical, and instrumentation that can be transferred into relevant Smart EDS as per COMPANY requirements. CONTRACTOR shall work in close coordination with package Vendor(s) so that all the prerequisites for Smart EDS relevant to the Vendor(s) packages are migrated successfully. The EPC Contractor shall be responsible and accountable for merging/migrating the Vendor packages into the Consolidated Master databases

The following disciplines items as a minimum shall be incorporated in the model:

Piping

- 3D Review Model of the existing facility will be made available to the successful bidder for TIE-IN reference, clash check and demolition purpose if applicable
- All process and utility piping including all piping components as per P&IDs.
- All escape routes (Primary and secondary) and maintenance ways.
- All vendor packages with piping inside package scope.
- All piping routes with drains/vents, all in-line field instrument, control valves and on-off valves & electrical attachments.
- All pipe supports (primary and secondary) shall be modelled as Physical support using Pipe Support assembly including volumes for special supports (Spring Hangers etc).
- All Fire Water systems including deluge systems with deluge valves, nozzles, etc.
- All process connections for instrument on piping and equipment as per P&IDs including Level Gauges etc.
- Valves including hand wheels, actuators etc. The modeling of the sizes shall be based on actual vendor data.
- Piping accessories such as strainers and all other piping specialty items.
- All safety equipment including safety showers and eye wash stations
- All Underground piping systems.
- Piping special supports including spring hanger supports, tie rods etc
- Expansion bellows subject to COMPANY approval
- Insulation outline considering actual insulation thicknesses.

- All major identification attributes shall be captured
- All operational access / maintenance handling ways for valves
- All primary and secondary safety access ways
- Comply with P&IDs to ensure system extraction of isometric for issue without any further manual update.
- CONTRACTOR shall model all lines of all sizes based on CONTRACTOR's P&IDs by suitably updating Smart 3D database.
- All future equipment with main future piping shall be modeled for space reservation and these shall be identified separately in 3D piping models.
- All isometrics and 2D General arrangement drawings shall be extracted and maintained including Issued for Design and As-Built within S3D

Machinery & Packaged Equipment

Equipment models shall show a reasonable representation of the equipment for presentation during design review, drawing extraction, interference detection, nozzle orientation drawings and reports, inter-discipline reference. Minimum contents shall be as follows:

- Parametric equipment (catalog equipment) shall be used whenever possible.
- Minimum properties to be keyed in for Equipment are Equipment Tag / Equipment Name, a description & insulation thickness. All these values should match the P&ID.
- Provide details for the properties of equipment (like material of construction, weight, pressure, diam., length T/L to T/\L).
- All Equipment, new, future and existing as applicable & above grade portion of foundations shall be modelled.
- Platforms and ladders attached to equipment and supplied by vendor.
- All nozzles.
- Equipment withdrawals drop out zones, man way, manholes and hatch access.
- Buildings and doors.
- Safety, operation, maintenance and access ways (including corrosion probe withdrawal areas), main escape routes and vehicle access.
- Miscellaneous steel and platforms.
- Major electrical equipment, control panels and junction boxes.
- Foundations supporting equipment from grade.
- Skid mounted equipment shall be modeled at the same level of detail as non- skid mounted equipment and with sufficient detail to enable access and maintenance clearances, interface connections, lifting and handling requirement.
- Fire Fighting fixed equipment (Hydrants, Monitors), Fire-fighting mobile equipment (Extinguishers), Hose Reel, Hose cabinet and safety shower/Eye wash etc.
- Ducts, platforms, support columns, and stacks.
- Davit, Deck crane, Monorail, EOT crane etc.
- The minimum required nozzle attributes are tag number, size, pressure rating, end prep and Schedule thickness where applicable.
- Ducts, platforms, support columns and stacks.
- All Packaged VENDOR items.

- Equipment envelopes for Maintenance / Access / Handling to be shown.

Civil / Structure

- Drainage and sewerage items including ditches routing outline and levels, manholes, catch basins, valve pits, joint boxes and pits.
- Steel items including beams, columns, bracings, fireproofing, all floors levels plating/grating, stairs, platforms, ladders, hoisting beams and handrails.
- Buildings including Outline and shape.
- Dyke wall Deflector wall, retaining wall etc.
- Steel Structures, pipe supports, crossovers, platforms, stair cases, etc.,
- Cutbacks for the Structural steel (Columns/Beams/Bracings) shall be made.
- The Weights and COG for the Structural items should be populated in the 3D Model.

Instrumentation

- Field instrument stands.
- Instrument Junction boxes and local control panels.
- Analyzers and Analyzer houses.
- Local Emergency Shutdown Switches and Hand Switches.
- Telecommunication Equipment both indoors and outdoors
- Air Volume Tanks/Bottles for Valves.
- Fire & Gas Devices like Detectors, Beacons, Sounders and Manual Alarm Call Points etc.
- Instrument Cable Tray/Ladders.
- All major identification attributes will be captured.
- Local Control Rooms.
- All electrical and instrument junction boxes.
- Local control panels, cabinets and skid mounted instruments.

Electrical

- Lighting fixtures and poles, junction boxes, cable routes, local control station, local panel, lighting and small power panel, welding socket outlets.
- Cable trays & ladders of sizes above 100mm width including supports.
- Sub-stations and Module Technical Rooms including equipment.
- Bus ducts.
- Outdoor transformers and Neutral Earthing Resistor (NER).
- Diesel Generator sets.
- Gas turbine power generation units
- Navigational aid systems (including solar power system).
- Inputs to Civil for electrical U/G facilities.

Fixed Fire Protection/Fire-fighting Systems

- Spray systems.
- Fire ring main routing, hydrants, block valves, fixed water monitors and all other monitors as per P&ID with necessary hose reels.
- Deluge valve manifolds and piping system.

- Firefighting, Safety and life-saving equipment like Extinguishers, Safety shower and Eyewash, Breathing Apparatus etc.

Data Checks & Cleanup

CONTRACTOR shall ensure that the delivered data is not corrupt and has been checked thoroughly for consistency before delivery. As a minimum, the following checks and cleanup shall be performed:

- All Temporary Hierarchy, drawings and views shall be deleted.
- Database Integrity Check shall be run on all databases and inconsistencies resolved.
- Database shall be purged of all deleted items; all flavors shall be deleted, and all the indexes shall be rebuilt.
- All To do list shall be cleared.
- 3D model shall be synchronized with Catalog
- Piping isometrics must be extracted successfully for all lines. Lines that fail to extract shall be corrected prior to delivery
- All drawings shall be extracted without error.
- All open actions shall be cleared

3D Model Review

CONTRACTOR shall use Intergraph's Smart Review software for interactively reviewing the S3D Model. 3D model review shall address as a minimum, accessibility, operability, and maintainability and constructability aspects of the PLANT.

CONTRACTOR shall conduct 3D model review workshops at 30%, 60% and 90% completion of the modeling which will be attended by COMPANY Representatives.

These review sessions shall be carried out in accordance with the procedure to be developed by CONTRACTOR and approved by COMPANY. CONTRACTOR shall prepare a comprehensive document clearly stating the scope of the 3D model reviews for approval by COMPANY.

Two weeks prior to the workshop, CONTRACTOR shall submit to COMPANY, hard copy of P&ID's and Island Plot Plans. During the review workshops sufficient copies showing the current and latest status of these drawings shall be available for hand-out to the participants as guidance in the workshop.

During the model design development CONTRACTOR shall conduct weekly coordination meetings which shall be attended by all concerned (including COMPANY), who contribute to the model development, to ensure coordinated development of the model.

CONTRACTOR shall provide the review files of the 3D model under development to COMPANY. Frequency to be agreed with COMPANY during initiation of FEED.

CONTRACTOR shall identify the tie-in, demolition portion in the 3D model with distinct color codes for all disciplines.

TIE-INS information must be reviewed in a daily basis in a separate model review to comply and create a complete tie-in schedule report.

CONTRACTOR shall modify the design based on the COMPANY'S model review comments at no cost to the COMPANY where the design is not in compliance with the applicable specifications, standards, codes, or regulations. However, this model review does not relieve CONTRACTOR responsibility from performing in accordance with the provisions of the CONTRACT.

Final Deliverables:

The final deliverables shall be in full compliance as specified in Company's Specification "Specification for implementation of Smart 3D, A0-ENG-I-SP-004 Rev.0" and COMPANY CADD Manual & Numbering Procedure

Smart Instrumentation

- CONTRACTOR shall utilize Intergraph's SmartPlant Instrumentation (SPI - Previously known as INtools) to initialize and develop the instrumentation database and generate the applicable project deliverables specified in this document in compliance with Company Specification "Specification for implementation of SmartPlant Instrumentation, A0-ENG-I-SP-003". CONTRACTOR shall use the Company provided SEED data as the basis and initiate a new SPI database for all EPC data including all vendor packages.
- COMPANY will provide software versions for SmartPlant Instrumentation during the execution of the project. However, CONTRACTOR shall comply with the requirement specified in the Company document "Specification for implementation of SPI, A0-ENG-I-SP-003" as a minimum during initialization of the project database.
- COMPANY will provide the formats (.PSR files) for Instrument index, IO list, Cable schedule, Junction box schedule, Instrument datasheets format (.PSR files) listed in Company Specification for generation of related project deliverables.
- In addition to develop the new instrument datasheet formats, layouts, symbols, templates and other required formats for the Instruments, analysers, F&G devices etc. in SPI (Using Info maker/SPI), it is the CONTRACTOR's responsibility to modify and finalize the all existing formats to meet project requirement. CONTRACTOR shall submit these formats as a part of "SmartPlant Instrumentation Implementation Plan" for COMPANY approval prior to developing the SPI project database. Also, all the fields mentioned in these format shall be populated completely by CONTRACTOR to define the project requirement.
- Instrument index shall include all primary and implied tags but not limited to wired instruments, local instruments, system generated tags, Interface(S) tags, alarms tags. Separate hardware and software I/O list shall be prepared for each system.
- CONTRACTOR shall develop data sheets for valves, process instruments, analysers, safety system instruments, fire and gas instruments and all other instruments as part of SPI database based on approved P&ID's/layouts and Instruments process data. Process datasheets for all instruments shall be populated in SPI. Multi-tag instrument datasheets are not acceptable.
- For instruments wherein the sizing calculation is required, CONTRACTOR shall populate the required data in SPI and generate the calculation report. These calculations shall be used for reference only.
- The loop generation shall be done using Enhanced report utility and CONTRACTOR shall generate the wiring and control system details completely in SPI without using external cad or macro-

embedded symbols/blocks. Control system information in loop drawing shall not be generated using User defined field (UDF) or user defined table (UDT).

- CONTRACTOR shall produce material take-offs for all items including bulk items. The bulk items shall include tubes, tube fittings, cables, trays, cable glands, junction boxes, installation items etc. The hook up templates shall be developed using SmartSketch. These templates shall have information related to tag numbers, hook-up item codes, description, material, quantity, etc. as per project requirement. CONTRACTOR shall follow Company item codes for all the hook up/bulk items. Company provided hook-up drawings should be used for reference only and not as TYPICAL or TEMPLATE for this project. The final hook-up deliverables shall be generated from SPI with associated tag list on each Template/Typical.
- For cable and JB schedules, these deliverables shall be generated in COMPANY approved PSR report format after data population in SPI.
- CONTRACTOR shall obtain all the required engineering data from SUBCONTRACTOR and SUPPLIER suitable for use with SPI. In case, SUBCONTACTORS or SUPPLIERS are not able to produce the required data in SPI, the CONTRACTOR will then have the responsibility to produce this data in SPI. It is CONTRACTOR's responsibility that all engineering deliverables including any Package Vendor/Subcontractor or Vendor to be generated from a single SPI database in accordance with SPI implementation plan.
- CONTRACTOR shall ensure that all the tags are consistent across the other SmartPlant Tools implemented in this project. CONTRACTOR shall extract reports to ensure consistency between these SmartPlant databases.

<u>Following deliverables shall be developed using SPI including, but are not limited to:</u>

- Instrument index
- Input / Output lists
- Instrument data sheets
- Instrument sizing calculations
- Cable schedules
- JB schedules
- Loop diagrams – Enhanced smart loop
- Installation / hook-up drawings
- Bill of Materials / MTO's
- Interconnection wiring diagrams
- Alarm and set point summary
- The above deliverables shall be generated separately for each system and for each unit .i.e. process control system, Safety systems, Fire & Gas system, Package systems, etc. these shall not be limited to wired instruments only but shall include all primary and implied tags, system generated tags, interfaces tags, alarms tags, etc.
- CONTRACTOR's SPI Administrator shall be responsible for the tasks related to administration, configuration, contractual requirements and coordination with various engineering disciplines. The SPI administrator shall also be responsible for liaising with COMPANY Application Specialist from ES-EIM division for all SPI related tasks.

- During the project phase, CONTRACTOR shall provide access to SPI database to Company personnel as per the requirement specified in the document A0-ENG-I-SP-003" to ensure the availability and quality of the data. CONTRACTOR shall also provide the complete SPI database backups at regular intervals to Company.
- CONTRACTOR shall meet the requirements for the database backup, quality audit, deliverables and project database final delivery as specified in the document "Specification for implementation of SmartPlant Instrumentation, A0-ENG-I-SP-003". Before the final handover, CONTRACTOR shall perform SPI Database integrity and data quality check audit in compliance with Company Specification. The final audit shall be staged before handover of the database. COMPANY will witness the entire session of these audits.
- After incorporation of audit findings, the SPI database shall be upgraded to the higher version, as specified by COMPANY later. The final version of the database will be notified to the CONTRACTOR before the final handover. The final as-built deliverables shall be generated from the upgraded database.
- Prior to data handover CONTRACTOR shall ensure that the data is error free and do not contain inconsistencies. CONTRACTOR shall resolve all database inconsistencies prior to handover to COMPANY.
- CONTRACTOR shall be responsible for facilitating the Company in uploading the data set at Company's site. CONTRACTOR shall resolve any issues which may arise when Company is evaluating the data, until the `Final Acceptance Certification' is obtained.
- Apart from the PROJECT deliverables, CONTRACTOR shall submit the following documents to meet the requirement specified in the document "Specification for implementation of SPI, A0-ENG-I-SP-003" as a minimum:
    - SmartPlant Instrumentation Implementation Plan
    - SmartPlant Instrumentation – Customization, Exceptions and Deviations Document
    - An oracle dump and ddti backup

9.12 GIS

Topography and Geotechnical investigations is required in areas of interest. The Survey data for existing disposal system is available with COMPANY ES-GIS Team for reference which will be made available to successful Bidders after the PROJECT Award for reference.

ENGINEERING CONTRACTOR shall review the available survey data and prepare the detailed report. ENGINEERING CONTRACTOR shall also identify and mark the areas where Topographic, UGD and Geotechnical information is considered insufficient and prepare the scope of work to appoint subcontractor and perform the required work as part the project scope.

Survey Works Preparation of Geotechnical, Geophysical, Topo and Under Ground (UG) detection survey Scope of Work (SOW), obtaining COMPANY approval, floating tender and managing subcontract, reviewing survey reports and implementation of recommendations into the define stage FEED Engineering works are in the scope of ENGINEERING CONTRACTOR's Geo-tech investigation, Topographic and UG detection survey.

a) Topographic Survey / UG detection survey

Requirement of topographic survey / UG detection survey is proposed. The SOW shall be prepared in line with AGPS-SP-CS-011 - Onshore Topographic and Underground Detection Survey.

ENGINEERING CONTRACTOR shall take into consideration of existing facilities and services by referring necessary drawings, and by performing required number of site visits and by having discussions with all the stakeholders of the existing facility. Accordingly detailed scope of work for Topographic survey / UG detection survey and drawing showing survey boundary shall be prepared as part of ENGINEERING CONTRACTOR's SCOPE OF SERVICES.

b) Geotechnical investigation

Requirement of Geotechnical survey is proposed. The SOW shall be prepared in line with AGPS-SP-CS-010 - Onshore Geotechnical Investigation.

Geo-tech SOW for survey to be carried out during FEED shall cover number and depth of boreholes and other necessary soil tests to define the soil parameters of required depth for the determination of settlement by PLAXIS-2D or 3D and other applicable conventional formulas and also to assess the formation of ground strata, any soft layers of soil, anomalies or cavity existing below proposed tanks in scope. SOW shall be developed by ENGINEERING CONTRACTOR before awarding the Geo-tech investigation subcontract. Change in location of bore holes due to any unforeseen / underground interferences and additional trial pits, as needed, shall be part of lump sum scope of the Geo-tech SUBCONTRACTOR.

9.13 Reviews & Studies

9.13.1 General

- ENGINEERING CONTRACTOR shall plan for all reviews and studies to verify design, safety, health and environmental protection, reliability, operability, maintainability and constructability aspects, COMPANY and/or designated independent third parties where applicable will participate in these Reviews. These shall include but not be limited to the following at FEED engineering phase:

SR. NO	DESCRIPTION	FEED
1	HAZOP	√
2	SIL	√
3	PHA REVIEW	√
4	HSE Layout review	√
5	CONSTRUCTABILITY REVIEW	√
6	PHSER 2	√
7	SIMOPS / MOPO Workshop	√
8	QRA	√
9	Design Review	√
10	ISDR	√

SR. NO	DESCRIPTION	FEED
11	HSEIA Screening	√
12	COMAH	√
13	Fire and Explosion Risk Assessment (FERA)	√
14	Escape Evacuation and Rescue Assessment (EERA)	√
15	Emergency Systems Survivability Analysis (ESSA)	√
16	H2S Zoning Study	√
17	Bowtie development & HSECES Identification (System Level)	√
18	Dropped Object Study	√
19	Environmental Baseline Study	√
20	Noise Study (Environment)	√
21	Noise Study & Contours (OH)	√
22	3D F&G Mapping Study	√
23	Power System Study	√
24	Material Selection Study	√
25	Corrosion Risk Assessment	√
26	Project Risk Assessment	√
27	Vent/Flare Dispersion(H2S,Flammable Gas & So2) study	√
28	Value Engineering Workshop	√
29	SHUTDOWN optimization workshops	√
30	CFD Study Report	√
31	Ergonomics and Human Factors Engineering	√
32	Fire Protection Adequacy study	√
33	Social impact Assessment Screening & SBS (Social baseline study)	√
34	G Green House Gas (GHG) Assessment	√

SR. NO	DESCRIPTION	FEED
35	Reliability Availability and Maintenance (RAM) Study Report	√
36	Plot Plan Review Workshop and report	√

- Design Review(s) shall be carried out at the ENGINEERING CONTRACTOR Home Office or COMPANY offices as required and approved by COMPANY.

- ENGINEERING CONTRACTOR shall make available for review by COMPANY or (by designated independent third parties) all Engineering documentation at, or before all stages of the Schedule, together with all such documentation that may be relevant.

- In addition to Project Reviews referred to above, COMPANY may perform or have performed quality audits of the Work as and when it deems necessary.

- ENGINEERING CONTRACTOR shall always keep available all records and aspects of the work and at all places where the work is being carried out for detailed auditing by COMPANY or any third party appointed by COMPANY for that purpose. ENGINEERING CONTRACTOR shall maintain, available for reference and COMPANY's use at all places where the work is being carried out, the latest issue of all applicable codes, norms, standards, drawings and specifications.

- Not less than FOUR (4) weeks' notice of the Project Reviews shall be given to the parties before commencing a meeting and a formal agenda agreed the subjects to be discussed.

- Soft copies of all analysis performed shall be made available to COMPANY during reviews when requested.

9.13.2 HAZID / ENVID and OHID (PHA)

- ENGINEERING CONTRACTOR shall conduct HAZID / ENVID and OHID review session to address the following points as minimum:
    a. Identify all possible HAZARDS
    b. Assess the risk based on probability and frequency of events based on COMPANY (6X6) risk matrix
- ENGINEERING CONTRACTOR shall prepare and issue a report detailing the findings and close out recommendations/actions of the PHA exercise for COMPANY review and approval.

9.13.3 Constructability Review

Constructability Review shall be carried out by ENGINEERING CONTRACTOR to identify the key construction issues and considerations on how sizes, layouts, weight, sequence and equipment delivery will affect the execution of the PROJECT. ENGINEERING CONTRACTOR shall develop the Construction and Installation Study report as per the design. This report shall be a pre-requisite to conduct the Constructability Review workshop.

ENGINEERING CONTRACTOR shall make optimum use of construction knowledge and experience in the planning and design of the new facilities as well as to establish the needs and requirements for plant, safety, maintenance, operating and start-up, in order to achieve the overall PROJECT objectives.

ENGINEERING CONTRACTOR shall involve experienced third-party or in-house expertise, with expertise in the type of facilities in the PROJECT for the Constructability Review Workshop.

ENGINEERING CONTRACTOR shall carry out a Preliminary Construction Risk Assessment in order to review the design induced construction risks and to choose the proper risk alternative. Accessibility w.r.t installation, lifting, cutting, welding, piling, blasting and surface treatment and selection of materials / chemicals shall be considered. COMPANY "HSE Guide" shall be followed for the Preliminary Construction Risk Assessment.

ENGINEERING CONTRACTOR shall also carry out a Preliminary Demolition Risk Assessment in order to review the design induced demolition risks and to choose the proper risk alternative. This risk assessment may be combined with the Preliminary Construction Risk Assessment.

ENGINEERING CONTRACTOR shall also carry out a Preliminary Operations & Maintenance Risk Assessment to review the design induced operation and maintenance risks and to choose the proper risk alternative.

The identified preventive measures to reduce the probability of incidents on construction site shall be implemented in the design, in order to promote satisfactorily HSE standards during construction phase.

The identified preventive measures together with the relevant safety reviews (e.g. PHA, HAZOP, FMEA, FTA, ergonomic and health risk assessments) shall contribute to safer operations and maintenance activities.

The Technical Reviews / Workshops and Presentations that shall support the Constructability Review include as minimum:

Reviews / Workshops	**Locations** (1)	**Facilitators**
HSEIA Screening / HSEIA	Abu Dhabi	Third party HSE Consultant (4)
PHA Review	Abu Dhabi	Third party HSE Consultant (4)
PHSER	Abu Dhabi	Third party HSE Consultant (4)
Design Review	Abu Dhabi	Independent chairman from ENGINEERING CONTRACTOR organization (5)
Layout Safety	Abu Dhabi	Independent chairman from ENGINEERING CONTRACTOR organization (5)
HAZOP	Abu Dhabi	Third party HSE Consultant (4)
ISDR	Abu Dhabi	Third party HSE Consultant (4)
SIL	Abu Dhabi	Third party HSE Consultant (4)

SIMOPS	Abu Dhabi	Third party HSE Consultant (4)
Bow-tie and HSECES identification	Abu Dhabi	Third party HSE Consultant (4)

HSE Reviews/Workshops and Facilitators

Notes:

1. ENGINEERING CONTRACTOR's office/premise
2. ENGINEERING CONTRACTOR shall appoint scribe for all Technical Reviews and VIP workshops.
3. Facilitator / Chairman shall be assigned by ENGINEERING CONTRACTOR for all HSE and Technical Reviews and VIP workshops.
4. Third-party specialist consultant to be engaged and sponsored by ENGINEERING CONTRACTOR. The CV of the chairman shall be pre-approved by COMPANY. ENGINEERING CONTRACTOR to submit at least 3 CVs of the relevant facilitator for COMPANY review and approval of suitable facilitator. Appointment of third-party chairman, where specified, and scribe shall be part of the ENGINEERING CONTRACTOR responsibility and part of PROJECT lump sum price.
5. Chairman shall be competent personnel independent of the ENGINEERING CONTRACTOR's project team. CV of the Chairman shall be pre-approved by COMPANY. In case competent personnel are not available within ENGINEERING CONTRACTOR's organization, ENGINEERING CONTRACTOR shall engage and sponsor a third-party Chairman on pre-approval by COMPANY with no additional cost impact.
6. ENGINEERING CONTRACTOR shall make all required arrangements and cover for all associated costs pertaining to the team's travel to / from Abu Dhabi, lodging, local transportation and all expenses.
7. ENGINEERING CONTRACTOR shall conduct a detailed constructability review to include but not to be limited to the following:

    a. Installation of new tie-ins and equipment

    b. Shutdown requirements / system isolation

    c. Material handling

    d. Obstruction to pipe routes, access routes and escape ways.

Constructability review shall be carried out in line with agreed tie-in methodology. ENGINEERING CONTRACTOR shall incorporate Constructability Review findings in the FEED engineering relevant documents.

9.13.4 Construction and Installation Study

ENGINEERING CONTRACTOR shall develop the Construction and Installation Study report as per the design. This report shall be a pre-requisite to conduct the Constructability Review workshop.

Physical constraints which could affect the construction and installation are to be identified and addressed.

A professional construction Manager shall be deployed to carry out this study. Multiple site visits and extensive review of the existing facilities are required to complete this study.

Tie-in and Integration Philosophy should be thoroughly reviewed and optimized where feasible such that production impact and overall plant shutdown duration shall be as minimum as possible.

ENGINEERING CONTRACTOR shall prepare a report addressing the following as minimum:

- Details of construction activities associated with the project, including clear step-by- step illustration.
- Study the integration methods for the proposed modifications in the field, substations and control room (DCS, ESD and F&G systems)
- Shutdown sequence of the units and shutdown duration including tie-in / modifications required for safety, electrical and control systems.
- Construction / tie-in proposals with minimum impact on production.
- Tie-in preparation / isolation requirements to install new piping.
- Identification of physical constraints that could affect the construction / installation and solution (space constraint around pipe, straight length in the piping GA, pipe thickness / physical condition of pipe).
- Reviewing the fabrication / assembly procedure to maximize ground level work.
- Identification of laydown areas that will be used during construction & installation and future maintenance.
- Requirement of maintenance access platforms, space constraints.
- Electrical and instrumentation hook-up and cable/tube routings.
- Construction crane approach for construction in congested area.
- Testing of affected piping and instrument systems.
- Constraints and possibilities of structural and foundations strengthening as required for installation of new facilities.
- Study methods of minimizing nuisance / health hazards during construction.
- Construction SIMOPS.

9.13.5 Value Engineering Workshop

ENGINEERING CONTRACTOR shall be responsible for organizing VE Workshop with COMPANY participation and schedule it at an early stage of engineering, such that outcome of the workshop can be effectively integrated in the design. ENGINEERING CONTRACTOR shall ensure participation of its senior experienced staff in the VE process. ENGINEERING CONTRACTOR shall implement all COMPANY approved proposals from the VE Workshop in the design.

ENGINEERING CONTRACTOR shall apply Value Engineering (VE) to evaluate whether a different approach may meet the same targets but at reduced costs.

During the VE Workshop a functional analysis shall be conducted, and all viable options shall be identified, assessed and selected through VE implementation phases. ENGINEERING CONTRACTOR shall follow a disciplined analytical method to reduce the CAPEX and OPEX by either combining or deleting unnecessary processing steps.

ENGINEERING CONTRACTOR shall provide the services of a specialist VE Workshop Facilitator independent of project. Following the workshop ENGINEERING CONTRACTOR shall produce a VE Report and implement the recommendations from the VE Workshop.

The VE workshop shall be conducted in accordance with COMPANY VE Guidelines, STD- 001-013-010. ENGINEERING CONTRACTOR shall develop project specific Value Engineering procedure based on COMPANY guidelines.

VE Workshop report shall be submitted for COMPANY review and comments. Thereafter a final report shall be issued by ENGINEERING CONTRACTOR. ENGINEERING CONTRACTOR shall issue VE Closeout Report addressing the close out actions on the value engineering proposals.

9.13.6 Piping Study and Layout Review

ENGINEERING CONTRACTOR shall conduct a piping routing and layout review and shall incorporate the assessment findings in the ENGINEERING design documents.

9.14 Adequacy Checks

ENGINEERING CONTRACTOR shall conduct site visits and collect data to asses and verify the adequacy of existing facilities/piping system, routing at different locations to accommodate the new requirement/tie-in, existing Instrumentation and package control system and prepare detailed adequacy study report. ENGINEERING CONTRACTOR shall propose solution if found not adequate.

ENGINEERING CONTRACTOR shall hire the service of SUB-CONTRACTOR for such specialized work. Any constraints shall be reported. ENGINEERING CONTRACTOR /SUB-CONTRACTOR shall coordinate with existing system vendors and propose suitable solutions.

ENGINEERING CONTRACTOR shall prepare and present the findings of the adequacy check in a report format accompanied all necessary drawings, calculation, etc.. for COMPANY review and approval. This adequacy check report shall also include the recommendations and proposed solution for COMPANY review and approval.

ENGINEERING CONTRACTOR shall perform FEED based on resolution/solution approved by COMPANY.

ENGINEERING CONTRACTOR shall propose solutions to overcome any inadequacies found in the existing systems, submit recommendations for COMPANY review and approval, upon approval ENGINEER shall further develop & prepare / incorporate in FEED design/deliverables.

9.15 Spring and Reliability, Availability and Maintainability (RAM) Modelling

CONTRACTOR shall develop the RAM study basis and objectives the following key aspects shall be within CONTRACTOR responsibilities:

1. RAM study should be conducted to predict the performance of the selected operating facility over its design life (typically thirty years) by determining its likelihood to meet specific reliability and availability targets. RAM study should determine the interdependence and impacts of individual equipment failure to support selection of the most optimum facility configuration.
2. The study should also identify relative culpability of constituting equipment and quantify the impact of alternative design configurations on availability and various other life cycle cost elements e.g. maintenance & inspection strategies, operating strategy variation, insurance spares availability etc. by using sensitivity analysis.
3. RAM study should be conducted in a holistic manner for the overall facility comprising of both the new/modified facility and the existing one. Model for the new/modified facility should be integrated with the model of the existing facility. Modeling of the existing facility shall be mandatorily included in the RAM SOW (Scope of Work) in case model of the same is not readily available.
4. RAM study should be conducted using DNV Maros software only as per Company standardization policy.
5. Typical inputs to RAM study, as shown but not limited to the list below, should be representative of ADNOC Offshore and are to be specifically agreed with the Company.
    a. Boundary definition
    b. Production profile
    c. Failure & repair characteristics of equipment
    d. Preventive Maintenance & Inspection protocols for equipment
    e. Shutdown / Turnaround requirements
    f. Start-up / Changeover delays, if any
    g. Ramp-up / Boosting capabilities
    h. Assumptions, Inclusions / Exclusions
6. Latest version of ADNOC Group Projects & Engineering Reliability, Availability & Maintainability (RAM) Study Guidelines [AGES-GL-16-002 dated Nov – 2021] shall be used as primary reference while conducting RAM study. Deviations/Conflicts, if any, shall be brought to the attention of the Company for effective resolution.
7. A specialist with proven expertise in conducting RAM study in ADNOC Group Companies is to be employed to conduct the RAM Study.
8. All details regarding RAM study including but not limited to Objective, Base case & Sensitivity cases, Target values, Specific inclusions/exclusions, Deliverables etc. shall be included in RAM – SOW (Scope of Work) document. This document is to be signed off before the commencement of the RAM study.

10. ENGINEERING SCOPE OF WORK

10.1 Process

ENGINEERING CONTRACTOR shall carry out all process engineering activities as well as prepare the drawings and documents specified in the scope of work, including but (not limited to) those identified in the subsections below, with the objective of meeting all the intended process requirements. ENGINEERING CONTRACTOR shall consider any necessary requirements for successful completion of the project as a part of the Scope.

ENGINEERING CONTRACTOR shall carry out all process engineering activities for the scope of work. As a minimum endorsement and validation of the NASR Phase 1 design concept, Tie-in's, process design rates, utilities requirements is to be done. As a minimum AGPS list of deliverables shall be followed for the FEED engineering work.

10.1.1 FEED Design scope

ENGINEERING CONTRACTOR to note that this PROJECT scope of work is related to DAS Minimum Facilities for NASR Crude Phase 1, which is a pre-execution of Package IV 'NASR Crude Alternative Scheme' to be Implemented later. Section 10.1.1.2 below provides detailed design scope for the 'NASR Crude Alternative Scheme' for the overall understanding.

10.1.1.1 Package IV 'NASR Crude Alternative Scheme

The "NASR Crude Alternative Scheme" is a scope among the side studies of current US LTDP 2.1/2.2 CONCEPT RESELECT project - to augment US plant capacity and allow the treatment of NASR Crude with no reliance on LZ LTDP1 project completion. A new standalone NASR train with minimal new facilities is required. The new NASR Crude train will be integrated with existing US plant at multiple tie-in points and will utilize the existing utilities (e.g., Steam, IA, Fuel gas, Nitrogen, Flare, Drain, Fire water, etc.) and auxiliaries (e.g., Power, I&C, Piperack, etc.).

Proposed train shall be designed for 150 MBD processing capacity (all process equipment). The design shall allow the upgrading i.e. conversion to full US train LP re-boiled scheme in future. However, note that Package IV itself does not include the future design activities that will be needed for that conversion – they will be placed into Package III when the ongoing work on them has completed.

The new train will include, but is not limited to, the following components:

- LP Separator
- ATM Separator
- Crude Charge Pumps
- Crude Charge Heater
- Steam Trap Package
- Crude Stripper/Stabilizer
- Product Pumps

The overall schematic of introducing a new train for NASR crude oil is shown below.

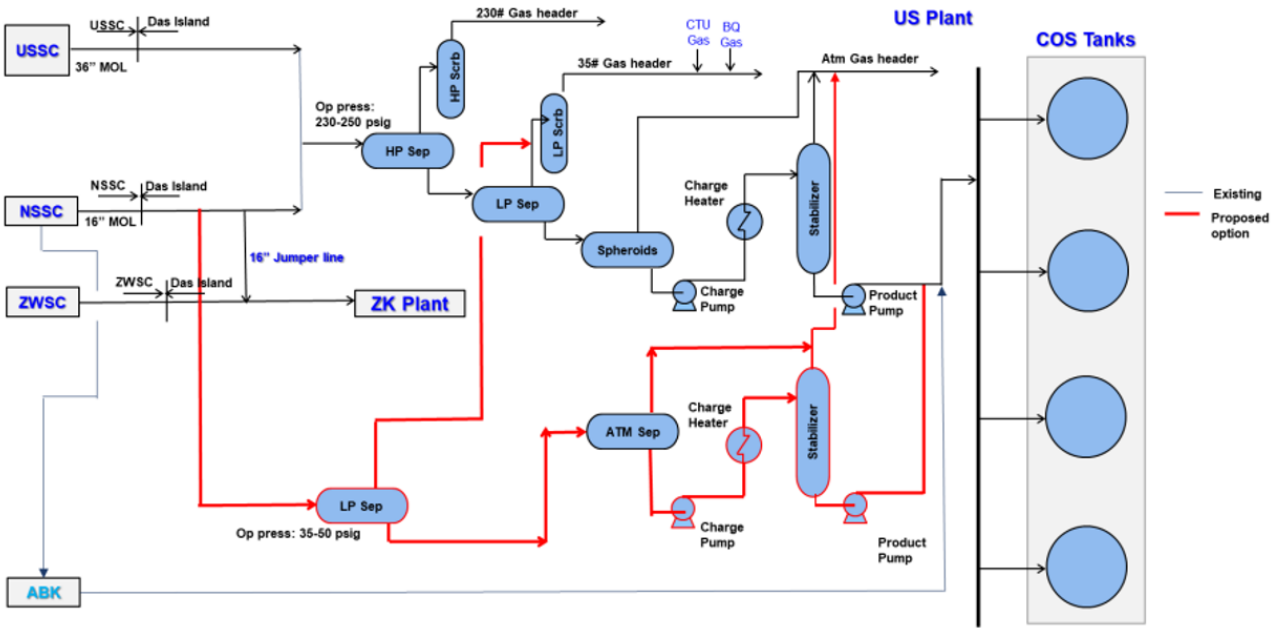


Figure 12: NASR Crude Alternate Scheme

The separated LP gas stream from new LP separator will be directed to the corresponding existing Scrubber (LP Scrubber) before being sent to the dedicated gas header at the LNG Plant on Das Island. Gas released form the new Stripper will be routed to existing ATM manifold.

The produced water from new facilities will be routed to the existing water treatment plant on DAS Island via existing US degassing drum.

10.1.1.2 DAS Minimum Facilities for NASR Crude Phase 1

As per AiP5 mandate, US LTDP-2.0 Project will ramp up US production to 390 MBD (by year 2027. To achieve a crude stabilization capacity of 150 MBD, initially utilizing cold stripping, with future adaptability for hot stripping at ZAKUM Plant as part of Phase 1.

The new NASR Phase 1 will include, but is not limited to, the following components:

- Crude Charge Heater
- Steam Trap Package
- Crude Stripper/Stabilizer
- Product Pumps

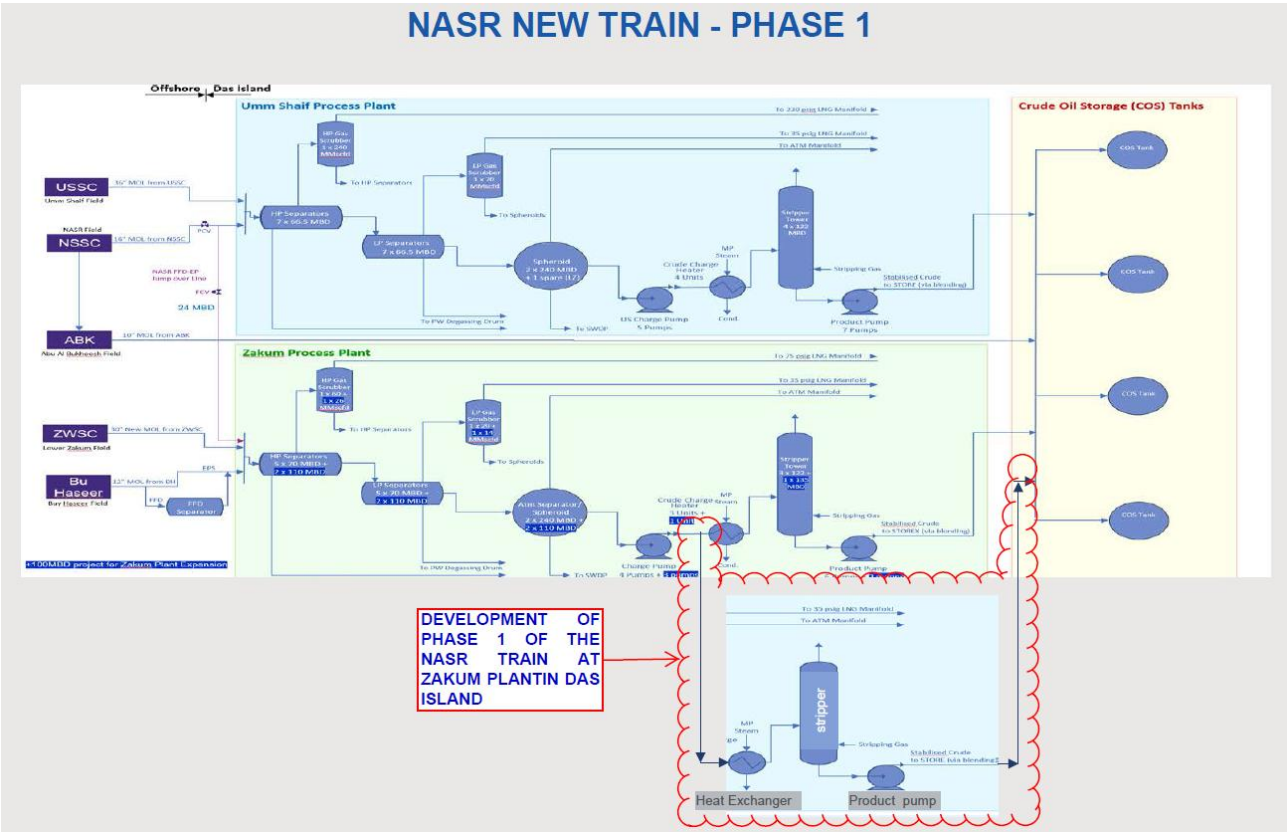
In the new phase 1 facility Crude oil take off is proposed from the ZAKUM charge pump discharge flowing through one New Charge Heater where crude oil will be preheated before entering the stripping column.

The Crude oil feed into the tower and is flashed in the vapor disengagement space. The resultant flashed vapours leave the column at slightly above the atmospheric pressure, while the oil descends through trays and our crude is stripped.

After stripping the crude in the towers, the sour gas passes through a demister pad before leaving the tower overhead and to be tie-in to the gas manifold. The stabilized and stripped crude flows from the bottom of the stripper is pumped to the crude oil storage tank area by a Stripped Crude Product Pumps.

The separated gas from new stripper will tie in into existing ATM manifold. ENGINEERING CONTRACTOR must work on the possibilities to connect NASR Phase1 stripper outlet to nearby possible header.

The following are the minimum utility tie-ins with the existing utilities at Das Island, Flare system, Fuel gas System, Steam, Condensate header, Nitrogen, Instrument air, Open Drain and Closed drain. The overall schematic of introducing a new minimal facility for NASR phase 1 crude oil is shown below:



9.1.1.2.1 New Charge Heater

The new processing train will have one New Charge Heater where crude oil will be preheated to 49 °C before entering the stripping column.

9.1.1.2.2 New Cold Stripping Unit

The new alternate processing train will have a cold stripping unit. The operational process of the new unit mirrors its existing stripping unit operations apart from having a vent drum for degassing and is detailed below:

The flow of crude begins by preheating with steam in the Sour Crude Charge Heaters until reaching approximately 49°C, all under temperature control. Crude oil feed into the tower and is flashed in the vapor disengagement space provided in the upper part of the stripper column above the 1st tray. The resultant flashed vapours leave the column at slightly above the atmospheric pressure, while the oil descends through trays, encountering a counter-current flow of sweet gas entering from the tower's bottom as stripping gas.

While flowing through the trays, sour crude is stripped by a light sweet hydrocarbon gas, which is introduced at the bottom of the stripper column. The hydrocarbon gas (sweet gas) aids in the removal of the residual H2S together with some of the volatile lighter hydrocarbons.

After stripping the crude in the towers, the sour gas passes through a demister pad before leaving the tower overhead and entering the 60" atmospheric gas manifold.

ENGINEERING CONTRACTOR to note that the cold stripping unit shall be designed for both Cold and hot stripping as required for the new complete New NASR train.

9.1.1.2.3 New Product Pumps

The stabilized and stripped crude flows from the bottom of the stripper is pumped to offsite storage in the crude oil storage tank area by a Stripped Crude Product Pumps.

New product pumps will be equipped with a suction strainer and high differential pressure alarm akin to existing ones. Flow will be measured on the discharge from the unit by a flow meter.

10.1.2 Process Engineering General

FEED Engineering Process Engineering Scope of Work (SOW) shall include, but not be limited to, the following:

- Develop all necessary engineering and detailed design for all support systems required for the proper functioning of all the equipment /packages of this project. If the ENGINEERING CONTRACTOR, deems it necessary to add additional item, then the ENGINEERING CONTRACTOR shall notify the COMPANY in writing that additional items are required prior to undertaking the design of the additional item.
- ENGINEERING CONTRACTOR shall ensure that the digital, AI technologies on each of the equipment are considered during the engineering phase and accordingly the process deliverables are prepared. The facilities in this scope shall be fully autonomous by design and shall be able operate and maintain with no manual human intervention.
- ENGINEERING CONTRACTOR shall do all sizes and rating of lines as a minimum.

- Perform process consistency checks and update, where necessary, all process information to ensure compliance with SPECIFICATIONS in order to meet the concept process design intent and ensure that the individual equipment and SYSTEMS are safe and operable.
- Obtain from the COMPANY the required line numbers, tag numbers, document numbers etc. in accordance with COMPANY procedures.
- Ensure all Project Philosophies are followed for all facilities including VENDOR and/or SUBENGINEER packages.
- Review all Process-related SPECIFICATIONS and detailed calculations to be carried out during engineering as required in order to meet the PROJECT objectives.
- Integrate all VENDOR and/or SUBENGINEER design information into the ENGINEERING CONTRACTOR's Process Design and submit the integrated design to COMPANY for review/approval
- ENGINEERING CONTRACTOR shall prepare separate construction and demolition P&IDs. Existing P&IDs shall be updated where necessary.
- Provide process input relevant items including Instruments/Valves.
- ENGINEERING shall verify line sizes and implement any required modifications after COMPANY approval.
- Review isolation and maintenance facilities to ensure that all lines can be prepared for maintenance in compliance with COMPANY procedures, SPECIFICATIONS, and philosophies.

10.1.3 Process deliverables

As a minimum AGPS to be followed for the process deliverables. Mainly Process design basis & Criteria, Process steady state hydraulics simulation, PFD, P&ID, Process Simulation Report, Heat and Material Balance Report, Process calculations, Philosophy Documents, Relief and Blowdown calculation report Data sheet, Process & Utility Tie-ins, Line List Any other deliverables not listed in AGPS deliverable list but required for FEED Design stage shall be considered.

ATTACHMENT 1 provides the minimum Process deliverables to be prepared for Process project scope. Any other deliverables not listed but required as per GPE-GE/GDL-001 required for FEED ENGINEERING stage shall be considered.

10.1.4 Reviews and Workshop

Below is the minimum review and workshop list for the FEED engineering.

P&ID review:

P&ID reviews shall take place prior to the HAZOP reviews, when sufficient design information is available on the P&IDs to allow a meaningful review.

Hazards and Operability (HAZOP):

ENGINEERING CONTRACTOR shall carry out the following activities as a minimum:

CONRACTOR shall use COMPANY Approved resource who independent of project to conduct the HAZOP Workshop.

HAZOP review shall be in accordance with Company Standards, DNOC HSE standard HSE-RM-ST04. Contractor to ensure that any risks identified due to AI and autonomous mode of operation, shall be considered during the HAZOP study.

Close out all HAZOP action items assigned to the ENGINEERING CONTRACTOR. All HAZOP action item resolutions shall be agreed with COMPANY prior to implementation.

10.1.5 Das Island SS#47 Surface Utilities

Refer to process description report (Doc. No. AD-90.0-G-00619) of current side study scope for relevant utility tie-ins

10.2 Health, Safety & Environment

The THSE scope of work is includes brief requirements upon which the CONTRACTOR shall carry out the FEED Engineering in accordance with COMPANY, National and International design codes, standards and Industry best practices for complete project HSE integrity.

ENGINEERING CONTRACTOR to carry out FEED for below listed brief scope:

1- Conduct all listed HSE studies / Sessions (PHA, PHSER, QRA, FERA, HAZOP, ISD, EIA, SIA …. ETC.) by approved consultant as per ADNOC AVL as complete HSEIA dossier considering implementing and closing all raised recommendations and action in HSE action tracking register and close-out report. HSE studies shall include the Risks and Hazards that may arise due to anomalies and failure of AI and Autonomous technologies.
2- Perform all FEED engineering for Firefighting system coverage for new facilities (water / foam) and necessary tie-in with existing fire water network with all necessary hydraulic and surge calculations, water demand calculation and required FF equipment specifications / datasheets, Piping & instruments diagrams, fire water network layout, deluge water spray coverage layouts, MTOs and MRQs in accordance with ADNOC and international standards.
3- Perform all FEED engineering for fire and gas detection system and notifications appliances, interfaces with ESD, DCS PA/GA systems for complete system integrity in line with F&G mapping study and other relevant studies (FERA, QRA, H2S and Gas dispersion studies …. Etc.) in accordance with ADNOC and international standards.
4- Perform all FEED engineering for escape routing, safety signs and safety equipment specifications / datasheets, MTOs and MRQs in accordance with ADNOC and international standards.
5- Perform all FEED engineering for passive fireproofing system including (fireproofing zones layout, specifications, fireproofing schedules) in accordance with ADNOC and international standards.
6- Conduct Hazardous area classification schedule / Layout for electrical system classification accordance with ADNOC and international standards.

10.2.1 Input from HSE Aspects and HSE Deliverables

ENGINEERING CONTRACTOR shall develop and provide input into the design of new DAS minimum facilities PHASE 1. ATTACHMENT 1 provides the list of reports / documents to be included as part of ENGINEERING CONTRACTOR deliverable during FEED Engineering Stage of the Project.

10.2.2 Input to Plot Location from HSE Aspects

ENGINEERING CONTRACTOR shall provide input to the project scope modification in Das Island as part of US LTDP-2 Phase 2.1 and Phase 2.2. project scope with respect to HSE Aspects. ENGINEERING CONTRACTOR shall comply with ADNOC and AGPS standards for the design of project scope modification in Das Island.

Project scope modification in Das Island and plant layout criteria and separation distances shall comply with AGPS-SD-HSE-001 and Piping Basis of Design (AD-26-M-20006).

10.2.3 SIMOPs Restriction/ Exclusion Zones

ENGINEERING CONTRACTOR shall carry out the QRA Study by engaging a COMPANY approved Third Party Consultant in accordance with ADNOC QRA Standard (HSE-RM-ST10).

QRA Study shall consider SIMOPS, in addition to process hazards, risk from non-process hazards including SIMOPS and impact to / from existing operations shall also be addressed.

Based on the QRA Study outcome, restriction and exclusion zone shall be established.

10.2.4 SIMOPs Matrix/ MOPO

ENGINEERING CONTRACTOR shall conduct SIMOPS review in accordance with ADNOC SIMOPS Standard (HSE-OS-ST06) and COMPANY standard A0-DR-P-GDL-001.

CONTRACTOR shall conduct a MOPO workshop involving COMPANY and develop SIMOPS/ MOPO matrix for typical SIMOPS activities envisaged as part of the Project. The objective of developing a MOPO is to perform various SIMOPS activities in safe and organized manner. MOPO template shall be agreed with COMPANY prior to the workshop.

10.3 Plant – Piping

10.3.1 FEED ENGINEERING SCOPE OF WORK

This section stipulates COMPANY's minimum requirements for FEED ENGINEERING Design of Piping system. ENGINEERING CONTRACTOR shall carry out FEED Engineering phase of this Project and shall prepare and Issue FEED Engineering documents / drawings separately. ENGINEERING CONTRACTOR shall prepare MR for Piping long lead Items including Valves and Specialty items during FEED Engineering which will be further updated and will be issued during Detail Engineering.

ENGINEERING CONTRACTOR shall adhere to the following project specific requirement as a minimum.

- ENGINEERING CONTRACTOR to perform site survey to finalise locations proposed during SELECT study for Phase I equipment such as Charge heater, Stabilizer and Product pumps. ENGINEERING CONTRACTOR to consider safety distances for locating equipment as per AGPS-SD-HSE-001.

- ENGINEERING CONTRACTOR to prepare Options selection report and Layout finalization report for the installation of new Equipment and facilities proposed as part of the FEED Scope.

- ENGINEERING CONTRACTOR shall propose the modifications required to install new facilities and Proposed Equipment installations.

This section specifies minimum requirements for the Piping Engineering as part of the FEED Engineering Scope of Work. Scope of work includes but not limited to the following:

a) ENGINEERING CONTRACTOR shall carry out all engineering activities for Piping based on the Select stage document / feasibility study report and requirements specified in the SOW for this PROJECT.

b) ENGINEERING CONTRACTOR scope shall include, but not be limited to, site survey, complete FEED piping engineering & design, review, verify & up-dating Documents and Drawings, produce new if is not available.

c) ENGINEERING CONTRACTOR shall prepare as a minimum, but not limited to the deliverables listed in APPENDIX-1, ENGINEERING CONTRACTOR to prepare all the necessary deliverables for completing scope of work in line with GPE-GE/GDL-001 and A0-ENG-Z-GDL-001

d) The Select stage documents and drawings/mark-up shall be used as a Preliminary basis by the ENGINEERING CONTRACTOR to develop based on FEED engineering and designs.

e) ENGINEERING CONTRACTOR to conduct Site surveys to study equipment/package location, routing of piping, Tie-ins, platform, structure, Pipe rack, support etc., and to verify space and layout requirements, to confirm feasibility of the proposed modifications / extension of existing facilities.

f) ENGINEERING CONTRACTOR shall note three locations AREA A, Area B & Area C proposed during Select stage for new NASR facility at DAS. However, ENGINEERING CONTRACTOR shall carry out engineering concluding all Studies required (including HSE Studies) and finalize suitable location / Area out of Area A, Area B and Area C (which were identified during Select stage) to install NASR Phase-I Facilities.

g) ENGINEERING CONTRACTOR shall also study and identify suitable locations to install new NASR Phase- I facilities considering optimized pipe route, Tie-in locations, access roads, Operation and maintenance requirements etc.

h) Site survey shall be carried out to collect the data related to the existing site conditions to properly and correctly identify the surface facilities, underground services before proceeding with the FEED.

- ENGINEERING CONTRACTOR shall note that the Select stage study / Feasibility study of Equipment location, Piping arrangement, documents are based on preliminary inputs which shall be reviewed, study, updated & firmed up based on final outcome.
- Any change in the design (i.e., location, Size, MOC, type, quantities, routing, arrangement etc.) due to change of design parameters/study shall be incorporated in the design as part of FEED scope without any Cost and Schedule impact on COMPANY.
- ENGINEERING CONTRACTOR to refer and verify Select stage study report / Feasibility study report, assumption, recommendation and if required same to be update/revise or prepare new for further engineering in consultation with COMPANY.
- Select stage study / / Feasibility study deliverables available are not detailed, ENGINEERING CONTRACTOR's Scope of Work includes all the Piping engineering design and detailing necessary for completion of work & Project, including but not limited to the following.
    1. To study/review/verify/finalize Equipment, Packages, Valves and Piping's Location, Size, MOC, type, quantities etc. and produce required documents based on Separation distance requirement, constructability, feasibility, access, escape, fouling with existing facilities, road crossing etc.
    2. To study/review/verify/finalize Piping routing, arrangement, location, Low point drain arrangement/dead leg/purging etc. and produce required documents.
    3. To study/review/verify/finalize location and feasibility of all Tie-ins.
    4. Supporting of Valves, instruments, and Piping arrangement etc. from existing/New structure/sleeper
    5. Access platform requirements to be checked and provided
    6. ENGINEERING CONTRACTOR shall study the Project requirement in reference to the COMPANY existing Specification and based on the same, addendum/new to existing Specification shall be prepared as per Project requirement
    7. Detail route study & Stress analysis not performed during Select stage study / Feasibility study, ENGINEERING CONTRACTOR to do route study / engineering and finalize the same in consultation with COMPANY.
    8. ENGINEERING CONTRACTOR to review requirements of Flow induced vibration (FIV) and Acoustic Induced Vibration (AIV) and if required same shall be performed by the ENGINEERING CONTRACTOR. Suitable mitigation measures shall be proposed and to be implemented in design by the ENGINEERING CONTRACTOR during FEED engineering.
    9. ENGINEERING CONTRACTOR to review and if required provide suitable material handling (Davit arm, Monorail etc…) arrangement for all elevated platforms considering proper drop zone area with road access
    10. The selection of piping materials shall be in accordance with Piping Material Classification (AAGPS-SD-MC-001/ AGPS-SP-PI-002/SP-1131/SP1000).

11. ENGINEERING CONTRACTOR shall review and if required prepare new or produce an addendum to existing COMPANY piping material specifications, other Specification, drawings, documents etc. to cover the scope of the entire Project piping work. ENGINEERING CONTRACTOR shall ensure the consistency for all piping and associated items materials, service and other data across the facility and shall follow the relevant international codes and standards and design the system accordingly.
12. The ENGINEERING CONTRACTOR shall be responsible for any new/additional specifications / drawings / documents / checklist that are required.
13. 3D modelling in Smart plant 3D and extraction of all relevant deliverables from the Model. Piping modelling shall be done as per approved piping classes.
14. Piping design shall consider as a minimum:
    a. Process flow
    b. Wind direction
    c. Provision for maintenance and lay down areas for Tanks, Pumps and other equipment, Piping and E&I.
    d. Safety of personnel/ facility and Evacuation. Provision of adequate escape routes and walkways.
    e. Maintenance, operations and handling requirements of valves and inline instrumentation.
    f. Natural ventilation and lighting are adequate.
    g. Process and safety requirements have been addressed.
    h. Provide a layout that has been done in a real and aesthetic manner, with piping routed as direct as possible, within the limits of good design practice, preferably in the banks or pipe way with due consideration being given to the preservation of escape routes and maintenance access. Dead legs in piping shall be avoided.
    i. Simultaneous operation (SIMOPS).
    j. Requirements for future facilities including their constructability.
    k. Address all additional modifications / new piping consequent to P&ID`s update based on any Process optimization / COMPANY requirements identified during the engineering phase.
    l. Piping GADs, MTO`s shall be prepared in line with P&ID requirements and the plot / space available.
    m. ENGINEERING CONTRACTOR shall review and update all drawings, documents, and stress analysis reports based on Preliminary VENDOR data.

- ENGINEERING CONTRACTOR shall develop the Piping Design Basis based on COMPANY specification AGPS-SD-PI-001 (Piping basis of design) as an input to common multi-discipline design basis to establish the Piping design criteria to be used for the FEED and Detail engineering activities and all aspects related to the existing facilities. The design basis shall be agreed with COMPANY prior to commencement of pre-engineering activities.
- ENGINEERING CONTRACTOR shall carry out Piping design, engineering and layout design as guidelines specified in AGES-SP-09-001 (Piping basis of design) and COMPANY Specification

AGPS-SD-HSE-001 (Facility Layout & separation distance guideline) and other applicable COMPANY / Project specification.

- Perform stress analysis for new piping systems and ensure no impact on existing piping or equipment nozzles.
- Pipe routing and pipe supports arrangements shall be engineered to ensure that loads exerted by piping on equipment/Tank/Pump nozzles and structural members are within safe and acceptable limits.
- Wherever existing pipe supports/structural members are utilized to support project scope piping, ENGINEERING CONTRACTOR shall review, verify, and ensure their adequacy to take the intended loads and engineer mitigations as necessary to complete the requirements of this scope.
- Piping design shall take into account the requirements to provide permanent adequate access for all instrumentation including instrument isolation valves, points such as piping low point drainage or high point venting, isolation spades/spacers or spectacle blinds for normal facilities operation and maintenance.
- ENGINEERING CONTRACTOR shall study all ongoing and other interface projects before concluding the piping design during the FEED stage. ENGINEERING CONTRACTOR shall prepare separate interface register during FEED which shall be further updated during Detail Engineering.
- ENGINEERING CONTRACTOR shall prepare layout study report for the options studied to finalize location of New facilities. Layout finalization report shall include the modification proposed to finalization of above proposed equipment.
- ENGINEERING CONTRACTOR shall update and finalize the plot plans / Equipment layout drawings based on Preliminary vendor data. Piping layout design shall carefully consider and satisfy requirements of Process, Safety and other HSE requirements.
- Piping general arrangement shall be finalized /designed by ENGINEERING CONTRACTOR with due consideration for accessibility, constructability, maintainability, operability, SIMOPS, mechanical handling, equipment / material transfer to designated lay-down areas, HSE considerations, Human Factor engineering etc. such to ensure the layouts are ergonomically acceptable. Piping systems and layouts shall adhere to COMPANY standards, specifications, international standards, established design methods and practices in accordance with all applicable codes.
- Piping design shall be carried out with due consideration to avoid shutdown of DAS facilities.
- Existing underground facilities in the vicinity such as utilities, fire water lines, cable works, pits, etc shall be identified and if required, suitable proposal to be provide for protection / realignment / modification or relocated to avoid clashes and to enable construction of new facilities.
- Header elevations/slope to be finalized meeting the process requirement and considering the existing facilities of hook-up pipework, Pipe support structure etc.
- Design of all pipework arrangement and other modifications shall be finalized strictly in accordance with latest revision of IFD P&ID's. Tag numbers of equipment, Instruments, piping special part items etc. shall be updated in relevant piping drawings as per P&IDs.

- ENGINEERING CONTRACTOR shall ensure that development of piping layout arrangement shall facilitate new modifications to match with the existing system without causing any hindrance to accessibility / maintainability of existing facilities in immediate vicinity.
- Outcome of Constructability workshop / HAZOP / HSE safety studies carried out during scope definitions / FEED phase and their recommendations & mitigation measures shall be incorporated into the FEED design.
- During FEED engineering, ENGINEERING CONTRACTOR shall conduct SITE surveys for physical verification and making corrections / incorporating the 'as-built' condition of existing facilities that are relevant to and which may have impact on the design of Project facilities. ENGINEERING CONTRACTOR shall carry out design of new facilities, considering the "as-built" conditions of existing facilities, update relevant PROJECT drawings accordingly before proceeding with finalization of design deliverables.
- ENGINEERING CONTRACTOR shall carry out site surveys to verify construction feasibility of new facilities including identification of any clashes or modifications required on existing facilities and prepare / update drawings & documents and execute additional tie-ins / pipe work / supports modifications as required and same shall be part of FEED scope of work. ENGINEERING CONTRACTOR site visit scope shall cover co-ordination with other ongoing projects for interface issues with respect to PROJECT facilities and for their construction.
- All new piping routing in the existing plant area including pipe racks / Pipe Corridor / Platform / Support shall be verified at site during FEED engineering and suitably designed according to the space availability. Any additional support / structure shall form part of FEED Engineering scope.
- ENGINEERING CONTRACTOR to review models / drawings / documents and deliverables of existing installed facilities and Interface projects and shall ensure that new facilities to be clash free with existing facilities.
- ENGINEERING CONTRACTOR shall obtain relevant controlled Project copies of 3D Model, Piping GA drawings etc. from COMPANY and update the tie-ins & pipe work modifications details on Piping GA drawings (Construction / Demolition). Same shall be applicable for any additional tie-ins required for successful completion of the PROJECT. ENGINEERING CONTRACTOR shall prepare piping elevations or section drawings/details as required for ease of understanding of layout in Piping GA drawings.
- Pockets and dead leg portions on piping works shall be avoided. Suitable drains shall be provided at low points where dead legs cannot be avoided. The drain root valve shall be located at minimum distance from the take-off point, in process drain connections. Drains to be connected to Blow case vessel and condensate trap as applicable. ENGINEERING CONTRACTOR shall also prepare dead leg register in line with ADNOC Code of practice and Dead Leg management (A0-IG-C-CP-001).
- ENGINEERING CONTRACTOR shall ensure operation and maintenance accessibility of Instruments, Valves, Spectacle blinds, Corrosion monitoring facilities by locating them at grade or provided with proper access platforms with stairs / ladders. Piping design / layout arrangement shall be carried out accordingly.
- ENGINEERING CONTRACTOR as a minimum shall verify and ensure that all maintainable items and valves are provided with adequate means for access, lifting and replacement activities.

- ENGINEERING CONTRACTOR shall prepare required piping demolition GA drawings capturing piping to be removed completely or partly, modifications of pipe works, and Tie-in works for requirement of PROJECT. All demolition items in the scope shall be moved to removed zone in PDMS model.
- ENGINEERING CONTRACTOR shall ensure that all drawings / documents including those generated by VENDOR / SUPPLIER shall be in compliance with COMPANY Numbering procedure A0-ENG-D-PRO-001, CADD Manual A0-ENG-D-PRO-002 applicable Project specific guidelines.
- During FEED Design, ENGINEERING CONTRACTOR shall review / update and finalize the COMPANY specifications including Piping material specification, Purchase specifications, valve data sheets etc. based on the FEED scope of work and related documents. ENGINEERING CONTRACTOR shall prepare new specifications, data sheets, special items list etc. as required, same to be comply with latest COMPANY Specifications.
- Based on service requirement (Sour, toxic/lethal, high pressure etc..) proper fluid category to be identified and suitable design section of applicable code to be followed in consultation with COMPANY, applicable requirement to be followed in engineering and documentation.
- Arrangement shall be designed as per Code ASME B31.3 with proper corrosion allowance and the CS/LTCS/SS pipes, also refer COMPANY applicable specification including A0-ENG-P-SP-003 Specification for Fabrication, Erection and Testing.
- ENGINEERING CONTRACTOR shall carry out pipe wall thickness calculation adhering with relevant Codes.
- ENGINEERING CONTRACTOR to prepare the new Material Requisitions for Valves and other Piping long lead items.
- ENGINEERING CONTRACTOR shall update and finalize the required piping standard details for vents, drains, instrument connection details etc. based on Piping Design & Construction Standard document that needs to be developed as part of FEED Design of Project. The standard details shall be in line with COMPANY standard specifications for both piping and instrumentation.
- ENGINEERING CONTRACTOR shall identify special piping Items. Individual data sheet and specification shall be prepared for each type of Piping Special Items.
- Design pressure and temperature of piping shall be in accordance with the PROJECT line list and specifications.
- ENGINEERING CONTRACTOR shall prepare piping material requisitions for the piping commodity materials, piping valves, piping special items, special supports (if required), etc., material requisitions for all piping commodities to be in compliance with the technical Specification/clarifications.
- The following points are to be considered by ENGINEERING CONTRACTOR, even if they are not covered in relevant specifications, data sheets.
- ENGINEERING CONTRACTOR shall update relevant documents in accordance with COMPANY specifications.

a. ENGINEERING CONTRACTOR shall ensure fugitive emission and its test requirement in data sheet of valves in hydrocarbon services as per COMPANY specification.

b. Minimize the number of connections as reasonably practical (flanges, fittings, gaskets, etc.).

c. ENGINEERING CONTRACTOR to note that Coating Specification for new & Existing Constructions of offshore and onshore structures shall be based on AGPS-SP-MC-003 – Painting & Coating Specification.

- For installation of new pipe/equipment works, any modification of existing facilities such as small-bore pipe re-routing, existing support modifications, electric light pole relocation, instrument junction box relocations etc. shall be identified as part of site survey and same shall be performed by ENGINEER.
- Dead legs and pockets in new piping shall be minimized and proper closed drain arrangement shall be designed by ENGINEERING CONTRACTOR .
- ENGINEERING CONTRACTOR shall be responsible for all specifications, procedures, scopes of work, reports, design basis, work packs and drawings that are required to complete the scope of work. Specifications, procedures, data sheets, etc. shall be developed as an addendum to COMPANY standards wherever applicable. All FEED phase deliverables shall be subject to COMPANY review and approval.
- Bolt removal envelopes. ENGINEERING CONTRACTOR shall pay special attention to minimum required distances for bolt removal from flanged joints and ensure that there are no “captive bolts” which require dismantling of other equipment/pipework and ensure no clashes with pipe supports or other equipment/pipework, such as elbows directly welded to flanges or pipe supports close to flange connections. These shall be checked to ensure that bolts can be fully removed from at least one side of flanged joint without clashing with support shoes and flanged branch connections etc. ENGINEERING CONTRACTOR shall consider the additional nut length required for hydraulic tensioning of bolting.
- Equipment/pipe work removal. ENGINEERING CONTRACTOR shall pay special attention to space requirements for equipment/pipework removal including required lifting devices i.e., lifting lugs on removable spools and mechanical handling facilities.
- All maintenance/operational volumes shall be modelled as soft volumes and clash checked in the SP3D model. This includes, as a minimum, laydown areas, secondary and primary escape routes, doors, material handling ways, monorail trolleys movement, hatches etc.
- Future equipment and piping shall be modelled in separate zone colors for ease of identification.
- Checks shall be made on a regular basis in order to ensure accuracy of the SP3D model, and documents extracted from it, clash detection, data base conflicts, etc.
- Model all instrumentation devices and equipment, 3D Modelling should not limited to shut down valves, control valves and junction boxes, cabinets, transmitters, F&G detectors etc.

10.3.2 Piping Design Basis

ENGINEERING CONTRACTOR shall provide input to Project Design based on job scope of COMPANY's piping design basis and shall define all new basic requirements based on the project. If necessary, an addendum specific to COMPANY's existing document or new document in line with COMPANY's requirements shall be prepared suitably to cover all project requirements.

ENGINEERING CONTRACTOR shall develop the piping design philosophy and piping layout philosophy based on all the project design parameters with the objective of fulfilling the applicable COMPANY requirements, existing piping systems, available specifications/standards, generic and specific design considerations

10.3.3 Piping and Material Specifications/Data Sheets

Based on the material selection report, Corrosion risk assessment and all other project specific requirements ENGINEERING CONTRACTOR shall prepare new or produce an addendum specific to existing COMPANY piping material specifications (including valve specifications and valve data sheets) to cover the entire piping scope for the portion.

ENGINEERING CONTRACTOR shall be responsible for any additional specifications required as per Project requirements.

ENGINEERING CONTRACTOR shall ensure the consistency for all piping and associated items materials and service across the facility.

ENGINEERING CONTRACTOR to study and recommend the piping materials. The selection of piping materials shall be in accordance with Piping Material Classification (A AGPS-SD-MC-001/ AGPS-SP-PI-002 / AGPS-SD-MC-002). ENGINEERING CONTRACTOR shall prepare project specific piping materials specification (including valve specifications & valve data sheets) and classification documents using applicable piping classes. Amendment sheets or new piping classes, valve specifications & valve data sheets, if required, shall be prepared based on COMPANY standards, international standards, wherever applicable.

Detailed piping material class shall be prepared for recommended piping material, covering all technical requirements in line with COMPANY guidelines. A detailed piping material specification shall be developed, which shall include all aspects of the design such as applicable codes, materials, wall thickness and stiffness class, design qualification and manufacturing, inspection & testing requirements, fabrication, installation and operation methods, ENGINEERING CONTRACTOR shall follow the relevant international codes and standards and design the system accordingly.

In case flexible Pipe material is selected as basic metallurgy for piping and piping components, ENGINEERING CONTRACTOR is required to hire the services of Material Specialist for preparing all the Flexible pipe engineering activities like, composition of flexible for pipes and piping items, thickness calculation, Specification, Piping classes, stress analysis, supporting details, testing etc. Also, Specialist to prepare the PC scope of work covering all the requirements of the flexible piping.

10.3.4 Piping Plot Plans and Piping General Arrangement Drawings

Piping General Arrangement drawings shall be based on the applicable piping design specifications. ENGINEERING CONTRACTOR shall critically study the site plot, review all the existing layouts, and develop the new piping routing considering all aspects and requirements of COMPANY. Alternative feasible piping routing shall also be studied and recommended for further review and approval by COMPANY. Further development of the layout shall be done in such a manner so as to enable the new modifications to match with the existing system without causing any hindrance to accessibility / maintenance of any associated existing facilities in the immediate vicinity. ENGINEERING CONTRACTOR shall carry out a detailed study of all the options. The layout shall include and suitably address all the requirements.

ENGINEERING CONTRACTOR shall develop fresh / new plot plans in Auto-cad to the suitable scale, showing all the new as well as existing facilities with safety distances for all the locations in case if existing Master drawings are not available.

ENGINEERING CONTRACTOR to ensure all new facilities as part of this PROJECT shall be designed and constructed within the existing area that is identified for this PROJECT. Requirements, if any, for additional space / modification shall be referred to COMPANY for approval.

Final layout arrangement shall be reviewed with site personnel by conducting walk-through. ENGINEERING CONTRACTOR shall organize its own personnel and include necessary presentations accordingly for carrying out the same at site.

ENGINEERING CONTRACTOR shall consider safety, operability, constructability, accessibility and maintenance as part of piping layout design. ENGINEERING CONTRACTOR shall as far as possible, make use of COMPANY piping standard details and pipe support details in the design of new facilities. Special supports shall be designed with approval from COMPANY.

ENGINEERING CONTRACTOR shall check the existing underground and above ground facilities and provide modification and demolition requirements.

ENGINEERING CONTRACTOR shall review as necessary and develop the plot plans, and general arrangement drawings based on latest equipment information, site surveys and any other design development stages

Plot plan and spacing shall be developed based on the following:

- Accessibility, Operability, Maintainability, wind direction, and Constructability.
- Process flow, Area Classification, and HSE considerations.
- COMPANY specifications, standards, procedures, codes of practice and international codes and standards.
- Safety of personnel / facility and evacuation.

ENGINEERING CONTRACTOR shall Prepare plot plans and general arrangement drawings (GADs) covering all new and existing equipment, piping, and all tie-in locations. layouts and piping GADs shall be extracted from the Smart Plant 3D model, to meet COMPANY Code of Practice (CP-00). Drawing grid numbering shall follow the guidelines of CP-00.

Piping GADs shall reflect all piping, valves, specialty items, tie-ins, instrumentation, pipe supports, and equipment (with tag numbers), Piping design shall take into account the requirements to provide adequate access, drainage, venting, and isolation facilities for normal facility operation, and maintenance.

Piping GAD shall reflect all line numbers, elevations, scope demarcation, Design code demarcation, along with all Equipment and Instrument tag numbers and SP item tag numbers.

ENGINEERING CONTRACTOR shall verify piping elevations marked in Piping GA drawings with actual site condition.

Where required ENGINEERING CONTRACTOR shall prepare piping elevations or section drawings for ease of understanding.

ENGINEERING CONTRACTOR shall prepare piping demolition drawings for all piping installations to be removed, partly removed or re-routed. Prepare demolition drawings for modifications and tie-ins.

The piping GAD to be developed during FEED Engineering stage shall be extracted from 3D model and all annotated in AutoCAD. However, ENGINEERING CONTRACTOR shall extract the same from 3D model with all annotations done in Smart Plant 3D model.

ENGINEERING CONTRACTOR shall note that Plot plans and Piping General arrangement Drawings and 3D model to be developed during FEED are based on preliminary vendor data.

The drawings shall clearly indicate all valves (including valve hand wheel orientations), piping, specialty items, tie-ins, instrumentation, pipe supports and equipment, all with tag numbers as applicable. Piping design shall consider the requirements to provide adequate access, drainage including hydro-test vents and drains, venting and isolation facilities for normal plant operation, and maintenance, as well as for hydrostatic testing of the pipe work.

Piping layout arrangement shall ensure valves, instruments, spectacle blinds that require frequent operation / maintenance to be located in such a way that they can be accessed from access platform and are properly supported.

While designing new piping layout ENGINEERING CONTRACTOR to ensure that the transfer of loads (due to installation of new loads) on to the existing supports, structures, equipment etc shall be kept as minimum as possible and in any case shall be within the acceptable limits.

Demolition of lines / equipment shall be indicated on separate demolition general arrangement drawings, demolition plot plans and demolition isometrics with BOM. New pipe supports shall be provided where existing pipe support arrangements are affected by the demolition.

All existing drawings relevant to the new piping installations shall be verified and updated to reflect actual site conditions and the lines being installed under the on-going projects.

Equipment Layout reference level shall match with the existing plant / platform benchmark. All level and marking shall be consistent with the existing level in that area. ENGINEERING CONTRACTOR shall prepare piping demolition drawings for all piping installations that to be removed completely or partly removed. The existing line(s) shall be rerouted in case found to be necessary.

10.3.5 Piping Stress Analysis and Supports

Pipe Stress Analysis of critical piping systems shall be performed in accordance with COMPANY Specification AGPS-SP-PI-004, Specification for Piping System Stress Analysis.

Critical line philosophy along with stress critical line list to be prepared. Particular attention is required to be given to tie-in with existing lines, their respective connections to the new facilities and their impact on the existing lines. ENGINEERING CONTRACTOR to include for stress analysis the part of existing connected line up to the first available anchor of equipment nozzle.

Existing lines that are to be re-stressed will require substantial investigatory work by ENGINEERING CONTRACTOR to establish the ‘as-is’ condition with regard to supports, anchors, allowable movements, allowable forces, etc.

All lines shall be stress analysed and final route shall be indicated on piping layout. The layout shall include the type of support as guide, axial stopper, resting or anchors. ENGINEERING CONTRACTOR will produce Stress Analysis Report for Lines.

For the Lines, the location and type of each support (guide, axial stopper, resting and anchor) shall be identified on the piping layout.

ENGINEERING CONTRACTOR shall make use of piping standard details and piping support details of the existing facilities in the design of new facilities. In case of insufficient details, ENGINEERING CONTRACTOR shall generate the required details for the project.

a) ENGINEERING CONTRACTOR shall carry out stress analysis of the affected systems adhering to the stipulations of AGPS-SP-PI-004, and with piping basis of design which includes stress analysis requirement for critical piping system and relevant design Code ASME B 31.3.

b) “Stress Analysis shall be carried out considering the latest revision of Stress critical line list’’.

c) The piping shall be designed in general considering flexibility for thermal expansion. The piping shall be designed and checked to ensure that the pipe size, configuration, mechanical strength, supports and restraints will prevent excessive stresses, loads and vibration.

d) During FEED engineering CONTRACTOR shall carry out detailed stress analysis for applicable these lines. Stress analysis shall be carried out using latest version of computer software CAESAR II & Nozzle allowable (If deemed required) of existing vessels shall be performed by PV Elite software.

e) Stress analysis of the modified / new lines shall be carried out by the ENGINEERING CONTRACTOR for the new system and parts of the existing lines, which are affected by tie-in.

f) For design of new lines connected to existing piping lines or existing equipment approach to attend will be the following listed:

- Tie-ins, as possible, will be designed to minimize the effects on existing piping.
- If new lines configuration, influence or could influence existing parts it is necessary to consider existing piping until fixed points or logical discontinuity. When these points are not available, verification must be extended until upstream and downstream connection points or the model shall be extended to minimum two change of direction to get the effect of thermal displacement.

g) An interim report containing all necessary information along with native CAESAR II files shall be submitted for COMPANY's review and comment before proceeding with final report.

h) ENGINEERING CONTRACTOR shall identify stress critical lines based on the guidelines and stress analysis requirements in piping basis of design and specification for stress analysis; Stress analysis shall be performed for all stress critical lines.

i) All calculations shall be carried out with respect to static, slug flow, wind and seismic conditions, etc. with load combinations as per Specification. ENGINEERING CONTRACTOR to evaluate pipe forces and moments on nozzle attachments to mechanical and rotating equipment (tanks, pumps etc.), confirm that these are within the allowable limits.

j) ENGINEERING CONTRACTOR to perform flange leakage calculation for applicable lines as per specification.

k) Where possible, new piping arrangement supports location shall match with existing support's locations to utilize the existing support structures. ENGINEERING CONTRACTOR shall ensure that the civil support structure shall be adequate to meet the loads of new piping arrangement. Support location plan and load details for the new piping arrangement shall be prepared by ENGINEERING CONTRACTOR

l) If the new piping is connected to existing piping, the existing piping up to the first anchor/equipment nozzle shall be included in the stress analysis scope.

m) ENGINEERING CONTRACTOR shall design and prepare primary support (Pipe Shoe, Guide, Limit stop, etc.) details for new piping arrangement based on Stress analysis calculation results.

n) ENGINEERING CONTRACTOR shall submit pipe stress analysis reports, hard as well as soft copies (native data files) for COMPANY review and approval.

o) ENGINEERING CONTRACTOR shall incorporate all the findings of the pipe stress analysis into the piping layout design, GA drawings, isometrics and pipe support details.

p) ENGINEERING CONTRACTOR shall design required Special pipe supports for the PROJECT based on stress analysis results and special pipe support details shall be included with the Stress Analysis Report. ENGINEERING CONTRACTOR shall prepare specifications and data sheets for any piping special supports items required to be bought from vendors and submit for COMPANY approval.

q) All existing structure where temporary and permanent pipe supports being connected shall be checked for adequacy for new piping loads by ENGINEERING CONTRACTOR.

r) ENGINEERING CONTRACTOR shall prepare other applicable calculations required for the PROJECT and submit for COMPANY approval:

- Hot tap calculations, if applicable.
- Adequacy calculation of existing support structure/pipe rack etc.

s) Pipe support locations with proper legend and identification tags shall be indicated on Piping GA drawings and isometrics for all sizes as per CADD Manual Part B A0-ENG-D-PRO-002.

t) ENGINEERING CONTRACTOR to develop pipe support standard in line with COMPANY support standard also include new supports if warranted by final stress analysis reports.

u) Where use of existing pipe support structures, Pipe racks etc. are considered, ENGINEERING CONTRACTOR shall ensure that supporting of the existing pipework is not compromised.

v) Existing lines that are to be re-stress analyzed will require substantial investigatory work by ENGINEERING CONTRACTOR to establish the `as-is' condition with regard to supports, anchors, allowable movements, allowable forces, etc. A comprehensive stress analysis of the complete system shall be carried out by the ENGINEERING CONTRACTOR for the new system and parts of the existing system which are affected by tie-in. The following shall be verified as a minimum.

- All relevant loading options including sustained, thermal, long term soil settlement, wind effects, flange leakage calculation, surge calculation.
- Analysis of various operating, maintenance, testing conditions.
- Allowable equipment nozzle loading.
- Stresses are within allowable limits as per relevant code & specifications.
- To guarantee stability against any external loading conditions.
- To avoid excessive reactions on restraints and anchors.
- Adequacy of the existing supports and design
- Additional requirement of spring supports and expansion below
- Flange joint leakage criteria and torque values in the flange joints.

w) Piping shall be supported by using supports designed to include allowances for the weight of pipe when full of water, attached unsupported components, wind, shock, insulation, and internal forces.

x) Where use of existing pipe support structures, Pipe racks etc. are considered, ENGINEERING CONTRACTOR shall ensure that supporting of the existing pipework is not compromised.

10.3.6 Calculation Notes

ENGINEERING CONTRACTOR shall provide calculation notes, basis and assumptions used in the calculations. The calculations shall include but not be limited to pipe-wall thickness, pipe rack loadings (if applicable), pipe support loading, road crossing calculation, tank bund calculations, fire water demand calculation, etc.

10.3.7 Pipe Supports

Pipe supports shall be designed in accordance with COMPANY Standard AGPS-SP-PI-006/ AGPS-DW-PI-001 latest revision.

ENGINEERING CONTRACTOR shall as far as possible, make use of piping standard details and pipe support details of the existing facilities in the design of new facilities.

The ENGINEERING CONTRACTOR shall perform existing support strengthening wherever required based on detailed design conditions. However, the existing piping supports at facilities (if proposed to be used) shall be checked for adequacy by ENGINEERING CONTRACTOR and shall submit structural design calculations for all pipe supports.

ENGINEERING CONTRACTOR shall make the necessary modifications of any existing pipework to accommodate the new facilities. The modified existing pipework drawings shall be submitted to COMPANY for approval.

Pipe supports shall be provided for all the lines as listed in P&IDs and piping GA's. The existing lines with tie-ins shall be verified for supports, if necessary additional supports shall be provided. For stress critical lines, pipe supports shall comply with the requirements of stress analysis reports.

All pipes' sizes and above shall be designed with appropriate shoe supports whether insulated or not, to mitigate corrosion issues, adhering to specifications. Non-ferrous, stainless steel and CRA piping shall be protected at pipe supports against galvanic and crevice corrosion. For small bore lines clamps and "U" bolts are to be provided.

10.3.8 Piping Standard connection details

ENGINEERING CONTRACTOR shall prepare the required piping standard connection details for vents, drains, all instrument connections, etc. involved in project scope of work.

10.3.9 Tie-in Package

ENGINEERING CONTRACTOR shall, at an early stage in the execution of the work, identify all possible tie-ins and prepare a “Tie-in Planning Package”. ENGINEERING CONTRACTOR to study all tie-ins and prepare the tie-in package and issue to COMPANY for review and approval. The tie in package shall include basic minimum information such as:

- Tie-in Methodology
- Sequence of work
- Shutdown requirements & necessary isolation/ inserting
- Safety measures and precautions
- Systems affected, etc.
- Bills of materials for the tie-ins
- Isometrics details, piping GAs, PIDs, any special supports etc.
- Demolition drawings
- Material requisitions for all materials required to be procured for tie-ins.

Scope shall include study to determine options for making tie-ins without requiring production shutdown, using existing isolation valves etc.

All required material requisitions for tie-in points should be considered separately and should be issued at an early stage to optimize the project schedule.

ENGINEERING CONTRACTOR to note that the Tie-ins to be carried out in sequential and phased manner to avoid plant shutdown.

a) ENGINEERING CONTRACTOR shall check the feasibility of the locations and fully design the tie-ins, including detail, general arrangement, and isometric drawings. Final locations and details must be agreed with COMPANY.

b) ENGINEERING CONTRACTOR shall identify all piping tie-ins required to accommodate the new facilities or for modification of existing facilities. The ENGINEERING CONTRACTOR shall endeavor to minimize field joints and avoid plant shutdowns.

c) ENGINEERING CONTRACTOR shall develop a tie-in schedule aimed at improving Select / Conceptual stage study /schemes, based on prevalent site conditions, HSE & schedule requirements. ENGINEERING CONTRACTOR shall be solely responsible for achieving tie-in as necessary including any modification required at tie-in locations. It is the responsibility of ENGINEERING CONTRACTOR to verify the number of proposed tie-in points identified during the Conceptual stage study and to comply fully with all requirements of the PROJECT.

d) If Early Tie-ins to ne envisaged, ENGINEERING CONTRACTOR to prepare following deliverables as a minimum part of Early Tie-in Package:

- Early Tie-in Schedule
- Early Tie-in Details
- Early Tie-in Material Take- Offs
- Early Tie-in Piping GA drawings (Construction and Demolition)
- Early Tie-in Isometrics
- Early Tie-in location Plan
- Early Tie-in Support Schedule
- Material Requisitions for Early Tie-in Valves (If applicable)
- Technical Bid Analysis for early Tie-in Valves (If applicable)

e) ENGINEERING CONTRACTOR shall be responsible for finalizing all tie-in details, including but not limited to the following:

- Exact tie-in location.
- Verification of process design and operating conditions, and integrity of the existing piping system.
- Method of isolation for tie-in operations.

f) ENGINEERING CONTRACTOR shall also study/finalize any additional tie-ins required for facilitating the erection and commissioning of new equipment / piping system and to enable successful completion of PROJECT, which is part of FEED scope of work without any cost and time implication.

g) Tie-in design shall strive to minimize disruption to plant operations. ENGINEERING CONTRACTOR shall develop the piping design to avoid shutdown work. ENGINEERING CONTRACTOR to consider isolation valves as per isolation philosophy in process service for tie-ins works.

h) Tie-in piping design may undergo revision due to the constructability and access point of view and also due to the line isolation methodology as decided by Site operation. All necessary changes & modification shall be part of FEED Scope of Work without any cost and time implication.

i) No variation under the terms and conditions of this CONTRACT will be considered as a result of ENGINEERING CONTRACTOR changes to the COMPANY proposed tie-in locations, notwithstanding the requirement of COMPANY approval of tie-in locations.

j) During FEED engineering and based on Site visit ENGINEERING CONTRACTOR to take care of following,

- If it is found by ENGINEERING CONTRACTOR, during progress of engineering, that few tie-ins cannot be implemented without shutdown, then, in such cases the "Early Tie-in package" should be prepared with all details stated above with approval of COMPANY.
- Hot work shall be avoided wherever practically possible. Tie-in drawings must show both new lines and existing line modifications required to complete the tie-in. The existing piping shall be inspected, and its integrity established at the tie-in location by suitable

k) ENGINEERING CONTRACTOR shall produce a Tie-in report fully verified as feasible and acceptable by site investigation and survey.

10.3.10 Piping MTO'S and Material Requisition

MTO shall be carried out by ENGINEERING CONTRACTOR to quantify piping and piping component bulks required to carry out for the entire scope. The MTO shall be segregated according to the various locations and the areas and finally as net final quantities.

ENGINEERING CONTRACTOR shall develop MTO to identify piping bulk materials, special piping material, pipe supports, isolation valves, spring hangers, expansion bellows, etc. Material take-off for tie-ins shall be prepared separately and included in the tie-in package.

ENGINEERING CONTRACTOR to prepare the Material requisition for all long lead items & Early Tie-in.

The material take-off for piping scope shall be extracted from the PDMS model. ENGINEERING CONTRACTOR should consider any additional quantities as contingency / allowance within the total material take-off.

ENGINEERING CONTRACTOR shall carry out stage wise piping material take-off (preliminary, intermediate & final) during engineering. ENGINEERING CONTRACTOR shall ensure that piping material take-off shall include adequate contingency / allowance for fabrication, erection, installation, testing & 2-year operation spares. The format, description and minimum information required in the document shall be in line with COMPANY requirement.

10.3.11 Piping Design Dossiers

ENGINEERING CONTRACTOR shall prepare comprehensive design dossier covering all the piping design aspects.

ENGINEERING CONTRACTOR shall submit the content list and proposal for dossier format for COMPANY approval. As a minimum, all design analysis and installation aspects shall be covered in the design dossier.

Any assumption made in the design analysis shall be properly explained and justified.

All the input, output and summary results shall be in a format such that they can be traced and reviewed effectively. Wherever possible, ENGINEERING CONTRACTOR shall prepare sketches indicating applicable parameters on these sketches.

As a minimum the design dossier shall include:

- Standards, codes and guidelines.
- Design Parameters.
- List of drawings, data sheets, piping specifications for design, fabrication, erection and testing stages, etc.
- Plot plans and key plans as applicable.
- Piping layout, Piping General Arrangement drawings along with nozzle schedule as applicable.
- Data sheets applicable for all valves and piping specialty items.
- Piping support details.
- Stress Analysis reports (flexibility, wind & seismic analysis report)/ flange leakage calculation reports, Engineering data for Pipe rack / Support structure loading data, Critical Line List.
- Tie-in schedule (in COMPANY format)
- Equipment Layout in plans, elevations & sections.
- Nozzle Orientation for all vessels/Equipment as applicable.
- Underground Piping drawings showing location of all piping headers & branches, funnels, manholes, catch basins, clean out pits along with structural & equipment foundations, as applicable
- Fire water piping drawings showing location of supports, break flanges, sprinklers, monitors, hydrants, hose reels, as applicable
- Special support drawings & Pipe support cleats on equipment.
- Updated NDT specification for all piping classes based on COMPANY spec.
- Final Piping Bill of material summary.
- Final Support material summary.
- Update of piping design basis.
- Piping fabrication specs.
- Preparation of new specification required not covered above.
- Calculations may be presented in separate calculation dossiers depending on the size of the analysis.
- The details of Special supports
- ENGINEERING CONTRACTOR shall be responsible for inter-discipline

- Material Requisitions for the following are to be provided.
- Piping valves
- Piping special part items
- Piping special support items etc.
- Long lead items
- Early Tie-in Valves

10.3.12 Piping - General Requirements

At Select study stage phase, following typical plot plans for **Umm Shaif Plant** on Das Island are developed:

Doc./ Drawing No.	Document / Drawing Title
AD-90.0-PL-625000	DAS Island New Umm Shaif Plant - Unit Plot plans
AD-90.0-L-270000	DAS Island New Umm Shaif Plant - Key Plan
AD-90.0-L-200006	DAS Island New Umm Shaif Plant - Tie-in Location Plans

ENGINEERING CONTRACTOR to note that the layouts are typical and preliminary. The layouts shall be detailed and updated as part of this PROJECT. Specific layout for each package/ facility shall be produced. These shall be updated considering the equipment sizes/ inter-separation distances, material handling requirements, tentative vendor data and other safety considerations.

Areas on the island shall be divided into different sectioned based on functions (such as processing area, utilities, sea water treatment/intake, etc) and detailed layouts for each area shall be developed.

10.3.13 Piping - Specific Requirements

Layout developed for Greenfield and brownfield scope during DEFINE and FEED Engineering Stage shall consider the following as minimum:

1. Location and orientation of the facility with respect to prevailing wind.
2. The relative location of equipment to meet process system and operational requirements.
3. Operability, Constructability, Accessibility, Maintainability, Mechanical handling, HSE, Ergonomics, etc. during piping layout design development and plot plan / equipment layout shall be designed accordingly.
4. Minimize impact of new facilities introduced as part of scope on to existing facilities to minimize modification works
5. Future space provision, as applicable
6. Boat approach considering the largest supply vessel/boats.
7. Accessibility for operation and maintenance including escape routes and lay-down areas for material handling.
8. Constructability aspects
9. ENGINEERING CONTRACTOR shall identify interfaces considering existing plant operation and any ongoing projects. These interfaces shall be recorded as part of interface register and necessary approvals from COMPANY in this regard.

10. ENGINEERING CONTRACTOR shall undertake site surveys to validate Plot plan / equipment layout / Piping General Arrangement layout drawings, proposed modifications, piping routing, construction, maintenance and safety access, tie-in points, etc. ENGINEERING CONTRACTOR shall be responsible for physical verification and making corrections / incorporating the As-Built conditions in piping drawings that are applicable to project scope. Necessary modifications / relocations required on the adjoining existing piping system, structures, instrumentation / panels, electrical, etc. because of the new facilities shall be clearly defined and captured in respective drawings.
11. ENGINEERING CONTRACTOR shall prepare piping elevations and section drawings where required for ease of understanding. Necessary Plot plans, Piping GA drawings (Construction & Demolition drawing separately), demolition Isometrics shall be prepared for brownfield piping Construction & Demolition scope, partly removed or rerouted. Piping layout drawings shall be developed as per Company code of practice A0-ENG-D-PRO-002.
12. ENGINEERING CONTRACTOR to identify and minimize the shutdown requirements associated with each Tie-In. ENGINEERING CONTRACTOR to identify connections that can be considered for Early Tie-ins and develop separate package for standard tie-in and early tie-ins.
13. ENGINEERING CONTRACTOR shall ensure that interlinked walkways and escape routes are provided between high level access platforms to allow cross access and egress between equipment and or Vendor packages without the need to descend to grade and re-ascend to other elevated platforms in proximity.
14. Layout / safety aspects meeting the international and AGPS guidelines.
15. ENGINEERING CONTRACTOR to review location of the plant with respect to existing flare. Contractor to propose necessary associated modification required to develop the plot area for new US plant.
16. ENGINEERING CONTRACTOR shall provide piping inputs for update of Weight Control Report for all the Facilities.
17. ENGINEERING CONTRACTOR shall develop during DEFINE enough details to address above aspects adequately and systematically. These facilities shall be included in project technical reviews like HAZID review, Constructability review, SIMOPS review, 3-D MODEL review, PHSER and other safety reviews.
18. ENGINEERING CONTRACTOR shall review and address open action points relevant to the above and those identified for DEFINE in the Design stage drawing/ documents & workshops e.g., Constructability Review, Design Review, Lessons Learned, Value Engineering etc.
19. ENGINEERING CONTRACTOR shall, without any limitation, prepare complete set of piping deliverable for each facility to meet the project scope requirement.
20. ENGINEERING CONTRACTOR shall study options to optimize the facilities in DEFINE phase. However, the design / layout shall be developed in such a way that stage wise installation is possible without modification of existing and new installed facilities at later stage.

ENGINEERING CONTRACTOR shall ensure that for all the new equipment & facilities, sufficient details are provided so that engineering can be performed with minimum optimization.

Final layout arrangement shall be reviewed with COMPANY personnel, all relevant stake holders by conducting walk-through model reviews. ENGINEERING CONTRACTOR shall organize its own personnel and include necessary presentations accordingly for carrying out the same at site.

10.3.14 List of FEED - Engineering Deliverables:

- ENGINEERING CONTRACTOR shall prepare all deliverables required per COMPANY document A0-ENG-Z-GDL-001 (guideline for project deliverables) & A0-ENG-D-PRO-001 (Numbering Procedure) and scope of work duly complying with COMPANY Drawing Office Practice (DOP) for their preparation.
- The deliverables listed in ATTACHMENT-1 and as outlined in this document are minimum shall be developed as part of the project and submitted to COMPANY.
- CONTRACTOR shall ensure that all Integrity Engineering requirements for both new facilities and existing facilities impacted by the project are fully considered, in accordance with COMPANY-referenced technical standard documents Integrity Requirements during Define Stage and Define Gate
- The following Integrity requirements shall be considered during (FEED) [Reference is made to AGAI-GDL-0029 ADNOC Group Asset Integrity in Projects Guideline]
- COMPANY's Risk Target & Acceptance Criteria shall be as per (A0-IG-R-PRO-002) Risk Based Inspection (RBI) Assessment Procedure.
- The design takes appropriate steps to minimize factors that can adversely affect degradation rates and increase the requirements for mitigation and monitoring.
- The design of the facility shall provide sufficient integrity to enable preservation, decommissioning, abandonment and dismantling to be carried out safely.
- Hazard identification and risk assessment process are fully utilized and incorporated in the design phase to meet operational integrity objectives.
- Materials of construction shall be selected so as to avoid deterioration or control the rate of deterioration to an acceptable level when considered over the design life as per ADNOC Group Companies approved standards.
- Anticipated deterioration shall be controlled by suitable corrosion management measures and where applicable, suitable monitoring shall be put in place.
- All integrity related deliverables shall be developed as FEED Minimum project deliverables
- Integrity Assurance requirements for impacted existing facilities shall be carried out as mentioned in this SOW.

10.4 Integrity Engineering

CONTRACTOR shall ensure that all Integrity Engineering requirements for both new facilities and existing facilities impacted by the project are fully considered, in accordance with COMPANY-referenced technical standard documents

Integrity Requirements during Define Stage and Define Gate:
The following Integrity requirements shall be considered during (FEED) [Reference is made to AGAI-GDL 0029 ADNOC Group Asset Integrity in Projects Guideline]:

- COMPANY's Risk Target & Acceptance Criteria shall be as per (A0-IG-R-PRO-002) Risk Based Inspection (RBI) Assessment Procedure.
- The design takes appropriate steps to minimize factors that can adversely affect degradation rates and increase the requirements for mitigation and monitoring.
- The design of the facility shall provide sufficient integrity to enable preservation, decommissioning, abandonment and dismantling to be carried out safely.
- Hazard identification and risk assessment process are fully utilized and incorporated in the design phase to meet operational integrity objectives.
- Materials of construction shall be selected so as to avoid deterioration or control the rate of deterioration to an acceptable level when considered over the design life as per ADNOC Group Companies approved standards.
- Anticipated deterioration shall be controlled by suitable corrosion management measures and where applicable, suitable monitoring shall be put in place.
- All integrity related deliverables shall be developed as FEED Minimum project deliverables
- Integrity Assurance requirements for impacted existing facilities shall be carried out as mentioned in this SOW.

CONTRACTOR shall consider the following for Integrity Management during FEED:
Integrity Requirements for Existing Facilities Impacted by Projects or Site Schemes

Define Stage (FEED):
Any projects or site schemes that interface existing facilities shall carry out [during Define Stage] the following studies concerning the impacted facilities **at and downstream tie-ins**:

1) **Existing Material Adequacy Check Study:**
   A Material Adequacy Check Study is essential when new project scenarios impact existing facilities to ensure that the existing materials can withstand the new operating conditions and prevent any integrity issues. Here are some specific scenarios when such a study is required:
   - **a Changes in Operating Conditions and IOW:**
     Significant changes in operating pressure, temperature, and IOW limits
   - **b Changes in chemical composition of service fluid(s)**
     Changes in the composition or conditions of service fluids [Risk Drivers such as Hydrogen Sulfide (H_2S), Carbon Dioxide (CO_2), Oxygen (O_2) …etc.) can significantly impact the integrity and lifespan of equipment by increasing the degradation rate or initiating new damage mechanisms, identifying and understanding these changing risk drivers is crucial for effective corrosion management and maintaining process safety.

c **Changes in Flow Rate**

Significant changes in Flow Rate (increased or decreased)

d **Introduction of New Processes**

When new processes or fluids are introduced that could interact with existing materials differently.

Material Adequacy Check Study for existing materials impacted by the project/ site schemes is a **Mandatory Requirement** to be developed during DEFINE Stage and update during Execute Stage.

2) **Existing facilities Integrity Risk Profile Review:**

The objective of this study is to evaluate and mitigate potential integrity risks arising from the project new scenario, ensuring the Asset Integrity and Process Safety (AIPS) of the existing facilities (including pressure Vessels, Heat Exchangers, COS tanks, ATM Tanks, Steel Structures, Civil Works land-based structures like foundations, sleepers, pipe supports…etc.). This study shall consider the following:

a **Increased Load or Stress:**

When the new project scenario increases the mechanical load or stress on existing equipment and structures.

b **Material Aging or Degradation:**

When the existing materials have aged or degraded and need to be evaluated for their remaining life under new conditions

Equipment Integrity condition shall be determined from the operating, inspection, testing history (using passports, latest inspection, lessons learnt, RCA…etc.).

Identification of Boundaries and Scope for Existing Facilities Affected by the New Project Follow the New Streams:

- Trace the Process Flow: Start from the tie-in point where the new stream interfaces with the existing facilities. Follow the flow direction of the new streams using updated P&IDs and PFDs.
- Define the Fluid Path: Understand how the new fluids (with their unique operating parameters) will travel through the system and map their path through the existing facilities.

Set Boundary Criteria:

- Physical Equipment Changes: Stop at equipment where operational conditions (pressure, temperature, or fluid compatibility) exceed their design specifications.
- Operating Parameter Mismatches: Identify points where the new stream's parameters significantly differ from the existing facility's design parameters.
- Material Adequacy: Evaluate materials along the flow path. Stop at boundaries where material upgrades or changes are needed to handle the new fluids.

Integrity Requirements for Material selection:

- Life Cycle Cost Analysis Report
- Materials and Corrosion Audit “MCA”:
- Corrosion Risk Assessment Study “CRAS”
- Integrity Requirements for Corrosion Monitoring System
- Integrity Requirements for Chemical Injection Package:
- Dead Legs Management in Projects:
- Integrity Requirements for Cathodic Protection:
- Integrity Requirement for Soil/Sand Infill Beneath the Tank Bottom:
- Integrity Requirements for External and Internal Coating
- Integrity Requirements for Insulation
- Integrity Requirements for Passive Fire Protection (PFP)
- Integrity Requirements for Non-Metallic Material
- Integrity Requirements for Pipe Supports:
- Integrity Requirements for Small Bore Connections Requirements
- Integrity Requirements for Well Integrity
- Integrity Requirements for Lifting in Project
- Integrity Requirements for Steel Structures
- Integrity Requirements for Inspection Baseline Survey for New Facilities (SOR)
- Integrity Requirements for Risk Based Inspection “RBI” Baseline Passports (SOR)
- Integrity Requirements for Civil Works in Projects

10.5 Instrumentation & Control

10.5.1 FEED ENGINEERING SCOPE OF WORK

The following sub-headers represent the specific Instrumentation & Control scope and activities for DAS Minimum facilities FOR Nasr Crude Phase 1, which is to be detailed and completed by ENGINEERING CONTRACTOR.

10.5.2 SOW & Activities

10.5.2.1 Utilized Existing Control Systems in Zakum LER (DCS, ESD, F&G & MMS):

Site visit survey shall be conducted to verify the adequacy of the existing Control systems in ZAKUM LER to accommodate the new Instrument, F&G and MMS signals due to NASR minimum facilities.

Adequacy study reports shall be issued by ENGINEERING CONTRACTOR & OEMs to verify the available spares in the existing Control systems in ZAKUM LER.

The primary operation, control and monitoring of Minimum facilities shall be carried out of Minimum facilities from the existing DAS Island CCR; existing HMI stations shall be updated / upgraded with all necessary hardware and software.

Utilizing the existing safety network from Zakum LER to CCR for new SIL certified ESD IO's and existing FGS Soft matrix shall be upgraded as part of this Project.

10.5.2.2 New ICSS (in new LER or ZAKUM LER Extension)

In case of existing control systems are inadequate; the design of new LER or ZAKUM LER Extension shall be in ENGINEERING CONTRACTOR scope of work

Minimum facilities for the DAS Island, shall be designed with new Integrated Control and Safety Shutdown System (ICSS) consisting of PCS system for monitoring and controlling of process signals, SIL-3 certified ESD system for safety shutdown for the critical signals, and F&G System for F&G signals. New ICSS shall be installed in new LER or in existing ZAUM LER (Building shall be extended to accommodate new ICSS)

ENGINEERING CONTRACTOR shall engage existing control system OEM ABB for the design, engineering of new ICSS system for new facilities as part of this Project.

The primary operation, control and monitoring of Minimum facilities shall be carried out of Minimum facilities from the existing DAS Island CCR, utilizing existing HMI stations with necessary hardware and software upgrade.

The existing system in CCR shall be integrated with new ICSS in new LER or ZAKUM LER (extended). The existing 800xA Plant Operation network (ONET) from DAS Island CCB shall be integrated/extended to ZAKUM LER for the integration of the new ICSS with existing DCS system by providing new Connectivity server or reuse the existing server in CCR and ZAKUM LER.

New safety network from new LER or ZAKUM LER (extended) shall be extended to CCR for new SIL certified ESD and FGS Soft matrix shall be provided as part of this Project.

ENGINEERING CONTRACTOR shall evaluate for the integration of new Fire & Gas system as part of ICSS with existing FGS System – Complicity Network through considering new OPC server in new LER or ZAKUM LER(extended)and further monitoring of new US trains shall be from FACTORY TALK FGS system in ESCR with required necessary software configuration.

ENGINEERING CONTRACTOR is responsible to review existing Alarm philosophy for the configuration of alarm management of new ICSS, in the existing Alarm Management servers with necessary hardware and software upgrade to cater requirements as part of this Project.

New Asset Management system is being provided as part of LZ LTDP-1 project, shall be utilized for new US trains with necessary hardware and software upgrade.

New MMS system shall be considered with System-1 server for future integration with existing Machine monitoring systems of the plant.

ENGINEERING CONTRACTOR shall liaise with existing OEMs to get concurrence for respective system modification, integration and submitted as part of adequacy report during DEFINE phase.
ENGINEERING CONTRACTOR shall conduct site visit during DEFINE phase in liaison with site operation & control team to understand the present operation & control philosophy to identify hardware and software upgrade requirements. ENGINEERING CONTRACTOR is responsible to perform detail adequacy study for the required hardware and software modifications to integrate new ICSS with existing Control system and FGS system by engaging OEM and UPS adequacy to cater power requirements of the new I&C equipment.

As identified during CONCEPT DESIGN phase, there is space constraint in CCR to install new equipment part of this project, ENGINEERING CONTRACTOR shall conduct site visit to perform adequacy check and provide solution either to accommodate with in available space or considering extension of CCR.

10.5.3 List of I&C Deliverables

ATTACHMENT 1 provides the minimum Instrument and Control systems deliverables to be prepared by ENGINEERING CONTRACTOR. Any other deliverables that are not specified explicitly but are imperative during FEED engineering shall be considered by ENGINEERING CONTRACTOR for successful completion of the project.

10.6 Telecommunications

10.6.1 FEED ENGINEERING SCOPE OF WORK

The new NASR Phase 1 facility in Das Island shall be provided with Telecommunication equipment which shall be extended to the Das Island Central Telecom Systems / CCB as applicable. Telecom systems which require integration / extension at Das Island include Process CCTV System, Field Telephones, Field PAGA speakers, Network connectivity, Radio coverage, Access control system for building, WiFi connectivity as minimum. Fiber optic cable connectivity for Telecom, Instrumentation and control system extensions shall also be provided. All telecom systems shall be connected to network management system (NMS) enabled by AI.

As per ADNOC's vision and strategy, all communication shall be enabled by 5G in future. Contractor to ensure that 5G provision is considered. The 5G provision shall be designed in a way that the future migration from existing telecom to 5G shall be technically seamless.

10.6.2 General

ENGINEERING CONTRACTOR shall note that the minimum scope / design requirements for the Telecom systems intended for the new project facility in Das Island shall be in compliance with AGPS requirements. ENGINEERING CONTRACTOR shall also be responsible for designing Tie-ins with the existing Das Island Telecom Centre and Das Central Control Building (CCB) for the applicable Telecom Systems. ENGINEERING CONTRACTOR shall further design the FO Cable network, both for the termination / interface of the project FO network between new Das Facility and for the terrestrial FOC interconnections with existing Das Facility, CCB and Telecom Centre. ENGINEERING CONTRACTOR shall develop the Das Island new facility FEED deliverables in accordance to COMPANY standard GPE-GE-GDL-001 .

10.6.3 Adequacy Studies

ENGINEERING CONTRACTOR shall carry out site surveys and perform all necessary adequacy checks along with involvement of respective system OEMs for various Telecommunication Systems in Das Island Control Building and Telecom Center Building. Design shall be based on the outcome of the adequacy study, and the status of OTHER / Ongoing Project(s) in coordination with COMPANY. As minimum the following project Telecom systems require study and interface with respective Das Island systems.

Telephone connectivity and LAN infrastructure shall be extended to the new project facility from the existing PABX and Business / IT LAN infrastructure respectively at the DAS Telecom building, design shall be finalised based on the adequacy report on existing systems. New PAGA (AI enabled) System for New facility shall be integrated with any future / Central PAGA System in Das Island, as per adequacy / interface study and Company philosophy. Dedicated Process CCTV (AI enabled) System, ACS considered for the new facility shall be interfaced with ongoing / Future Das Island CCTV, ACS as per prevailing system status and in consultation with Company. UHF plant radio coverage as well as Paging system coverage for new Das Facility shall be extended from the existing Central Systems in Das Island – interface details shall be finalised by ENGINEERING CONTRACTOR. All Modification works as part of the project to be carried out on existing Telecom equipment, shall be through equipment OEM.

10.6.4 Engineering Deliverables

ENGINEERING CONTRACTOR shall refer to GPE-GE-GDL-001-R00 Standard Engineering Deliverables for Projects for the list of ENGINEERING Deliverables; Prepare and issue FEED Engineering of all Telecom deliverables including Layouts , calculations, schedules, datasheets, Tie in schedule ..etc and scope of work for all Required modifications for existing Telecom systems, as applicable through equipment OEM.

ENGINEERING CONTRACTOR shall be responsible for any other deliverable required for the Telecom scope, even if not given in this deliverable list.

10.6.5 Engineering Studies

ENGINEERING CONTRACTOR shall carryout coverage studies for CCTV (AI enabled), PAGA (AI enabled), WiFi and UHF to determine the location and quantity of Telecom equipment for the new facilities. ENGINEERING CONTRACTOR shall be responsible for any other Study / Engineering required for the Telecom scope, even if not given in this list. Contractor shall also include provision of 5G connectivity in future.

10.6.6 Interface/Tie-in scope

ENGINEERING CONTRACTOR shall be responsible for providing the Das Facility Telecom equipment design inputs to the TSI Scope of Work. Further ENGINEERING CONTRACTOR shall provide the necessary assistance for evaluating TSI proposals and finalizing / acceptance of the TSI equipment & services for the Das new facility scope.

10.7 Autonomous Oil Field – Digital and AI

10.7.1 FEED ENGINEERING SCOPE OF WORK

ENGINEERING CONTRACTOR shall define the functional and technical specifications, define scope parameters for new digital and AI solutions finalized as part of digital design thinking workshop and shall prepare the list deliverables.

ENGINEERING CONTRACTOR shall identify list of interfaces required between new digital and AI solutions and interface requirements to transfer the data between new digital and AI solutions and existing digital and AI solution.

ENGINEERING CONTRACTOR shall identify the list of digital and AI technologies from Attachmnet-5 that are applicable to new Das Island Facilities as part of the Project.

The Digital and AI requirements have been provided in the form of Checklist as provided as Attachmnet-5 of this document. The checklist is divided into 3 sections as below,

1. Foundational Requirements
   This includes the process instruments and control loops that are required for remote monitoring and control of the equipment. These foundational requirements are mostly part of standard engineering packages. Contractor to ensure that all instruments required to monitor and control the process parameters shall transmit the real time data to Central/remote Control Room. Please refer to Attachment-6 (Surface Facilities Instrumentation Guidelines) for the critical instrumentation mandatory for each equipment.

2. AI Requirements
   This includes the AI solutions/software that are required to operate the facilities' autonomously. Contractor shall refer to Attachment-5 and identify the AI applications that shall fit for the scope of Das Minimum Facilities. Contractor shall perform the value analysis of each applicable AI solutions (software/hardware) and get Company's consent before finalizing the Applications/solutions. Contractor to prepare the SOW for detailed engineering, Procurement and Construction for each of the selected AI solution.

3. Integration with Digital Factory
   ADNOC Offshore has an "In-House" setup for Digital and AI solutions that are being used for monitoring, analysis and diagnostics of static and rotary equipment. Several AI applications are being used based on real time data to monitor the equipment efficiency and perform prescriptive maintenance and process optimization.

10.8 Static Equipment

This section defines minimum requirements for Static Equipment engineering as part of the Front End Design ENGINEERING Contractor shall develop the Front End design based on the approved study endorsements.

Static Equipment Engineering Scope of WORK during FEED shall include, but not be limited to the following:

a. Prepare Design basis, Specifications, Mechanical Data sheets and Material requisitions as part of Front end design.

b. ENGINEERING CONTRACTOR shall review applicable AGPS specifications and make addendums to the specifications when any addition, deletion and/or modification is required. Where AGPS specifications does not exist for any type of equipment and packages then, independent project specification shall be prepared for that equipment / package.

c. Prepare and issue all documents and drawings included in “List of Deliverables” for COMPANY approval. Also develop any additional Specifications, Drawings and Documents that are required to complete the intended PROJECT SCOPE OF WORK.

d. Conduct site survey and data collection as necessary.

e. Known uncertainties, HOLDSs or considerations with respect to equipment and packages shall be resolved. Closely work with COMPANY for all design documents approval.

f. Prepare Mechanical Equipment List (input to Project Equipment List) as per COMPANY approved template.

g. Participate in HAZOP, 3D Model Reviews, and other multi-discipline reviews/ workshops or audits and .incorporate the recommendations/ findings in design deliverables.

i. Prepare Material Requisitions of Equipment for Inquiry and issue for COMPANY review and approval.

j. Prepare ‘Scope of Work’ for detailed Engineering, Procurement and Construction. This Scope of Work shall also include the scope of Engineering Sub-Contractor (engaged by PC) to execute detail engineering activities post PO placement of LLI items.

k. Review VENDORS and/or SUBCONTRACTORS technical bids received from Procurement & Construction Contractor and perform Technical Bid Evaluations. This shall include issuing necessary technical clarifications and conduct clarification meetings with VENDORS and/or SUBCONTRACTORS ensuring that technically suitable technologies conforming to the SPECIFICATION requirements are provided.

l. Participate in HAZOP, 3D Model Reviews, and other multi-discipline reviews/ workshops or audits and incorporate the recommendations/ findings in design deliverables and implement in the design.

m. Coordinate and resolve engineering queries with equipment VENDORS and/or SUBCONTRACTORS.

n. Establish criticality rating for equipment. Review & Approve VENDORS and/or SUBCONTRACTORS testing programs and procedures and participate in the inspection and testing of the equipment and packages, per COMPANY approved ITP.

o. Provide inputs for preparation of the Plant Operation, Maintenance and Safety Manual.

p. Prepare and issue inputs to support Project Close-out Reports and documentation.

10.8.1 Deliverables

ENGINEERING CONTRACTOR shall generate the documents and drawings as per the deliverables list attached in this SOW document and Company specification GDL-001. It shall include the following as a minimum.

ENGINEERING CONTRACTOR shall prepare Mechanical specifications, Datasheets Material requisitions and Technical Bid Evaluation for all Long lead items (LLI) mechanical items. The level of detail on these documents shall allow for enquiries to be placed for equipment immediately after completion of the FEED.

10.8.2 Design Basis

ENGINEERING CONTRACTOR shall prepare Mechanical Design Basis incorporating all the design criteria, environmental & safety requirements and shall align with ADNOC Group Practices and Standards (AGPS). COMPANY approved design basis shall form basis for further engineering of mechanical equipment

The applicable codes, standards, specifications and statutory regulations are listed in ITT. ENGINEERING CONTRACTOR shall use other relevant international codes and standards when so required to meet the technical requirements to furnish the guarantee for the specified performance and integrity of the equipment

10.8.3 Equipment Specifications

ENGINEERING CONTRACTOR shall review applicable AGPS specifications and make addendums to the specifications when any addition, deletion and/or modification is required. Specifications shall be prepared in accordance with the Project's standard units of measurement.

Specifications shall furnish all the design requirements commonly applicable for the category of the equipment covered by the specification.

Where AGPS specifications does not exist for any type of equipment then, independent project specification shall be prepared for that equipment. New specifications shall be issued in format, to be agreed with the COMPANY

10.8.4 Equipment Data Sheet

ENGINEERING CONTRACTOR shall prepare Mechanical Data Sheets for all equipment.

Process basis of design and related process data sheets shall be considered as guideline documents for finalizing the mechanical equipment design parameters

Extent of detail shall include material selection, minimum thicknesses, footprints, weights, nozzle sizes, flange ratings, etc. For shell and tube type heat exchangers, full thermal designs shall be performed and the data sheet shall include TEMA type, number of tubes, tube size, tube thickness and tube sheet layout, no. of passes, etc.

Datasheets shall be prepared in accordance with the Project's standard units of measurement. Data sheets shall be supplemented with preliminary design calculations to demonstrate critical component's thickness.

Each piece of equipment shall be specified by the CONTRACTOR on an equipment data sheet indicating min / normal / max flows (based on turn down and rated flow), pressure, temperature along with design conditions, etc.

When equipment (Stripper Column) sizing and performance guarantee are based on proprietary vendor components like trays/ demister / inlet device / distribution baffle etc., then ENGINEERING CONTRACTOR shall conduct enquiries and clarifications with relevant specialist vendors to ensure the corresponding equipment data sheet shall have sufficient information for purchasing.

10.8.5 Material Requisition

ENGINEERING CONTRACTOR shall be responsible for producing Material Requisitions (MRQ's) for all Long lead items (LLI) equipment (as required by project). Material Requisitions shall be prepared in COMPANY agreed template only.

10.8.6 Technical Bid Evaluation

ENGINEERING CONTRACTOR shall prepare and submit for COMPANY approval, TBE for all Long lead item (LLI) equipment. The contractor shall liaise with bidder and shall fill the TBE based on the input of the vendors. Technical Bid Evaluation report shall be submitted along with the technical bids, clarifications & other communications with the bidders.

10.8.7 Equipment Specific Requirements

Pressure vessels and Shell & Tube type heat exchangers shall be designed and constructed per ASME Section VIII, Div. 1 or Div. 2 as applicable and AGPS specifications. Equipment shall be ASME code stamped and registered with National Board of Boilers and Pressure Vessels. Shell and tube type heat exchangers shall be designed per Sub-section UHX of ASME Section VIII, Div. 1 and TEMA Standard.

NACE compliant and HIC resistant and tested materials shall be used for all the pressure vessels and heat exchangers. PWHT is also required.

10.9 ROTATING EQUIPMENT

This section shall be read in conjunction with other parts of the scope of work and any related contract documents. Services shall be provided accordingly.

- Currently, the planned rotating equipment is stripped crude pumps. However, FEED shall be carried out for any other equipment as evolved as part of FEED.
- While carrying out the FEED, ENGINEERING CONTRACTOR shall follow ADNOC AGPS, COMPANY's other applicable specifications and standards. Where not available, good engineering practice shall be followed however, for such nonavailability of AGPS, CONTRACTOR to discuss with COMPANY for agreement as to how to proceed
- Where required, ENGINEERING CONTRACTOR shall prepare addendum to AGPS to cover the project specific requirements. Any deviation to AGPS shall be formalized through Technical Deviation Authorization form Design change authorization etc.
- ENGINEERING CONTRACTOR shall prepare all the requisite engineering deliverables such as data sheet for pump, mechanical seal, specification for Pump, Material Requisitions etc. The data sheet shall be prepared in respective API format.
- The specification and material requisition shall cover the sufficient details to procure the pump along with its driver and instrumentation etc. from the Vendor.

- While carrying the FEED, ENGINEERING CONTRACTOR shall obtain preliminary Vendor information to feed to various other disciplines for size, weight, material handling, driver rating, utility requirements etc. The pump selection shall ensure that it is suitable to operate in parallel to existing pumps if specified on Process data sheet. The Pump operations shall have the auto start/stop and switchover capability in addition to real time monitoring of process parameters. Pump shall be equipped with machine monitoring system (MMS) to monitor the health, provide advisory and self corrective action.

- All documents prepared by the ENGINEERING CONTRACTOR shall be reviewed and approved by COMPANY.

- CRS shall be submitted along with documents in next revision.

- Being brown field nature of project, if required, site visit will be done and thereafter site visit report shall be issued to COMPANY for review and agreement to proceed further.

- Any existing documents required, ENGINEERING CONTRACTOR shall identify the same and obtain from COMPANY.

- Participate in various studies, workshops etc. to provide rotating equipment related input as required.

- When all the requisite deliverables for MR are approved, the MR along with its attachments shall be floated to ADNOC Approved Vendors and obtain the firm technical proposal.

- Vendor's proposal shall be reviewed with multidisciplinary approach and if required, queries shall be raised.

- Where required, meeting shall be conducted with vendors and finally technical evaluation report shall be issued which shall be reviewed and approved by COMPANY. The format for the TBE shall be agreed with COMPANY in advance.

- Unless agreed with COMPANY, ENGINEERING CONTRACTORshall ensure that minimum three technically acceptable vendors are there in the TBE.

- All the deviations proposed by Vendor shall be reviewed and agreed and the agreed deviation list shall be part of TBE. All the Vendor's proposal, TQ, MOM shall be part of TBE.

- Refer ATTACHMENT 1 for minimum deliverables to be prepared.

10.10 Electrical

10.10.1 FEED ENGINEERING SCOPE OF WORK

This section gives an overview of the electrical scope of work and defines minimum requirements for the electrical engineering to be carried out as part of FEED Engineering and other associated activities for DAS Minimum facilities FOR Nasr Crude Phase 1, which is to be detailed and completed by ENGINEERING CONTRACTOR.

The minimum requirements for this Facility are provided in this Scope of Work and Appendices. ENGINEERING CONTRACTOR shall develop from these basic requirements all necessary design, engineering for FEED Engineering,

10.10.2 Site Survey and Data Collection

Site visit survey shall be conducted to verify the electrical interfaces with existing facilities (electrical substations) to verify all the electrical tie-ins with existing electrical equipment which shall interface with including but not limited to (11kV Switchgears – 3.3kV Switchgears – 415V Switchgears, distribution panels,....etc) through utilizing of the existing spare feeders or the additional of new cubicles. in addition to survey the locations for the existing and the new electrical cable routing trenches,..etc.

10.10.3 Electrical Distribution

For NASR minimum facilities, the selected power supply option for the new facility in DAS Island is using the existing Utility Substation.

ENGINEERING CONTRACTOR shall study utilizing/modification of the spare power feeders on 11KV switchgear 90-SWGR-788 OR by the addition of two new cubicles feeders in Utility Boiler S/S-09 to feed the new pumps . ENGINEERING CONTRACTOR to refer to the attached Single Line Diagram for the 11kV Switchgear AD221-06.5-E-40025.

Electrical Modification works to be carried out on all existing equipment through equipment OEM.

Necessary control & protection cabling shall be modified as per the existing requirement.

10.10.4 Power Supply Facility for Minimum Facilities at Das Island

ENGINEERING CONTRACTOR shall investigate through the site visit all required electrical interfaces for 11kV & 3.3kV & 415V power supplies for all electrical loads in addition to the required power supplies for normal and emergency lighting systems. ENGINEERING Contractor to check the available spare breakers and sufficient power in the nearby normal and emergency DBs to feed the new lighting system after performing the required adequacy checks otherwise new power DBs for all electrical loads including but not limited to normal and emergency lighting, …etc to be provided.

ENGINEERING CONTRACTOR shall re-identify the emergency power requirement of minimum facility loads in the nearby substations. If available, contractor shall carryout adequacy check of existing nearby EDG for emergency power supply.

ENGINEERING CONTRACTOR shall re-identify the UPS power supply requirements in the nearby facilities. If available, contractor shall carry out adequacy check of existing nearby UPS power supply. If not exist ENGINEERING CONTRACTOR shall consider a new UPS system to feed all new UPS loads based on the minimum facilities.

FEED engineering design shall include any modifications/upgradations on existing equipment/facilities that are required for seamless integration of existing systems at existing facilities.

10.10.5 Electrical Load List /Adequacy Check

ENGINEERING CONTRACTOR shall update the existing Adequacy check report document (AD-90.0-E-40120) performed during the select study stage to check the adequacy for all interfaced electrical equipment based on the minimum facility loads.

Load List document no. AD-90.0-SCHDB-400004 shall be updated to reflect the new total loads based on the minimum facility requirements.

10.10.6 ECMS System Modification

ENGINEERING CONTRACTOR shall carry out ECMS Modification of existing ECMS in Utility Boiler Sub-station for the modification scope in the existing 11KV switchgear 90-SWGR-788 or 3.3kV Switchgear and modification shall be carried out through the ECMS existing OEM.

10.10.7 Lighting system

A complete normal and emergency lighting system shall be provided for the new minimum facilities including but not limited to (DBs – LED fixtures - poles, junction boxes, cabling, trays, etc).

10.10.8 Earthing system

A complete earthing system shall be provided for the new minimum facilities including but not limited to (earthing rods, pits, earthing bars, cables, etc). The new earthing grid shall be connected to the existing earthing network.

10.10.9 Cabling and Wiring

Design of Electrical cables (power & control) for feeding the Electrical loads for new process areas. The cables shall be laid in existing/new cable electrical cable trenches and on pipe rack cable trays.

10.10.10 Material Requisitions and TBEs

Prepare material/ purchase requisition for all new items (motors, Distribution boards, cables, bulk material, cable trays,.etc). Requisitions shall include well-defined inspection certification requirements, level of inspection, extent of inspection, commissioning assistance requirements etc.

10.10.11 Electrical Bill Of Materials

ENGINEERING CONTRACTOR shall prepare Material take offs for all itemized electrical equipment and bulk items but not limited to the following: Under itemized electrical equipment Electrical cables, bulk material like cable glands, junction boxes, cable lugs etc, cable tag, route markers, cable warning tape, cable ties, lighting system, earthing system, cable trays, LCS, MCTs,….etc shall be considered.

10.10.12 Other Engineering Deliverables

- ENGINEERING CONTRACTOR shall prepare the scope of work for all electrical equipment modifications (11kV/3.3kV switchgears, LV Switchgears, ECMS…etc) and place the order to OEMs for modification work.
- ENGINEERING CONTRACTOR shall perform power system study calculation (LF-SC-MS-HA) using ETAP software for the new electrical system after adding new loads.
- ENGINEERING CONTRACTOR scope of supply shall include relay setting coordination study to define the new relay settings for all additional new loads.
- Participate with Electrical input in the 3D model.
-

10.10.13 List of Deliverables

ATTACHMENT 1 provides the minimum electrical deliverables to be prepared. Any other deliverables not listed but required as per GPE-GE/GDL-001 required for FEED ENGINEERING stage shall be considered.

10.11 Civil & Structural

10.11.1 FEED ENGINEERING SCOPE OF WORK

This section gives an overview of the Civil & Structural scope of work and defines minimum requirements for the Civil & Structural engineering to be carried out as part of FEED Engineering and other associated activities for DAS Minimum facilities FOR Nasr Crude Phase 1, which is to be detailed and completed by ENGINEERING CONTRACTOR

The PROJECT scope includes design and detailed for FEED engineering for the following facilities:

1. Equipment supporting structure

2. Pipe racks

3. Pipe sleepers

4. Equipment foundations

5. Pipe supports

6. Electrical/Instrument/Telecom trenches including road crossing /duct banks etc.

7. Extension of Zakum LER building

8. other misc. items like paving / roads required for access including demolition of existing paving /road as applicable

The FEED engineering scope includes design and detailed and 3 D modelling of above and other facilities to complete the scope of work and also liaising with other departments to complete the scope work. Following works is envisaged in the FEED:

- Site visit of the area to collect the data and appraise of the plot plan and verify sufficiency of the area
- 3D modelling of the facilities based on approximate sizing provided by other disciplines.
- Basic design of above facilities to complete the FEED including calculation notes and drawings.
- Preparation of SOW for the geotechnical /Topographic and underground detection survey of the area under consideration.
- Preparation of UG composite drawings

- Preparation of FEED MTO for Civil/Structural works.
- Preparation of Cost estimate for Civil/Structural Items.
- Preparation of SOW for EPC works

The deliverable list for FEED will be as follows :

Sr. No.	Description	Remarks
1.	Civil Design basis	
2.	Civil/Structural specifications (Addendum to AGPS)	
3.	Design calculation for Pipe racks	
4	Design calculation for Equipment supporting structure	
5.	Design calculation for equipment (Stripper column/Pumps /heat exchanger)	
6.	Design calculation for Extension Zakum LER	
7.	Pipe rack drawings	
8.	Equipment supporting structure drawings	
7	Equipment foundation drawings	
8	UG composite drawings	
9.	Zakum LER extension drawings (Arch and Str. Drgs)	
10.	Road/Paving drawings	
11	SOW for Geotechnical/ Topographical /UG detection works	
12	Civil/Structural MTO	
13	Input to SOW for EPC scope of work	

10.11.2 Logistics & Transportation Study in Das Island

ENGINEERING CONTRACTOR to carry out Transportation study to transport the Equipment to the proposed project site taking into consideration the existing facilities and the feasibility of transportation on the most suitable route. A detailed UG detection survey shall be carried out for the entire route of transportation to detect and establish the presence of existing UG pipes/cables and avoid any potential hazard of clashes. For details refer Logistics & Transportation Study Report Doc. No. AD-37.0-C-33004.

10.11.3 List of Deliverables

ATTACHMENT 1 provides the minimum civil deliverables to be prepared. Any other deliverables not listed but required as per GPE-GE/GDL-001 required for FEED ENGINEERING stage shall be considered:

10.11.4 Buildings

This section gives an overview of the Architecture scope of work and defines minimum requirements for the architecture engineering to be carried out as part of FEED Engineering and other associated activities for DAS Minimum facilities FOR Nasr Crude Phase 1, which is to be detailed and completed by ENGINEERING CONTRACTOR

ENGINEERING CONTRACTOR shall design the following buildings and produce deliverables in line with the List of deliverables:

- LZ Outstation

11. SUBCONTRACT MANAGEMENT

In general, subcontracts are not allowed unless it is for specialist work or study that cannot be performed by ENGINEERING CONTRACTOR. ENGINEERING CONTRACTOR shall provide the table listing of SUBCONTRACTs works or study as part of the tender submission along with proposed parties responsible for the performance of the work.

Subcontracts for the performance of any portion of the WORK shall be only in accordance with the subcontracting plan contained in ENGINEERING CONTRACTOR's bid, and only after ENGINEERING CONTRACTOR has received written authorization from COMPANY that ENGINEERING CONTRACTOR may subcontract that portion of the WORK.

All Subcontracts services to be carried out by COMPANY approved Sub-contractors selected from COMPANY approved Contractor List (ACL).The selection and short-listing of SUBCONTRACTOR'S is ENGINEERING CONTRACTOR responsibility. In case of any shortage of bidders for any reason, ENGINEERING CONTRACTOR shall propose, with proper justification, additional bidders from their list and/or from lists of other ADNOC group of companies. A project-specific pre-qualification audit exercise shall be arranged by ENGINEERING CONTRACTOR including desktop and field reviews. All costs related to the project specific pre-qualification exercise shall be part of the Lump Sum.

CONTRACTOR to submit the Technical Bid Evaluation (TBE) report of minimum Three (3) technically compliant Bidder for COMPANY approval. CONTRACTOR to obtained prior approvals from COMPANY for such Sub-contractors.

For major SUBCONTRACT award, a final meeting prior to award is to be held with the successful bidder to avoid any misunderstanding or ambiguity in an effort to minimize any possible post award claims and disputes.

12. DRAWINGS AND DOCUMENT SERVICES

12.1 General Engineering Requirement

1. ENGINEERING CONTRACTOR shall produce the drawings and documentation according to the requirements set out in COMPANY Standard.

2. ENGINEERING CONTRACTOR shall review and validate correctness of applicable drawings through physical verification at site and update where necessary.

3. ENGINEERING CONTRACTOR shall maintain all technical data, calculations, and records in such a form as to permit a technical audit to be carried out at any time.

4. ENGINEERING CONTRACTOR shall furnish soft editable and/or pdf copies of project deliverables along with drawing list for each submission made to the COMPANY.

5. Soft copies of 'final' submission shall be made in five sets on DVD's, using writing software, preferably EZ DVD creator or Nero Burning ROM, ensuring proper closing of the DVD after writing it. Additionally, full-size prints of the final drawings (number as specified elsewhere in the contract), shall be provided.

6. COMPANY will allocate tag numbers for all major and minor equipment required for the Project. ENGINEERING CONTRACTOR shall be responsible for completion of applicable forms for requesting allocation of tag numbers.

7. In addition to hardcopy requirements stated elsewhere in the Contract; electronic copies of all the manuals specified in the deliverables shall be provided in the following formats:

    - The manuals shall be presented in MS Office format. Any embedded drawings can be in any format acceptable as per COMPANY standard. The function of this documentation manual is to enable updating of the same in case of change.

    - The manual shall also be presented in Acrobat PDF format (Portable Data Format). Each manual shall be combined to resemble the Hardcopy manual. The PDF manuals shall be indexed (linked) to at least the Table of Content level. The function of this manual is to enable publishing of the same.

    - All documentation shall adhere to COMPANY list of abbreviation and documentation and shall use COMPANY specific format for tags and lines.

8. ENGINEERING CONTRACTOR shall maintain all technical data, calculations and records in such a form as to permit a technical audit to be carried out at any time and thereafter shall make documents available for at least five (5) years, from the date of Completion of the Contract. The procedure for archiving shall be developed by ENGINEERING CONTRACTOR and shall be subject to COMPANY review/acceptance.

9. ENGINEERING CONTRACTOR shall provide all documentation for new equipment, facility registration in CMMS.

10. Provide listing of normal operating spares recommended by Vendor & COMPANY, unit price, etc. in COMPANY SPIR/e-SPIR format and documentation for its registration in CMMS.

11. Provide asset register and CMMS documentation for registration of new facility items in line with and compliance with COMPANY procedure for CMMS Requirement for Asset Register Data Build-up

Project Stage. Procedure (A0-OP-R-PRO-005) and templates will be provided later.

12. For all inspectable items, provide documentation for equipment registration in CMMS with all required mandatory documents as per APPENDIX-E of COP for Integrity Assurance in Projects (A0-IG-Z-CP-001). NEW TEMPLATE various sheets with prefix “INSP-“shall be used to prepare the data and cross reference the document numbers.

13. Provide data sheets/details in accordance with COMPANY standard for CMMS (Computerized Maintenance Management System).

14. Project document control / management and its compliance to specification for Information Management in Projects (A0-ENG-I-SP-001) including MDR/VDR templates.

15. Handover of final documentation in compliance to Final Dossier Handover Procedure (A0-ENG-D-PRO-003).

16. Ensure CMMS data requirements are included in the VENDOR document requirement lists (VDRLs) of all purchase order specifications.

17. A list of all new equipment shall be submitted to COMPANY TSD/ID department in order to segregate them and identify the inspectable items

18. The static mechanical equipment shall be identified from the equipment list and provided to COMPANY TSD/ID department and their relevant documentation, drawings and records shall be validated accordingly

19. Baseline inspection data is to be submitted to COMPANY TSD department and is to be uploaded into CMMS per COMPANY Specification A0-IG-Z-SP-001.

20. CONTRACTOR shall maintain a comprehensive documentation system which shall include a test dossier or set of such dossiers specifying site inspection and testing requirements for new equipment and their relevant records

12.2 Document Control Procedure

1. ENGINEERING CONTRACTOR shall prepare a detailed document control procedure for COMPANY approval and include the formats for COMPANY comment(s) and approval.

2. Approval by COMPANY of ENGINEERING CONTRACTOR’s documentation shall not in any way relieve ENGINEERING CONTRACTOR of any of its responsibilities or liabilities under the Contract.

12.3 Master Deliverable Register

1. ENGINEERING CONTRACTOR, within two (2) weeks of the EFFECTIVE DATE of the Contract, shall provide a Master Drawings and Document Register for all documentation proposed for the the project FEED ENGINEERING. The Register shall indicate for each drawing/document as a minimum and format/content shall comply with COMPANY specification for information management in projects (A0-ENG-I-SP-001).

    - Discipline
    - Drawing/document number
    - Title

- Class
- Revision control
- Forecast date of issue
- Forecast date by which comment/approval is required
- Actual date of issue
- Percentage completion
- Weight factor

2. The Master Drawings and Document Register shall be routinely updated (weekly) and provided to COMPANY as per the project procedures with weekly/monthly progress reports.

12.4 Delivery of Drawings and Documents

1. During the execution of the Contract, copies of documents required for approval / review / information shall be submitted by ENGINEERING CONTRACTOR in electronic format.

2. The exact requirement for document distribution to COMPANY shall be set out by ENGINEERING CONTRACTOR in Document Distribution Matrix to be included in the coordination/document control procedure as applicable. COMPANY shall review/approve the Document Distribution Matrix. It is the responsibility of ENGINEERING CONTRACTOR to prepare the project coordination/document control procedure, which should also incorporate requirements stated in this document.

3. Full size prints of all final drawings shall be provided (number of prints shall be as listed in coordination/document control procedure).

4. Documents/drawings for each location shall be developed separately.

5. ENGINEERING CONTRACTOR drawings and documentation shall be delivered in book form. The books (grouping together drawings, indices, lists and schedules) shall be of loose-leaf design and with a rigid cover that gives protection for the complete set of copies reduced to A3 format. Each volume shall not be more than 75 mm thick.

6. ENGINEERING CONTRACTOR at the end of the engineering activities of the project shall handover to COMPANY five (5) sets of DVD (s) ROM containing the CAD & PDF files of all the project drawings/documents, together with complete documentation list in the COMPANY approved format as per COMPANY standard, Final Dossier Handover Procedure (A0-ENG-D-PRO-003). For copying electronic files to DVD, the following points shall be adhered:

    - Normal DVD writing software is to be used, preferably, EZ CD creator or Nero Burning Rom. It is to be ensured to select to CLOSE the DVD after writing to it.
    - The written DVD shall be tested on a standard computer with no DVD writer hardware installed, to ensure that the content can be read properly.
    - Each DVD shall be labeled with ENGINEERING CONTRACTOR's name, project number/ title, contract number and date of issue or as per COMPANY standard for final handover documentation.
    - Classifications of documents are as follows:

- Class 1 – CONTRACTOR may not proceed with the associated part of the WORK prior to receiving COMPANY Approval.
- Class 2 –CONTRACTOR may proceed with the associated part of the WORK if not comments have been received from COMPANY after the due date.
- Class 3 – CONTRACTOR may proceed with the associated WORK

Class 1	Class 2	Class 3
Design Basis, philosophy documents, plot plans, scope of work, layouts, PFD, P&ID, Bid/Commercial evaluation documents (TBE/CBE), RFBA, LLI data sheets, Design calculations /reports, design drawings and planning package, construction & commissioning tender scope of work	Specifications, MRs, data sheets, site visit reports, Construction drawings, isometrics, project procedures, ITPs, Supplier GAs and supplier documents	MTO, fabrication drawings, cable schedules

12.5 Document Review and Turnaround

The following durations shall be taken into account for the schedule:

Ref. No.	Deliverable Type	1st Submission Working Days	Subsequent Submission Working Days
1	Technical Queries	5	5
2	Class 1 documents	5	5
3	Class 2 and Class 3 documents	5	5

Related activities shall only be considered complete once the documents are APPROVED by the COMPANY.

12.6 Document Distribution

During the execution process of the Contract, copies of documents required for approval / review / information shall be submitted by ENGINEERING CONTRACTOR. The exact requirement for document distribution to COMPANY shall be set out by ENGINEERING CONTRACTOR in Document Distribution Matrix to be included in the co-ordination/document control procedure.

12.7 Electronic Copies

Final submissions of hardcopy deliverables in five sets shall be accompanied with an electronic equivalent prepared as per COMPANY's standard and as follows:

1. Each individually bound/stapled/filed/collated hardcopy Volume, Manual, Book or Dossier complied as per the Contract, ENGINEERING CONTRACTOR supplies to COMPANY, shall be called the Hardcopy Manual.
2. As part of the final document deliverables, and unless specifically excluded in the Contract/PO, for each hardcopy manual that ENGINEERING CONTRACTOR supplies to COMPANY, there shall be an equivalent deliverable in electronic format which shall be called an e-Manual.
3. The specifications and approval of the e-Manuals are governed by COMPANY standard.

4. Unless the e-Manuals are specifically approved by COMPANY, the Contract document deliverables shall not be considered approved.

13. FINAL DOCUMENTATION

Final FEED documentation shall be in compliance with COMPANY standard, Final Dossier Handover Procedure (A0-ENG-D-PRO-003). The format, indices and extent of the content of the final documentation handover index and deliverables will be subject to COMPANY written approval prior to the start of compilation. ENGINEERING CONTRACTOR shall arrange a meeting with COMPANY, discuss, clarify doubts and agree the contents/formats to be followed.

Final documentation prepared by ENGINEERING CONTRACTOR and submitted to COMPANY in five sets shall include, but not be limited to, the following documents. Contractor to consider the AI enabled document repository and retrieval system.

13.1 Design Report

ENGINEERING CONTRACTOR shall prepare and issue a detailed FEED design report after the Issued for Design documents.

13.2 Design Manual

ENGINEERING CONTRACTOR shall prepare and issue a design manual after the Issued for Design documents/drawings.

The design manual shall include but not be limited to:

- Drawing / Document index
- Specifications segregated per discipline
- Project philosophies segregated by discipline
- All data sheets segregated per discipline
- All documents segregated by discipline
- All drawings segregated per discipline
- All calculation notes segregated per discipline
- Process simulation models, PDMS models, etc.
- All study reports

13.3 Operating, Maintenance and Safety Manuals (As applicable)

ENGINEERING CONTRACTOR shall develop and submit to COMPANY for approval, a proforma (table of contents, format, presentation, graphics, style, proposed content etc.) for the preparation of systematic (process, utility, instrumentation, electrical, telecommunication, materials handling and all other necessary) operating procedures/manual.

The Operations and maintenance manuals & procedures shall be in digital format. These manuals & procedures shall be uploaded in Company's existing digital platform and LMS (learning management system). Contractor to consider the AI enabled document repository and retrieval system.

These operating procedures shall cover black start-up, cold start-up, start-up after maintenance, hot start-up, normal operations, planned shutdown and emergency shutdown and blow down as applicable for the WORK. They shall also cover all temporary operations, including temporary pigging (and safe and environmentally acceptable disposal of pigging fluids).

ENGINEERING CONTRACTOR shall prepare an Operating, maintenance and safety manual. Preparation of this document shall require close interaction with COMPANY in order to ensure conformity with COMPANY requirements. ENGINEERING CONTRACTOR and specialized SUBCONTRACTOR(s) shall, therefore, jointly develop the Table of Contents in line with ADNOC standard (A0-OP-M-SP-001, Specification for Operating Manuals & Operating Procedures) and obtain COMPANY approval prior to proceeding with the WORK.

ENGINEERING CONTRACTOR shall prepare the basis for operating manuals for further development and completion of final manuals by a specialist SUBCONTRACTOR. ENGINEERING CONTRACTOR shall award a subcontract to "an experienced" SUBCONTRACTOR, subject to COMPANY approval, for preparation of electronic operating manuals incorporating interactive capabilities between drawings, texts and technical data.

The following are general contents of a typical manual, reference is made to A0-OP-M-SP-001. A soft copy of COMPANY'S typical available manual will be provided at ENGINEERING CONTRACTOR'S request.

Operating Manual

- Design basis
- Process description –including inlet and outlet products composition and flow rates, material balances and design conditions, consumption of chemical products, Process flow sheets and P&ID's and equipment lists.
- Control Systems – Control Strategy, Local Control/Equipment Rooms, DCS, ESD, F&G, and Cause and effect Charts
- Instrument Summary- Design Conditions, Control Valve Summary, Alarm and Trip Settings, Orifice Plate Summary.
- Troubleshooting
- HSE- HSE Protection Policy/Philosophy, Loss Prevention/Environmental Protection and Waste Management, Hazards in Workplace, General Safety Procedures, Material Safety Data Sheets, Safety Equipment Distribution Map, Escape Route layout and Relief Valve Settings

Operating Procedures

- Initial start-up

- Start-up after power failure or black start
- Start-up after major maintenance to steady operation
- Start-up from standby to normal operation
- Start-up after ESD1, ESD 2, and ESD 3 shutdowns
- Shutdown from normal operation for major maintenance
- Shutdown from normal operation to standby
- Emergency shutdown ESD 1, ESD 2, and ESD 3
- Chemical refilling

Further, Contractor to refer the existing Operating, maintenance and safety manual of DAS Umm Shaif plant.

The existing operating, maintenance and safety manual shall be updated as an addendum in the FEED stage related to scope of work.

14. LIST OF ATTACHMENTS

ATTACHMENT 1	Engineering Report (CED Job No.: C1292)
ATTACHMENT 2	Uweinat Cap Gas to ABK - TABK Study Report (TR21-TMT-ADNOC-MEM-0003)
ATTACHMENT 3	AGPS DOCUMENT CROSS REFERENCE TO AGES STANDARDS
ATTACHMENT 4	Company project control requirements - A0-PSD-PCT-PRO-004 Rev.1
ATTACHMENT 5	ADNOC Offshore Autonomous Operations Compliance Checklist for LTDP projects
ATTACHMENT 6	ADNOC Smart Upstream - Surface Facilities Instrumentation Guideline – ASU-WP2-DS-HQ-SF-IAF-04-Aug 2023
ATTACHMENT 7	List of deliverables